

MINDMATE: A FLUTTER-BASED MENTAL HEALTH APP FOR UNIVERSITY
STUDENTS WITH EMOTION TRACKING AND AI-POWERED VIRTUAL
ASSISTANT

LAU YUN XI

UNIVERSITI TEKNOLOGI MALAYSIA

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TS. DR. JOHANNA BINTI AHMAD
SENIOR LECTURER
FACULTY OF COMPUTING
UNIVERSITI TEKNOLOGI MALAYSIA
81310, SKUDAI, JOHOR
TEL: 012-8184435
EMAIL: johanna@utm.my

SIGNATURE OF STUDENT

SIGNATURE OF SUPERVISOR

030708-07-0520

MATRIX NUMBER

DR. JOHANNA BT AHMAD

NAME OF SUPERVISOR

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Signature

:



TS. DR. JOHANNA BINTI AHMAD
SENIOR LECTURER
FACULTY OF COMPUTING
UNIVERSITI TEKNOLOGI MALAYSIA
#1310, SKUDAI, JOHOR
TEL: 012-8184435
EMAIL: johanna@utm.my

Name of Supervisor : JOHANNA BT AHMAD

Date : 13 MAY 2025

ON-LINE RECOGNITION OF DEVELOPING CONTROL CHART PATTERNS

LAU YUN XI

A thesis submitted in fulfilment of the
requirements for the award of the degree of
Bachelor of Computer Science (Software Engineering)

Faculty of Computing
Universiti Teknologi Malaysia

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DEDICATION

This thesis is dedicated to my father, who instilled in me the importance of learning not for grades, but for the sake of knowledge itself. It is also dedicated to my mother, who taught me that any challenge, no matter how large, can be tackled one step at a time.

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ABSTRACT

This study focuses on the development of MindMate, a Flutter-based mobile application designed to support university students' mental health. The app is aimed at addressing the increasing mental health concerns among students by providing an easy-to-use tool for emotional tracking, insights, and support. The stakeholder, UTM (Universiti Teknologi Malaysia), expressed concerns about the rising number of students experiencing mental health issues, especially during stressful academic periods. MindMate helps alleviate this by allowing users to track their emotions daily, view their emotional trends over time, and interact with an AI-powered virtual assistant for guidance. The app uses Firebase for user authentication, data storage, and real-time interaction. The MindMate app involves two types of users: students, and administrators. Students can log their emotions, interact with the AI chatbot, and receive mental health insights, while administrators manage user accounts and system data. The app is built using Flutter, Dialogflow for AI-powered chatbot functionality, and Firebase firestore for storing user data and logs. The development follows a Plan-driven methodology with a focus on structured planning and documentation, and UML diagrams were employed during the requirement and design phases. The project was tested using black-box testing and usability testing to ensure its effectiveness and ease of use.

ABSTRAK

Kajian ini memberi tumpuan kepada pembangunan MindMate, aplikasi mudah alih berasaskan Flutter yang direka untuk menyokong kesihatan mental pelajar universiti. Aplikasi ini bertujuan untuk menangani kebimbangan yang semakin meningkat mengenai masalah kesihatan mental di kalangan pelajar dengan menyediakan alat yang mudah digunakan untuk penjejakan emosi, pandangan, dan sokongan. Pemegang taruh, UTM (Universiti Teknologi Malaysia), telah menyuarakan kebimbangan mereka mengenai peningkatan jumlah pelajar yang mengalami masalah kesihatan mental, terutamanya semasa tempoh akademik yang stres. MindMate membantu mengatasi masalah ini dengan membolehkan pengguna menjejaki emosi mereka setiap hari, melihat corak emosi mereka dari masa ke masa, dan berinteraksi dengan pembantu maya berkuasa AI untuk mendapatkan panduan. Aplikasi ini menggunakan Firebase untuk pengesahan pengguna, penyimpanan data, dan interaksi masa nyata. Aplikasi MindMate melibatkan dua jenis pengguna: pelajar dan pentadbir. Pelajar boleh mencatatkan emosi mereka, berinteraksi dengan chatbot AI, dan menerima pandangan mengenai kesihatan mental mereka, manakala pentadbir menguruskan akaun pengguna dan data sistem. Aplikasi ini dibina menggunakan Flutter, Dialogflow untuk fungsi chatbot AI, dan Firebase Firestore untuk menyimpan data pengguna dan log. Pembangunan aplikasi mengikuti metodologi Plan-driven yang memberi tumpuan kepada perancangan dan dokumentasi yang terstruktur, dan diagram UML digunakan semasa fasa keperluan dan reka bentuk. Projek ini diuji menggunakan ujian kotak hitam dan ujian kebolehgunaan untuk memastikan keberkesanan dan kemudahan penggunaan aplikasi.

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LIST OF ABBREVIATIONS

AI	- Artificial Intelligence. In MindMate, used to support the chatbot virtual assistant for mental health conversations.
API	- Application Programming Interface – A set of rules that allows systems to communicate with each other.
Chatbot	- A conversational interface powered by AI, designed to interact with users and provide emotional support.
Dialogflow	- A Google cloud-based service that enables conversational AI, used for chatbot interaction.
ERD	- Entity Relationship Diagram. A data modeling technique that shows the relationships between different data entities.
Firebase	- A cloud-based platform by Google that provides backend services like authentication and database management.
Flutter	- An open-source UI toolkit developed by Google for creating natively compiled applications from a single codebase.
Gamification	- The application of game-design elements such as rewards, streaks, or badges in non-game contexts to increase engagement.
OAuth	- Open Authorization – A secure protocol for token-based user authentication (e.g., Google or Facebook login).
SDS / SDD	- Software Design Specification / Software Design Description. A document that outlines how software will be designed and structured.
SRS	- Software Requirements Specification – A document that outlines detailed software system requirements.
STD	- Software Test Documentation. A document that describes how the app is tested to make sure it works correctly.
UI	- User Interface – The layout and interactive elements through which users interact with an application.
UML	- Unified Modeling Language. A standardized modeling language used for visualizing system design (e.g., use case, class, and activity diagrams).
UTM	- Universiti Teknologi Malaysia – The institution under which this project is conducted.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Mental health is now an important concern of student wellbeing recognition, particularly at higher education learning institutions where academic pressures, social transitions and personal challenges are present (WHO, 2021). Students at the university are somehow affected by anxiety, depression and stress but due to stigmas, high demand and institutional barrier; students do not find the normal mental support (Hunt & Eisenberg, 2010). Being digital natives, students are progressively resorting to mobile technology to meet their emotional needs, so mental health apps should be seen as a valuable way of aiding (Firth et al., 2017).

MindMate is an application that should assist university students in their mental health by tracking emotions and providing virtual assistance with the help of artificial intelligence. The app was created on the Flutter framework and enables the users (students) to record their everyday feelings, breathing and mindfulness exercises designed according to their moods, and communicate with a smart assistant that will support their experience in real time. It is targeted at enabling students to build emotional strength and educating them to deal with stress rather than react to it privately and conveniently and in an engaging manner.

1.2 Problem Background

University students have increasingly earned mental health as a matter of great concern, particularly when they have to interact with a new academic, social, and

personal responsibility environment in higher school settings. During this time, students are usually subjected to different sources of stress which could include school work, financial pressures, homesickness, time management and pressure from peers. These exert a great deal of pressure and distress on the emotional level and psychological level when not dealt with.

Academic stress is a prevalent factor to mental health disorders among Malaysian students. According to Hasim et al. (2023), engineering students under study in public universities were examined, and poor sleep quality and high levels of stress were determined to affect emotional well-being directly. In Universiti Teknologi Malaysia (UTM), students in their first year of study usually have problems in settling a new environment, whereas their fourth year students face the onset of pressure in thesis writing as well as confusion of finding a job and performance expectations of a jobseeker.

Although UTM offers formal counselling services to enhance the well-being of students, most of them are reluctant to seek help. Students do not choose to use professional support due to barriers which may include social stigma, limited slots in which students can get counselling, fear of confidentiality and time limits (Eisenberg et al., 2007; Gulliver et al., 2010). The interviews given to UTM students (Appendix C) also indicated that the majority of students know that they are emotionally stressed yet they shy away seeking the help of counsellors as they fear stigmatization, or they have to wait long durations to be served by the counsellor or they have no time to attend the sessions. Both first-year and final-year participants felt that they needed discreet, quick, and empowering mobile tools letting them be able to communicate emotions publicly and privately. To limit these drawbacks, a mobile application, which can be used more readily by students in need than mindfulness, is the offering, i.e., MindMate. The application combines real-time emotion monitoring system, an AI-based virtual coach that will engage in motivating conversation, and gamified self-care tasks. By making the support anonymous, non-judgmental and individualized, MindMate fills the gap that might exist between formal mental health support and the everyday emotional support students need in their lives. Its design will be to encourage

proactive mental health practices, diminish stigma, and stimulate self-introspection in a digital space that forms a comfortable and enjoyable environment for the user.

To eliminate such restrictions, MindMate, a mobile app, can be suggested, containing a more convenient and accessible to students' method of emotional support. The app incorporates real-time emotion monitoring, an AI-based virtual behavior support coach that people can use to chat motivating conversations to stay focused, and gamified exercises to do self-care. Because it facilitates anonymous, non-judgmental, and personalized care, MindMate disconnects formal mental health services and daily emotions of students. Its structure will foster active mental health practices and de-stigmatizing mentality and will also facilitate self-reflection in an approachable and interactive computerized setting.

1.3 Project Aim

To design and develop a mobile application to maintain the mental health of the students of the university with the features of monitoring emotions and an AI assistant within the framework of Flutter.

1.4 Project Objectives

- (a) To analyze the requirements for developing MindMate: A Flutter-Based Mental Health App.
- (b) To design MindMate: A Flutter-Based Mental Health App with an emotion tracking feature that helps users identify, monitor, and reflect on their emotional states in relation to academic demands.
- (c) To implement a virtual assistant driven by AI that depending on user's emotion history, offering them personalized coping mechanism, motivational support and writing prompts.

- (d) To evaluate the effectiveness of the app in helping students manage stress and improve emotional awareness.

1.5 Project Scope

The scopes of the project are:

- (a) The MindMate app is a cross platform using Flutter to develop mobile applications that run on the iOS and Android platforms.
- (b) MindMate app allows users to log and monitor their feelings including happy, anxious, stressed, overwhelmed and focused with the help of a daily mood entry.
- (c) The assistance available in MindMate app is AI-based and offers real-time response with NLP through Dialogflow. It will provide inspirational messages and soothing ideas.
- (d) MindMate App will keep and process user emotion history by weekly summaries and by plotting of mood curve to aid students on reflecting their progress.
- (e) MindMate app offers such gamification as streaks counter and badges to encourage frequent use.
- (f) MindMate app will not provide professional medical diagnosis or treatment.
- (g) MindMate app will not replace licensed therapists or mental health professionals which requires professional human expertise.
- (h) MindMate app will not offer real time human chat support.

1.6 Project Importance

The development of MindMate the need to build a stigma-free and easy approach to mental health care with personalization. Relying on the combination of emotion tracking and AI-powered conversations, MindMate enables students to develop self-care, awareness of their mental health, and welcome feedback without judgments and outside pressure.

This app contrasts with other fixed self-help applications; it develops with the input of the user, hence remained relevant and attractive (Torious et al., 2018). Besides, its evolution not only serves the cause of digital mental health but also shows how mobile tools could be used to take emotional wellness in a purposeful and effective direction (Holmes et al., 2020).

It has the potential to be used more widely in institutions as well, as it will give universities an optional and scalable tool that will help to establish emotional awareness and self-regulation.

1.7 Report Organization

In this section, the structure of the entire report will be elaborated upon. This report is divided into 5 chapters namely Introduction, Literature Review, System Development Methodology, Requirement Analysis and Design and Conclusion. Chapter 1 is an introduction of MindMate. In Chapter 2, the author has discussed the theoretical overview of the application through comparing and contrasting the existing application with MindMate and analysing the similarities and differences between the two. In chapter 3, the author addresses the chosen methods, technologies and system requirements that were employed in development of MindMate. This section on system design of the project talks about the architecture of the system like system requirements, database design and system user interface in Chapter 4. Chapter 5 summarizes what has been accomplished in the entire system and what the future work should be done in order to enhance the system feature.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The availability of mobile apps that help one with mental support is on the increase as they are aimed to help individuals under the stress, anxiety or depression. As smartphones and AI technology keep improving, features such as mood tracking, virtual assistants and even mini games start to pop up in many applications and keep the users occupied. This chapter examines some of the apps which are already on the market, how they perform and what is lacking in them. It also reveals the difference of MindMate and why it might be more useful to our university students.

2.1.1 The Typical Stressors in University Students

Students in the university are faced by countless stressors which influence their mental and academic capacity. These stressors are normally related to each other and can recur at the same time:

- **Academic Stress:** This is the most common factor that results in stress and includes pressure to have good grades and good performance in examinations and other course requirements (Misra & McKean, 2000; Hasim, Zainal, & Jaffri, 2023).
- **Financial Stress:** Fear about the cost of education, maintenance of day-to-day life and financial assistance may cause apprehension and reduce concentration (Robotham, 2008).

- **Social and Pressure:** Students undergo difficulties with establishing new relationships, peer comparison, keeping social status, which all lead to emotional distress (Beiter et al., 2015).
- **Time Management Stress:** Lack of success in maintaining the balance between academics and personal life and other activities may cause the feeling of being overwhelmed (Beiter et al., 2015).

It is of essence to unravel such stressors so as to formulate a system that would support the true emotional needs of students.

2.1.2 Non-Complex Counselling of the Student Stress

There are numerous evidence-based approaches applied by counsellors in assisting students who are struggling with stressful challenges to their psyche. They are focused on better emotional regulation, decision-making, and mental health on the whole:

- **Cognitive Behavioral Therapy (CBT):** CBT is used to enable the individuals to recognize and dispute irrational or nonproductive ways of thoughts and substitute them with a more constructive way of thinking (Reavley & Jorm, 2010).
- **Mindfulness-Based interventions:** These include activities such as breathing exercises and meditation that have the effect of letting the students learn how to concentrate on the present state of being and thus also minimizing stress (Shapiro, Brown, & Astin, 2011).
- **Motivational Interviewing (MI):** This is a client-specific orientation form of counselling on increasing motivation to change through addressing and eliminating ambivalence (Naar-King & Suarez, 2011).

- **Journaling and Self-Reflective:** Journaling therapy enables human beings to contemplate feelings and monitor progress of their mental wellness (Baikie & Wilhelm, 2005).
- **Positive Reinforcement:** This is a strategy of seeking to reinforce positive behavior in a bid to promote mental health practices and long change (Ryan & Deci, 2000).

These strategies are successful in clinical as well as self-help settings and therefore they are fit to be used as a component of mobile applications.

2.1.3 MindMate incorporating the use of Counselling Approaches

The MindMate application uses digital technology, to imbibe the essential elements of counselling strategies into the application making the user experience a personalised and stigma-free mental wellness platform.

- **CBT Techniques:** The AI chatbot imitates CBT strategies by informing users with motivational messages, as well as assisting them in reframing negative thinking on the basis of mood diaries (Fitzpatrick, Darcy, & Vierhile, 2017).
- **Mindfulness:** It includes activities like the breathing process and relaxation prompts especially in cases where stress patterns are repeated.
- **Motivational Interviewing:** The chat bot mimics the simple use of the MI methods, modifying its behavior depending on the emotional patterns of the user with the passage of time.
- **Journaling:** keeping an emotion log and the chat history is a version of expressive writing and it enables the student to be able to take notings on their past experiences (Kauer, Reid, & Sanci, 2009).

- Gamification for reinforcement: Despite the app being, in general, a positively valued support service, it still bases its self-care support system on digital reinforcement methods by giving badges and streaks in the event of regular self-tracking emotions (Lister et al., 2014).

These attributes add to an intelligent support system which reflects some aspects of professional counselling but is easily accessible and appealing.

2.2 Case Study

This section provides the description of the current state of mental health support to students and indicate the necessity to implement the creation of MindMate. It reveals the issues of the existing system and how the app can assist students to refer more to the monitoring of their wellbeing. The knowledge here is founded on what students and the university supporting services are doing.

2.2.1 As-Is Process – Manual university counselling service

In most modern university environments, formatted mental-health care is an almost exclusive responsibility of professional counsellors. World Health Organization (2022) and Eisenberg et al. (2007) state that counselling is meant to help students cope with stress, anxiety, depression, and any academic-related pressure. Generally, a counselling interaction involves:

- Evaluating the emotional situation of the student by means of oral conversation or distribution of structured questionnaires.
- Engaging the student in reflective conversation that was directed to explain and work out the feelings of the learner.
- The encouragement of motivation, building coping skills, and the facilitation of self-care processes.

- Helping promotes a long-term practice of self-awareness as well as regular emotional checks.

Despite these interventions, the dominant counselling approach is facing a lot of problems:

- Low Availability: Because of the high number of people in need, the waiting time can be quite long (Hasim et al., 2023).
- Stigma: The fear of being judged can make the students fail to seek help (Gulliver et al., 2010).
- Non-consistent Monitoring: Emotions do not remain tracked in between sessions.
- Accessibility Barriers: The students can be uncomfortable with face-to-face contact or would not have time to meet at indicated hours.

In as much as the counselling services offered by UTM provide guidance through well-trained personnel, a number of gaps in their operations restrict their usefulness. The time it takes to get an appointment is frequently days to weeks and in times of severe stress, students will not receive immediate assistance. Moreover, no continuous emotional monitoring takes place implying that the mood changes or a persistent negative mood during the period between sessions are not tracked.

These results are supported by qualitative assessment on student interviews (Appendix C) among the majority of the students voiced they simply did not know how to make an appointment to use counselling services or were disheartened by the stigma and conflict of time schedules. By doing this, then emotional problems might not be attended to until it is too late. This points out as an imperative to have a personal, more immediate, and discreet, and student-focused mental health support tool.

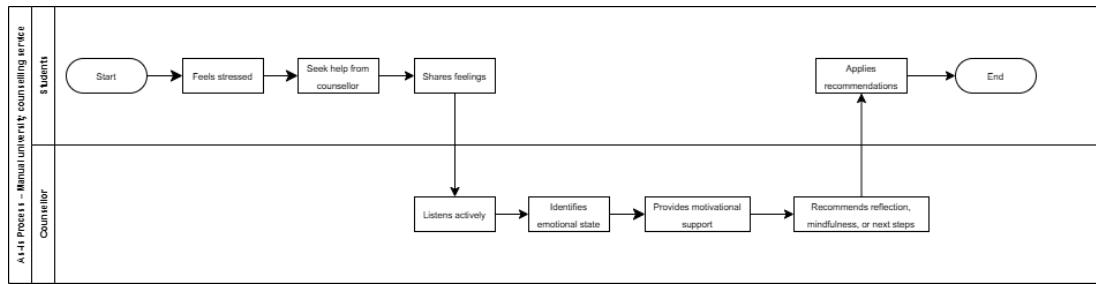


Figure 2.2.1.1 Swimlane diagram for as-is process

2.2.2 To-Be Process – MindMate App

MindMate is a drive aimed at providing an open, non-stigmatic platform that is conveniently available as a digital solution that selectively mirrors some elements of the counselling process through the technology. The platform will allow the students to:

- Record daily mood via mood icons and intensity levels
- Read real-time, AI-driven feedback through the Dialogflow virtual assistant
- Participate in supportive conversations, consisting of basic counselling components
- Obtain motivational tips and journaling ideas, based on the logged mood
- Estimate their emotional tendencies over time to develop self-awareness
- Enjoy gamification features (badges, streaks) to motivate daily use.

All of the features provided by MindMate are aimed to be a reaction to the drawbacks of the existing system of support. As an example, mood tracking in real time means that changes in moods will be continuously monitored between sessions. AI chatbot, which has been designed with simple counselling techniques, provides context-specific feedback promptly, and this makes students feel heard and supported without having to wait on a formal basis. Moreover, to help develop the routine, gamification and daily engagement streaks are employed, which allows introducing emotional awareness in a more palatable style.

In contrast to services provided in the past, MindMate can be used at any day and time, does not involve a restriction of location and provides an interface that does not judge, which are major components identified by students that promote frequent use. According to the recent research findings, when developed based on the psychological theory, digital tools, such as AI-based mental health applications, would be sufficient to reproduce certain advantages of the counselling process (Fitzpatrick et al., 2017; Torous et al., 2018).

Through a combination of an emotion-tracking approach, AI-supported conversational support and gamification, MindMate offers 24-hour, user-friendly mental health support as an adjunct to more formal counselling services and enables students to play a proactive role in maintaining their well-being.

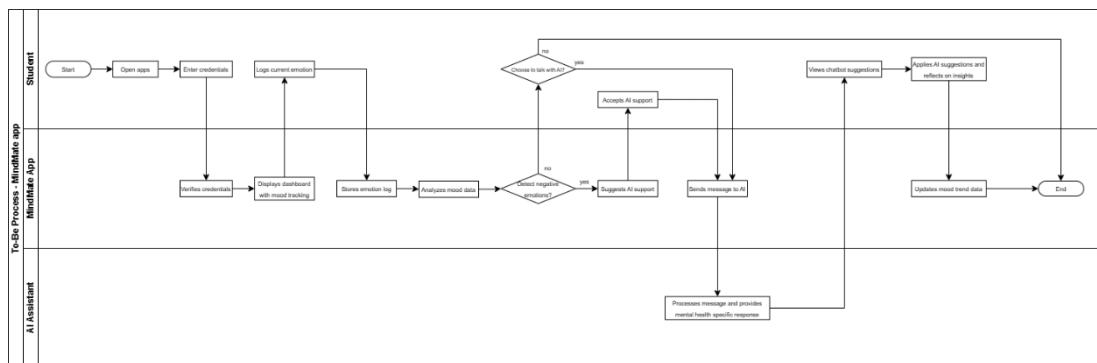


Figure 2.2.2.1 Swimlane diagram for to-be process

2.3 Current System Analysis

There are many mental health apps available, and the problem is that they concentrate on one area. As an example, Headspace is also famous due to the provision of meditation and breathing exercises, which are useful in eliminating stress. The recent addition by Headspace is a companion called Ebb that is an AI that offers

personalized content and empathetic support. It does not yet provide emotion tracking or gamification capabilities, though, which encourage prolonged engagement.

Some apps such as Woebot apply AI to communicate with users. It is more effective in developing a conversation, but users should not be able to monitor how their feelings develop over time, and it has nothing to monitor moods and offer users some features to maintain over time, such as streaks or rewards.

MindMate is an attempt to integrate the most interesting features of all these apps together: emotion tracking, an artificial intelligent assistant, mood summaries and gamification. This increases its appropriateness regarding students who require counselling and encouragement.

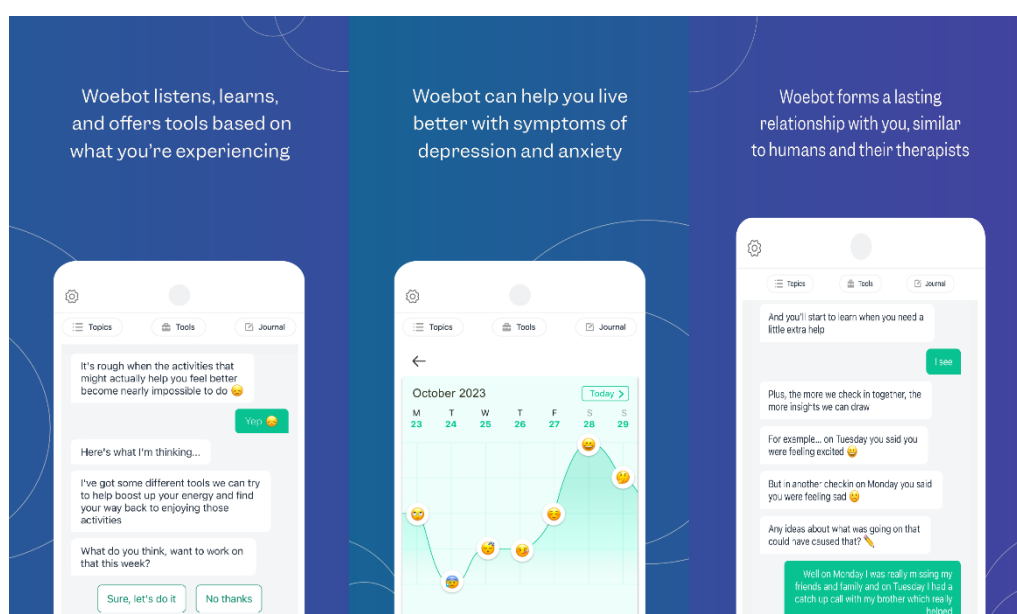


Figure 2.3.1 Interface of Woebot app

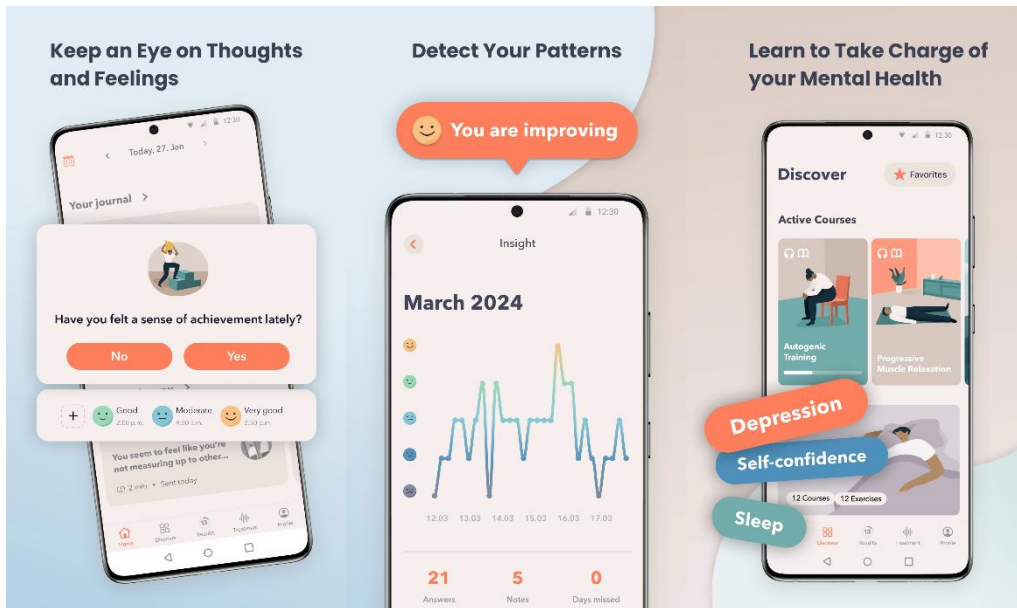


Figure 2.3.2 Interface of MindDoc app

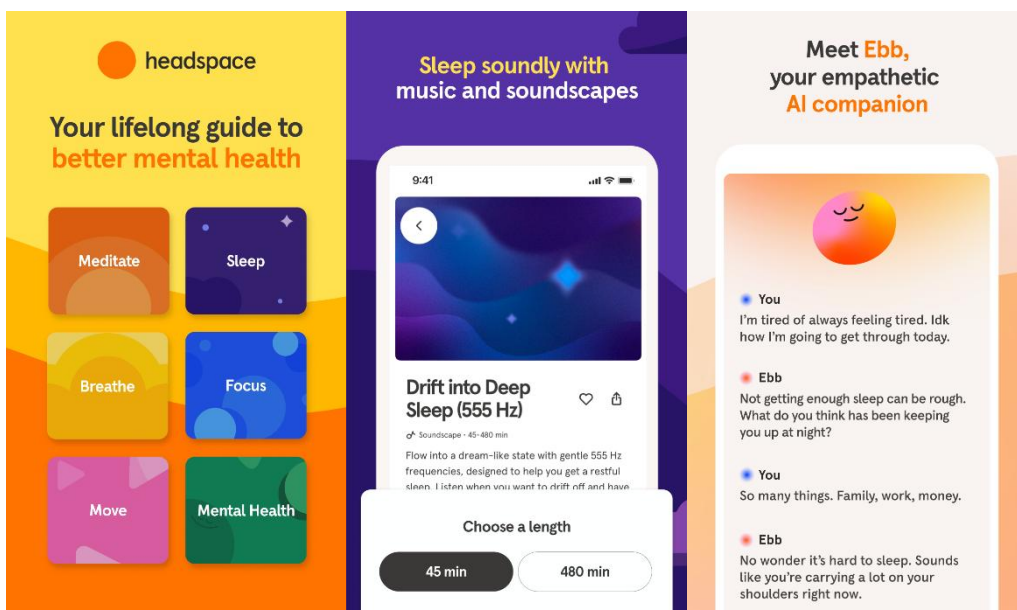


Figure 2.3.3 Interface of Headspace app

2.4 Comparison between existing systems and MindMate

The table below will compare some of the popular mental health and wellness apps to the proposed MindMate system. The comparison is based on the features like emotion monitoring, AI chatbots, gamification, and user engagement opportunities.

Table 2.4.1 Comparison between existing systems and MindMate

Features	Woebot	MindDoc	Headspace	MindMate
Emotion tracking with daily mood input	NO	YES	NO	YES
AI-powered chatbot support	YES	NO	NO	YES
Gamification	NO	NO	NO	YES
Personalized suggestion based on mood	NO	YES	YES	YES
Breathing or mindfulness exercises	LIMITED	LIMITED	YES	YES
Mood trend visualization	NO	YES	NO	YES
Cross-platform support (Android & iOS)	YES	YES	YES	YES

2.5 Literature Review of Technology Used

This section explains the primary technologies behind the development of MindMate. The app was developed with a system integration method, utilizing several

already existing frameworks and services to provide such functions as emotion tracking, work with chatbots, and storing data in secure storage. Each of the tools has been chosen for its ability to integrate with cross-platform development, user friendly and its accessibility.

2.5.1 Flutter

Flutter is a mobile development framework that is cross platform, developed by Google. It enables developers to create efficient applications with Android and iOS based on same code. Flutter also has a broad collection of pre-made widgets and adheres to the reactive pattern of programming, allowing easier creation of responsive and dynamic user interfaces. All elements of user interaction with the front-end are being designed using Flutter as it was in MindMate.

2.5.2 Dart

The first major programming language in Flutter is dart. It endorses object programming and has a clean syntax to develop mobile applications. Dart is utilized to manage the behavior of the app, keep navigation, and data between screens. It also implements logic of such features as the mood submission and work with Firebase and Dialogflow APIs.

2.5.3 Firebase

Firebase is a cloud platform, which can be applied to manage back ends functions including authentication and database storage. MindMate buys the Firebase Authentication to provide the secure access via Google or Facebook accounts. All the logs of user emotion, the history of using, and progress are kept within the Firestore

database. The role of Firebase is to make sure that data related to the users is saved in one place and can be effectively used in real time.

2.5.4 Dialogflow (via API integration)

Dialogflow is a conversational artificial intelligence tool that is a cloud platform created by Google and popular among the development of smart chatbots. The use of Dialogflow in the MindMate application is an API that enables text-based, in-application support in real-time.

Unlike a general-purpose chatbot, the Dialogflow instance used on MindMate is preferably tailored in such a way that it models the scenario of simple digital counselling. Dialogflow offers a user with contextual answers based on emotion that he has logged into so that it can facilitate self-reflection and motivation as well as emotion awareness. It is like the way human counsellors give verbal support and mechanisms of coping in sessions.

The prevalence of the concept of using Dialogflow in mental health applications will be confirmed by the latest research. A recent experiment organized by BMC Psychology (2025) showed how AI chatbots can provide important conversational support, decrease stigma, and increase accessibility concerns to people who are reluctant to use conventional counselling. Likewise, the implementation of mental health Supports within River Publishers (2025) and Library Imaging (2025) were quite successful and driven by Dialogflow-based mental health chatbots that help users to cope with anxiety, stress and emotional challenges.

MindMate uses Dialogflow to:

- Offers of AI-powered inspirational prompts and suggestions
- Deliver unique responses that rely on emotion-tracking records

- Provide immediate, available and confidential support emotionally in a stigma-free environment

The strategy of MindMate as a self-help application reinforces its role in providing traditional counselling services to students, enabling them to voice their thoughts and treat their mental health on their own.

2.5.5 Gamification

The concept of gamification in turn helps keep users occupied and form positive habits. Streaks are applied as a feature to keep people using the app every day, and badges and acknowledgment in prestige tracking achievements are used to understand what the user feels about the game. In inculcating the positive habits of checking on emotions, these features provide roundabout assistance to the mental health awareness just as the reinforcing methods championed by human counsellors.

2.5.6 Cloud-based API

MindMate interacts with third-party services (Dialogflow and Firebase) with the help of cloud-based APIs. This enables real-time processing of user and responses of chatbot without any hitches. Scalability can also be achieved through the deployment of cloud APIs, whereby updating of app logic and data on conversations can be carried out remotely without necessarily making users download updates.

2.5.7 Security Elements

Any mental health application should be concerned with security. MindMate has the following security features:

- End-to-End Encryption: Any data sent to the user or to the app is end-to-end encrypted, thus securely sending sensitive data such as mood entries and chat history.
- OAuth Authentication: The secure form of log in is established through OAuth, which enables authentication of user through trusted sources (Google and Facebook), according to which they do not need to create new passwords.

2.6 Chapter Summary

This chapter has discussed the problems of contemporary traditional mental health counselling in students and the impediments to it like the stigma, accessibility and irregular emotional surveillance. The literature emphasizes that these limitations generally fail to allow students to obtain timely support or any continuous support, which leads to even more serious mental health outcomes. Besides, some of the most popular mental health-supporting apps are analyzed and compared in this chapter. It is obvious that most apps are useful to relax or provide some dialogs with chatbots, but they fail to be entertaining, engaging, and student-life-oriented.

The idea behind the MindMate app was to present an addition to current mental health solutions, offering a way to conduct emotional self-monitoring and motivation on a daily basis on a mobile device that is AI assisted. MindMate offers a stigma-free approach to mental health backed by user-friendly emotional support, an AI-based conversation assistant created with Dialogflow, and gamification elements. In contrast to typical wellness applications, MindMate shares specific features of the counselling experience: it is created to support emotional recognition, external motivation, and self-recognition. Most research studies on the efficacy of AI-backed chatbots using chat platforms to deliver real-time interventions in the mental health sphere have proven successful in supporting people reluctant to access face-to-face counselling (study by BMC Psychology, 2025).

Technologically, the development stack consists of an effective and scalable platform that MindMate uses like Flutter, Dart, Firebase, and Dialogflow. The agile

mechanisms of gamification, in addition to the robust data security provided by end-to-end encryption and OAuth authentication, guarantee the user engagement and protection of privacy.

The combination of these methods gives a simple, active means of assisting students in maintaining their mental health, solidifying the effectiveness of technology as a mental well-being activator among college campuses.

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

This chapter covers the methodology used in development of the MindMate mobile application. In section 3.3, a description of detailed steps and procedures that were followed in each phase of the chosen methodology is elaborated. The kind of technologies that are used along with the tools that are used during the development are also mentioned, and the specific hardware and software are mentioned under the system requirement analysis section.

3.2 Methodology Choice and Justification

The Waterfall model was selected to work with the creation of the MindMate application. It is a classical development model of software, and it is sequential in nature. What is also exciting about it is that every phase of the project is accomplished before proceeding to the other since each phase must be completed meeting all the standards, so it is a better option when the project has very clear and stable requirements and the changes required in such a project during development are minimal.

Waterfall model was chosen in the case of the MindMate since the aims and scope of the project were understood at the beginning of the project. Since the development occurred in an educational environment and there were few changes invoked by the stakeholders, the Waterfall methodology has offered the required structure to careful documentation, validation at each relay, and a well-structured development drinking route. The method served also to guard against having all the project deliverables done in an unstructured and uncontrolled way.

3.3 Phases of the Chosen Methodology

The MindMate mobile application was created under the methodology of Waterfall that is sequential and organized. The phases were followed properly, i.e. one was completed first before the second one started, there was no misconception regarding the scope and deliverable. There are six major phases which are as follows:

3.3.1 Requirement Analysis Phase

The first stage consisted of both functional and non-functional requirements, identified in the students based on an In the first stage, both functional and non-functional requirements were obtained among the students based on a combination of interviewing students, studying the cases, and researching the literature on the needs of universities on mental health. The features which were identified and documented included emotions synchronized, interaction with AI chatbots, mood insight, gamified tasks, and anonymous support.

The product of this step was the Software Requirement Specification (SRS) that documented:

- System Scope and boundaries
- User roles (e.g. student, admin)
- Definition of Use cases
- Interface requirements
- Non-functional and functional constraints

The SRS acted as an agreement to proceed to further stages where all features should be justified and in line with the needs of the user.

3.3.2 System Design Phase

This phase defined the architecture and structural flow of the system. The backend of the MindMate system is a NoSQL database Firestore by Firebase. The structure is structured in such a way that it has collections and subdocuments that facilitate scalability, real-time synchronizations and modular access. The description of the core collections and the way they support the functional components of the system is offered at section 3.3.2.1 until 3.3.2.10.

The product of this step was the Software Design Document (SDD), that comprised:

- View of system architecture
- The data structure and ERP of firebase
- Interface flow maps and navigation maps

3.3.2.1 users Collection

Stores user profile data and is referenced across all other user-specific collections.

Field	Type	Description
userID	string (PK)	Unique ID for each user
userName	string	Full name of the user
userEmail	string	Email used for login
userPassword	string	Encrypted password
age	number	Age of user
gender	string	Gender identity

status	string	Current user status (e.g., active, deactivated)
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3.3.2.2 emotionLogs Collection

Logs the user's emotional states and intensity for daily tracking.

Field	Type	Description
emotionLogID	string (PK)	Unique log ID
emotionType	string	e.g., happy, sad, stressed
intensity	int	Value 1–10 based on user input
emotionLogNotes	string	Optional notes added by user
emotionLogTimestamp	datetime	Time the emotion was recorded
userID	string (FK)	Reference to users collection

3.3.2.3 chatbotSessions Collection

Records chatbot conversations between the AI and user.

Field	Type	Description
-------	------	-------------

sessionID	string (PK)	Unique session ID
startTime	timestamp	When the chat started
endTime	timestamp	When the chat ended
messages	array	List of message strings exchanged
userID	string (FK)	Linked user

3.3.2.4 activities Collection

Stores gamified or suggested self-care activities like journaling or breathing exercises.

Field	Type	Description
activityID	string (PK)	Unique activity ID
title	string	Activity name/title
activityType	string	e.g., journaling, breathing
activityStatus	string	e.g., completed, skipped
duration	string	Estimated/actual time taken
completionTime	timestamp	When user completed activity
activityNotes	string	Optional reflection note
userID	string (FK)	Owner of activity log

3.3.2.5 gamificationHistory Collection

Tracks reward-based activity history to encourage habit formation.

Field	Type	Description
gamificationHistoryID	string (PK)	Unique achievement ID
pointsEarned	int	Points gained
achievementUnlocked	string	e.g., “Streak Master”
achievementTimestamp	datetime	Time of achievement
userID	string (FK)	Linked to user

3.3.2.6 notifications Collection

System-generated reminders or recommendations to the user.

Field	Type	Description
notificationID	string (PK)	Unique notification ID
notificationMessage	string	Content shown to user
notificationStatus	string	e.g., read, unread
notificationTimestamp	timestamp	Date/time sent
userID	string (FK)	Linked user

3.3.2.7 userSettings Collection

Stores personalization preferences like language and theme.

Field	Type	Description
settingsID	string (PK)	Unique setting ID
darkMode	boolean	UI theme preference
language	string	Preferred language
notificationEnabled	boolean	Opt-in for reminders
userID	string (FK)	Linked user account

3.3.2.8 usageReports Collection

Tracks general usage data and system feedback metrics.

Field	Type	Description
reportID	string (PK)	Unique report ID
reportType	string	e.g., trend, engagement
reportData	string	Summary or metrics
generatedAt	datetime	When generated
generatedBy	string	Admin ID

3.3.2.9 admins Collection

Stores user profile data and is referenced across all other user-specific collections.

Field	Type	Description
adminID	string (PK)	Unique admin ID
adminName	string	Full name
adminEmail	string	Login credential
adminPassword	string	Encrypted password
role	string	e.g., admin

3.3.2.10 aiAssistants Collection

Manages the identity or version of the AI model/chatbot.

Field	Type	Description
assistantID	string (PK)	Unique AI version ID
assistantName	string	e.g., MindMate v1

3.3.3 Implementation Phase

It was implemented on Flutter technology with Dart as a programming language. The frontend involves UI screens to log one of four emotions, chatbot communication, and draw insights.

It has the backend built using Firebase which includes:

- Secure login Authentication
- Data storage Firestore
- The chatbot based on AI is Dialogflow.

The modules were executed depending on the priority during the SRS. Version control and cooperative handling of the code were carried out on GitHub.

3.3.4 Testing Phase

Test was performed following the implementation to consider stability and the usability. This was a stage that consisted of:

- Unit Testing and Integration Testing: All the parts were tested independently (e.g., emotion logging, chatbot response, activity tracking) and integrated to check flow consistency.
- User Acceptance Testing: The working prototype was demonstrated and tested among three students at the university: user friendly, system responsiveness, applicability of chatbot recommendations.
- Usability Testing: Performed with the help of the operative application. The response by the chatbot was measured in terms of its tone and attitudes by a mock counsellor.

The test results were recorded in one Software Test Document (STD) including:

- Acceptance tests and test results
- Bugs identified and fixes
- Feedback of users

3.3.5 Deployment Phase

The app marked as a success in the testing process was implemented in the target devices via Android emulator and physical Android phones to provide demonstrations.

Firebase is used to:

- Host backend (Firestore, Auth, Dialogflow)
- Run live analytics and queries on Firebase Console
- Usability Testing: Performed with the help of the operative application. The response by the chatbot was measured in terms of its tone and attitudes by a mock counsellor.

3.3.6 Evaluation and Maintenance Phase

The last phase included the examination of the MindMate system in the relation to its initial goals and expectations of the users. Feedback from testers and user were retrieved to determine:

- Minor bugs
- Usability areas of improvement
- Recommendations on feature extension

These feedback were then included into a series of minor revisions before final report was to be submitted. Despite it being a prototype, the system was developed to a stable state, which is presentable and fits in the further development.

3.4 Technology Used Description

Table 3.4.1 shows the list of technologies and tools used to build the MindMate mobile application.

Table 3.4.1: List of technologies and tools used

Technologies	Description
Flutter	Applied in the development of the cross-platform mobile application both on Android and iOS operating system.
Dart	The code or language in writing the application and controlling its systems.
Firebase	Firestore database to store mood logs and user data, Firestore used as authentication and Firebase Analytics.
Dialogflow	It is used to develop the chatbot offering support to the individuals through emotions.
Figma	Employed to plan the app positioning, as well as generate consumer interface prototypes.
Draw.io	It is applied in formulation of diagrams such as data flow charts and system designing.
GitHub	Believed to save and manager the source code and changes in place of development.

3.5 System Requirement Analysis

The section represents the software and hardware requirements in the development of the system in the system requirement analysis.

3.5.1 Hardware Requirements

Table 3.5.1.1 below shows the type and description about the hardware requirements involved in this system.

Table 3.5.1.1: Hardware Requirements

Hardware	Description
Laptop	Core i5, or greater, RAM 8, 64-bit OS, ideal hardware on which to code, test and simulate.
Android Smartphone	Android 9.0 minimum. OS, 3 GB RAM - perfect to test MindMate on Android.
iOS Device	iOS 13 or higher Apple device to make iOS compatible.

3.5.2 Software Requirements

Table 3.5.2.1 below shows the type and description about the software requirements involved in this system.

Table 3.5.2.1: Software Requirements

Software	Description
Operating System	Windows 11
Flutter SDK	For building the cross-platform mobile application
Dart SDK	Programming tools needed for Flutter
Firebase Console	Platform used for login, data storage, and analytics
Dialogflow Console	Used to manage the chatbot responses

Visual Studio Code	Text editor used for writing and testing code
Figma	Used to design the app interface
draw.io	Used to make system diagrams
GitHub	Used for version control
Google Chrome / Android Emulator	Used to test the app during development

3.6 Chapter Summary

In this chapter, a detailed description of the methodology used in the design and development of the MindMate mobile application is given. It also clearly explains the steps that are undertaken at each step of the chosen method, the activities that are performed at each step and the results that are obtained at any one step. It also explains the technologies and tools used during the project in addition to talking about the system requirement analysis that lists the hardware and software requirements needed to ensure a successful implementation.

CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

This chapter focuses on the analysis of requirements and system design of the MindMate mobile application, continuing with the identification and discussion of the functional and non-functional requirements. The proposed system can be explained by some diagrams of Unified Modeling Language (UML): use case diagram, activity diagram, domain model, class diagram, component diagram, package diagram. There are also the development of system architecture, database architecture, as well as the interface design. To document more in detail, please refer to Appendix D (Software Requirements Specification (SRS)) and Appendix E (Software Design Description (SDD))

4.2 Requirement Analysis

Both functional and non-functional requirements of MindMate mobile application were elicited by interviewing university students in Y1S1 and Y4S2. Two interviews were conducted to gather insights:

- One with a **Year 1 Semester 1 (Y1S1)** student, new students that could be undergoing adjustment difficulties at a very early stage.
- One with a **Year 4 Semester 2 (Y4S2)** student, as an example of experienced students who could suffer stress factors related to the final projects, job hunting, or imminent graduation.

The interviews used open-ended questions about emotional experiences of participants at the university, the specifications of the mental health support tools that they prefer, and their opinions on what types of tools they would like to have (e.g.,

mood trackers, chatbots, or motivation elements). Findings were recorded in form of detailed transcripts and summarised in an analysis (see Appendix C).

4.2.1 Functional Requirements

Functional requirements are the main functions that an application should be able to provide to its users. These requirements were obtained by direct user feedback, systematic literature searches, and exploration of available mental health apps. In Figure 4.2.1.1, the core use case diagram of the MindMate is presented, whereas Table 4.2.1.1 presents a brief overview of the related functionalities. Each use case is described in detail in Appendix D.

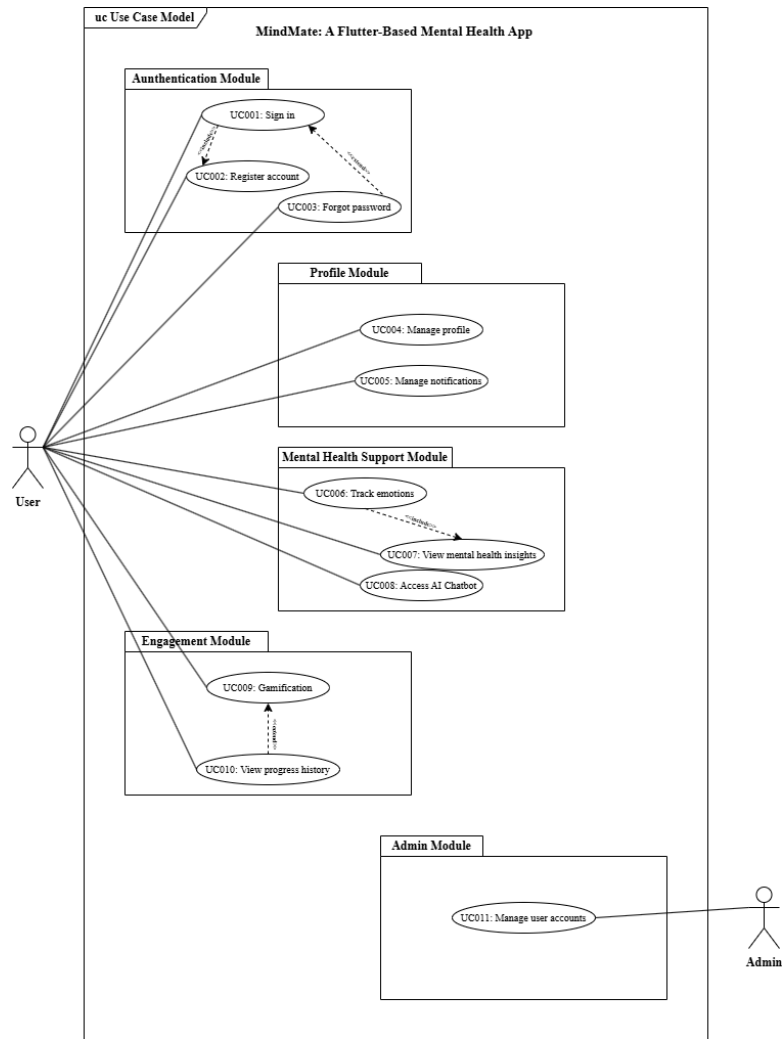


Figure 4.2.1.1: Use Case Diagram for MindMate

Table 4.2.1.1: Description of Module and Functions for MindMate

Module	Function	Description
Authentication Module	UC001 : Sign in	Enables users to use Google to log into the app.
	UC002 : Register account	New users can create an account to start using the app.
	UC003 : Forgot password	Helps users reset their password if they forget it.
Profile Module	UC004 – Manage profile	Users can update their name, age, or other personal details.
	UC005 – Manage notifications	Users can turn notifications on or off based on their preference
Mental Health Support Module	UC006 – Track emotions	Users log their daily mood using one of the emotion options.
	UC007 – View mental health insights	Shows users a summary of their emotion trends.
	UC008 – Access AI chatbot	Allows users to chat with the AI assistant for motivation or reflection.
Engagement Module	UC009 – Gamification	Gives users badges and streaks for using the app regularly.
	UC010 – View progress history	Shows users their past usage history and achievements.
Admin Module	UC011 – Manage user accounts	The admin can manage and monitor user data from the Firebase console.

4.2.1.1 Actor Description

There are two actors in the MindMate system: the user and admin.

4.2.1.1.1 User (University Students)

- The target audience of the system in question will consist of students of the university who have basic computer literacy. They are the people who are, generally, used to mobile apps and can easily perform simple activities such as authentication, choice of options, and mood history retrieval.
- While some students may be proficient in computer technologies, others may lack confidence. Accordingly, the program has been made to be easy to use even for those who may not be very tech-savvy.
- In addition, some students may not be open to speaking with counselors, or they may experience practical challenges that prevent them from participating in in-person or virtual counseling sessions. Consequently, it is thought to be advantageous to incorporate a chatbot since it offers psychological assistance in a discreet and user-friendly manner.

4.2.1.1.2 Admin

- The administrator is typically the system maintainer or developer. In this project, the project owner will use the Firebase and Dialogflow platforms to examine the system's backend.
- The administrator won't utilize the mobile app in the same way as other users. Rather, they will use Firebase to manage user accounts, data, and, if necessary, system issues.
- The admin is in charge of upgrading and maintaining the system, the administrator is required to possess technical expertise with tools like Flutter, Firebase, and Dialogflow.

4.2.1.2 Use Case Description

Each use case describes how the user interacts with the app. One example is shown below for UC001 – Sign in. More use case descriptions can be found in the SRS document (Appendix E).

Table 4.2.1.2.1: Use Case Description for UC001 Sign in

Use Case ID	UC001
Use Case Name	Sign in
Description	Allows users or admins to log in using Google or email/password authentication.
Actor(s)	Primary: User, Admin Secondary: Google Authentication API, Firebase Authentication System
Pre-condition(s)	The user or admin must be registered with the system.
Normal Flow(s)-NF	Scenario 1 : User - Sign in via Google 1. The user clicks the "Sign in with Google" button. 2. The system redirects to the Google authentication page. 3. The user enters their credentials and confirms. 4. Google API returns a success response. 5. The system logs in the user and navigates to the dashboard. Scenario 2 : User - Sign in via Email/Password 1. The user selects "Sign in with Email". 2. The user enters their email and password. 3. The system validates the credentials using Firebase Authentication. 4. When log in success, the user is navigated to the dashboard. Scenario 3 : Admin – Sign in via Email/Password 1. Admin accesses admin login panel. 2. Admin enters email and password.

	3. System authenticates admin via Firebase. 4. Admin is taken to the admin console.
Alternative Flow(s) - AF	AF1. User or admin cancels login. AF2. User or admin chooses "Forgot Password" (redirects to UC003).
Exception Flow(s) - EF	EF1. Invalid credentials entered. EF2. Authentication system is unreachable or fails.
Post-condition(s)	1. Authenticated user is navigated to their dashboard. 2. Authenticated admin is directed to the admin panel.

4.2.1.3 Sequence Diagram

The sequence diagram models the communication between the user and the system. Figure 4.2.1.3.1 shows the sequence for UC001 – Sign in. Other diagrams are available in Appendix E.

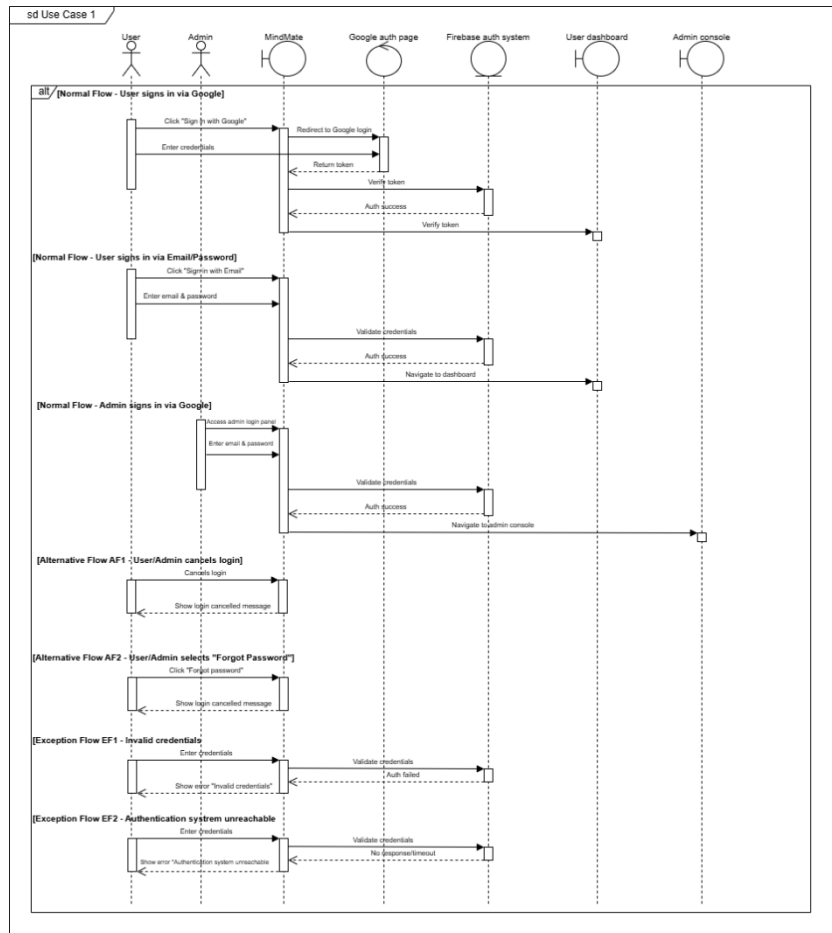


Figure 4.2.1.3.1: Sequence Diagram for UC001 Sign in

4.2.1.4 Activity Diagram

The activity diagram demonstrates the steps of a use case. Figure 4.2.1.4.1 shows the activity for UC001 – Sign in. Other diagrams are available in Appendix E.

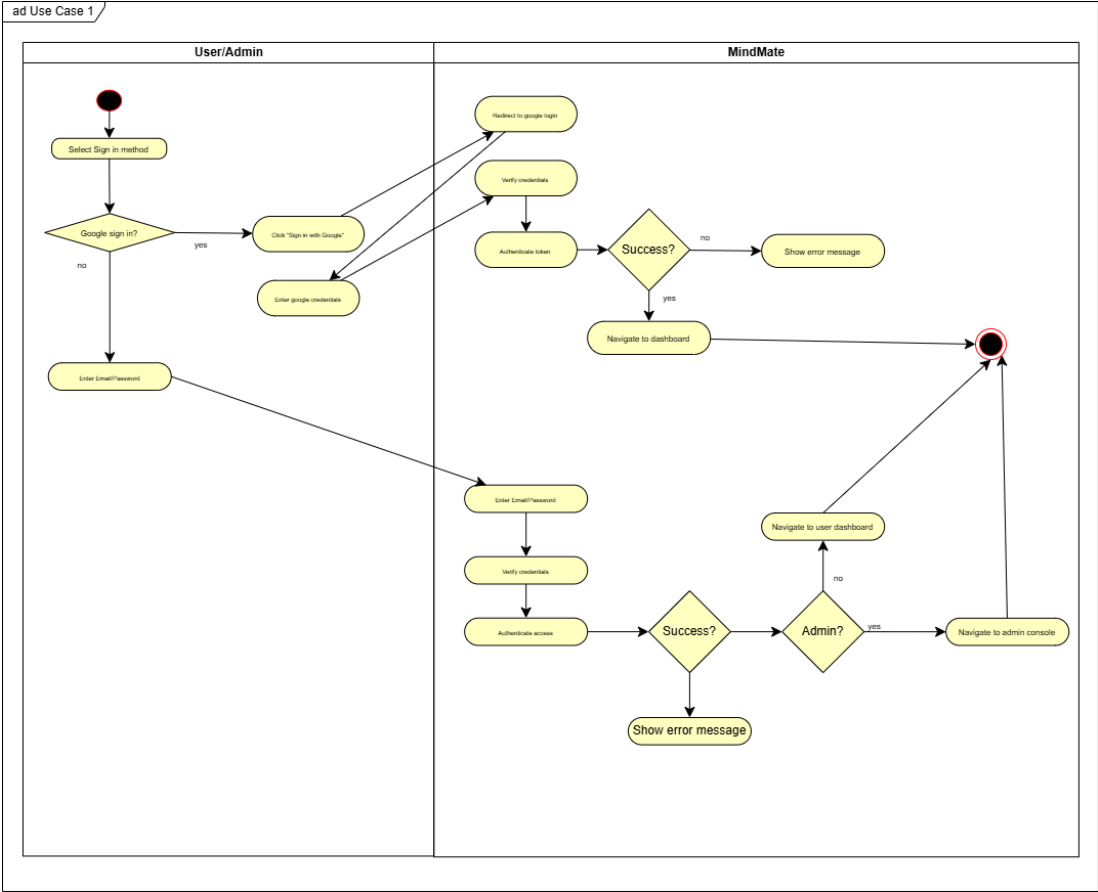


Figure 4.2.1.4.1: Activity Diagram for UC001 Sign in

4.2.2 Non-functional Requirements

Non-functional requirements define how the system should perform. Table 4.2.2.1 lists the non-functional requirements for the MindMate app.

Table 4.2.2.1: Description of Module and Functions for MindMate

Non-Functional Requirements	ID	Description
Performance	NFR-001	The authentication of user credentials (via Google or email) shall take no longer than 5 seconds.
	NFR-002	The system shall respond to mood logging actions within 2 seconds under normal network conditions.
	NFR-003	The system shall return chatbot responses within 3 seconds during typical usage.
	NFR-004	All insights or history data must be retrieved and displayed in less than 5 seconds.
	NFR-005	The application load time shall not exceed 10 seconds on devices with at least 2GB RAM.
Design Constraint	NFR-006	All UI text shall be displayed in English only.
	NFR-007	The application shall follow Material Design

		principles and support both light and dark modes.
	NFR-008	The system shall be developed using Flutter and optimized for mobile platforms.
Security	NFR-009	Only authenticated users shall be allowed to access MindMate's main functionalities.
	NFR-010	The system shall store all sensitive data in Firebase using AES-256 encryption
	NFR-011	The chatbot shall not collect or store any personally identifiable information (PII).
Usability	NFR-012	The system's main features shall be accessible within 3 taps from the home screen.
	NFR-013	The system will be used by users without needing extra training.
	NFR-014	Accessibility elements like enhanced text and color contrast settings must be supported by the interface.
Compatibility	NFR-015	The application must use the Flutter 3.19 SDK or later to run on both iOS and Android.

	NFR-016	The system will easily interface with Dialogflow CX APIs and Firebase Authentication.
Portability	NFR-017	Any mobile device with iOS 13.0+ or Android 8.0+ and a steady internet connection should be able to run the app without any issues.
Availability	NFR-018	Except for planned maintenance, the system will be accessible 24/7.
Maintainability	NFR-019	The codebase of the application will be modular, enabling the addition of new features with little modification.
	NFR-020	Each bug fix or update shall not take longer than 24 hours to deploy under normal conditions.
Reliability	NFR-021	The system shall complete its core operations (login, tracking, chatbot) 99% of the time without failure.

4.3 Project Design

The mobile application in MindMate has been designed to use Model-View-ViewModel (MVVM) pattern that is incorporated in three-tier layered architecture. Figure 4.3.1 gives the diagrammatic representation of the structure of this structure, it separates the system into three layers Presentation Layer, Business Logic Layer and Data Layer.

In the application, the Model is the Data Layer which acts as an intercession between the application and the cloud-based storage, namely Firebase Firestore, and performs data activities including logging of emotions, profile management, chatbot history, tracking of gamification progress.

The ViewModel is in Business Logic Layer and acts as the middle-man between the information and the user-interface. It processes raw data into a form that can be presented on the screen and controls the state of the interface; it also arranges input validation and carries out asynchronous communication with back-end services that are offered by Firebase and Dialogflow APIs.

The View as a part of the Presentation Layer is the user interface that appears in Flutter. It reflects the changes passed by the ViewModel and provides a responsive and interactive user experience, which incorporates interfaces of tracking emotion, interaction with a chatbot, and insights reviewing.

In combination, MVVM architecture and the 3-tier layered architecture encourage separation of concerns, make the system easy to maintain, promote scalability, and easy testing on an individual level.

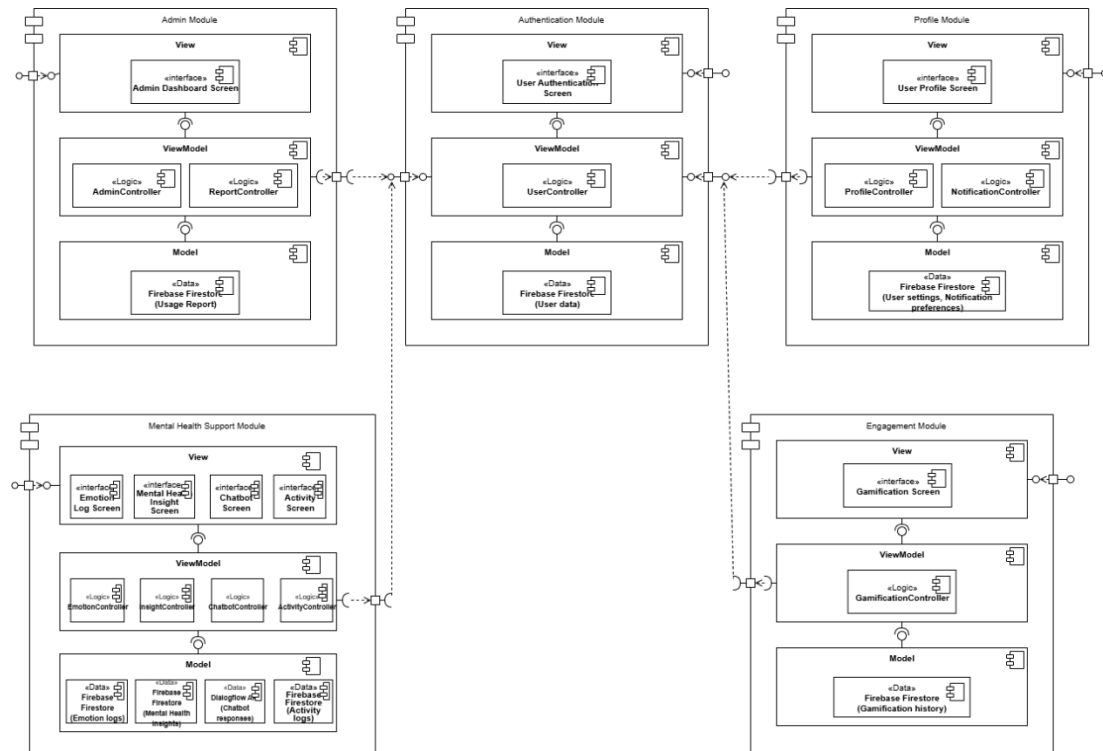


Figure 4.3.1: Architectural Style of MindMate

4.4 Database Design

This mobile application is run on a database-based architecture that most efficiently maintains and stores the basic information upon which its main features of emotion tracking, chats with a bot, gamification process, and user-profile management depend. Design rule concentrates on consistency of data, minimization of redundancy and ability to scale so as to meet the requirements of both efficiency in performance and storage.

The Entity Relationship Diagram (ERD) as shown in Figure 4.4.1 provides the breakdown of the important collections and their relationship. It illustrates the entities Users, EmotionLogs, ChatbotSessions, GamificationHistory, Notifications and UsageReports.

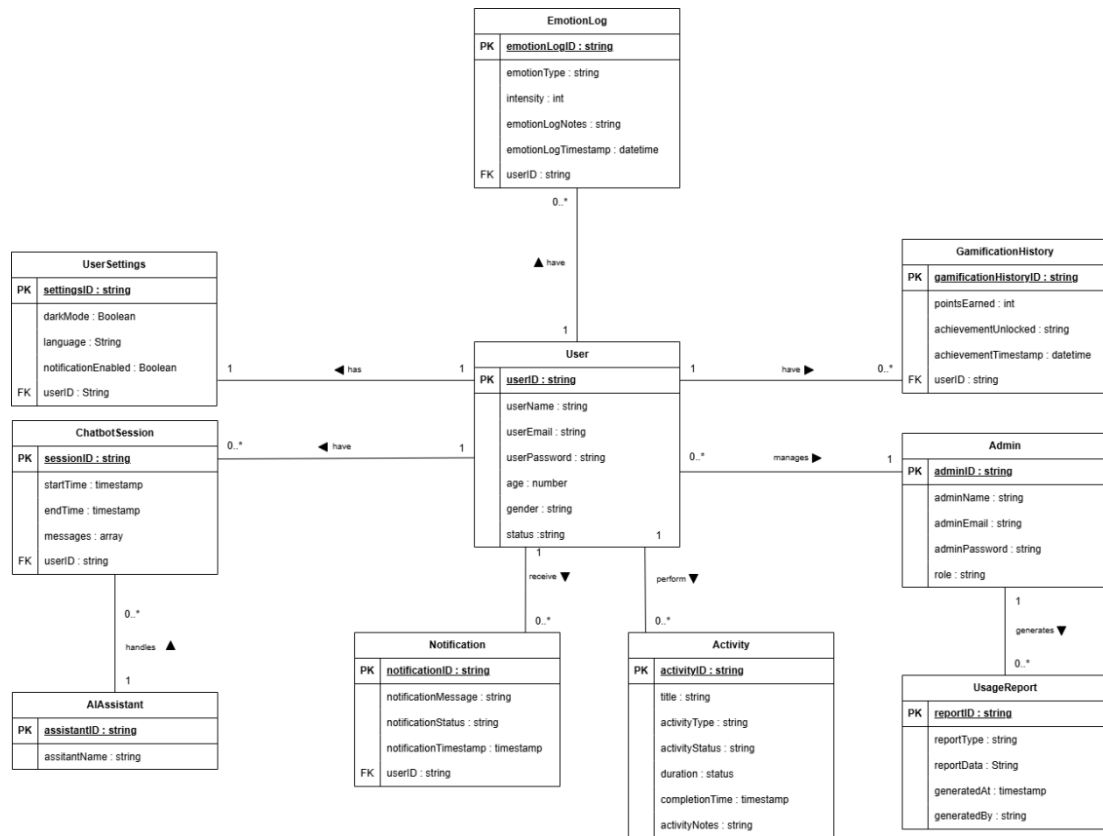
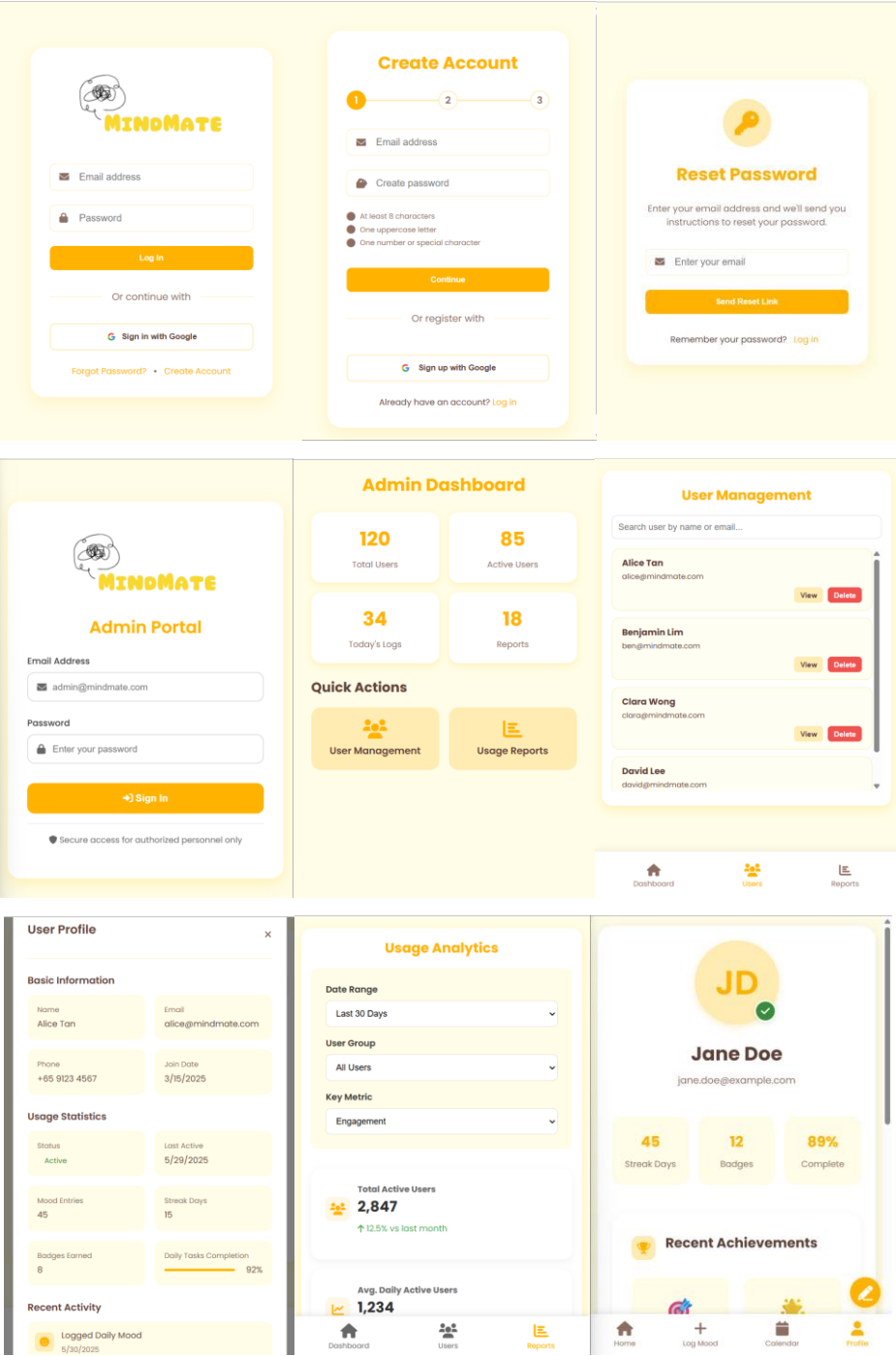


Figure 4.4.1: Entity-Relationship-Diagram of MindMate Database Design

4.5 Interface Design

The user interface presents the visual realization of the MindMate mobile application and is specifically designed to be engaging and user-friendly, thus contributing to the inclusive activity among the entire target audience of the population, including the university students, as well as experiencing a range of emotional states. One of the main features of core wellness is its immediate availability and well-defined visual cues, role-based dashboards and simplified workflows which keep the overall layout clean and minimalistic. As shown in Figure 4.5.1, the design of the screen is calm, inviting, however, at the same time, all the main elements are brought to the foreground to promote a smooth and easy flowing experience and maintain user attention.



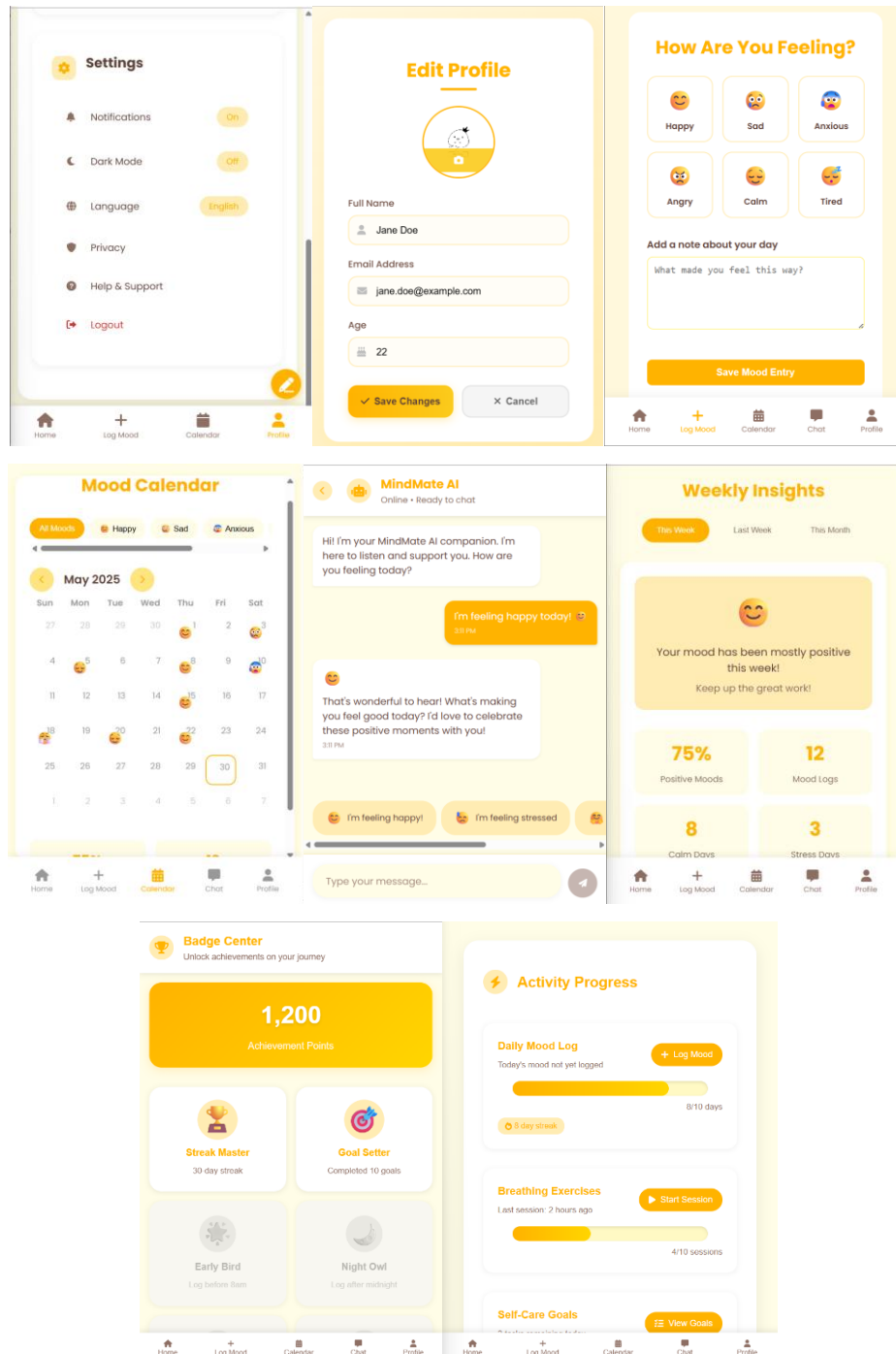


Figure 4.5.1 : MindMate app user interfaces

4.6 Chapter Summary

This chapter explains architectural planning and design of the mobile application of MindMate based on the utilization of the UML diagrams and an Entity-Relationship Diagram (ERD). A brief specification of the functional and non-

functional requirements is given so that the developers faithfully follow the system. MindMate uses Model-View-ViewModel (MVVM) architecture with a three-layer structure in order to leave the code clean and promote future code maintenance. Such design simplifies the continuous updates and modularization. The app will connect to: Firebase as the storage place of user-data and Dialogflow as the place of emotion tracking and automated responses of chatbot.

CHAPTER 5

CONCLUSION

5.1 Introduction

A synthesis of the concept of the MindMate project, which is a Flutter-based mobile application, aimed at helping university students to monitor their emotional states and use AI-powered virtual assistant is provided in the chapter. It assesses the achievement of the stated goals of the project, determines the main strengths of the system and possible spheres of its further improvement.

5.2 Achievement of Project Objectives

The development of MindMate successfully achieved all the main objectives set out at the beginning of the project.

First, analyzing the requirements for developing MindMate, has been successfully completed. The features should be supported by the application has been defined by performing interviews with the students and carrying out the complete research to obtain the key list of specifications that should be incorporated in the application, such as the module of tracking the emotions, the use of the chatbot powered by artificial intelligence and the ability to visualize the trends in emotions. These requirements are now enshrined in the Software Requirements Specification (SRS) at Appendix D and this forms a good base on which acting can be done.

The proposed system of designing the system where there is a feature of tracking emotions has also been covered. Appendix E contains the Software Design Document (SDD), which describes the architecture, database schema, and interface design and the logic behind the interaction through the emotion-tracking part. The

discussed module is a tool that is supposed to help users detect, track, and confirm their states of emotion in the context of academic requirements, thus providing helpful information on mental health.

Although the initial stages of the implementation of the AI-powered virtual assistant and the software testing that are supposed to come along with it are still in the process of being completed, the foundations of the said goals have been laid down. The mechanism of how the rich personalised journaling prompts, such as specific reinforcement and coping strategies will be provided by the virtual assistant has been specifically addressed. This functionality will continue to be enhanced by development and soon to be seen through tests.

The active implementation and testing of the application will lead to its evaluation in respect to the role it plays in stress management and the promotion of emotional awareness. The system preparation is to be used by the users and expects to get some important feedback when the system is put into operation..

To summarize, the core work has moved forward as it should be, even though most of the implementation and testing is still to be done, the project is currently on its way to meet its listed goals.

REFERENCES

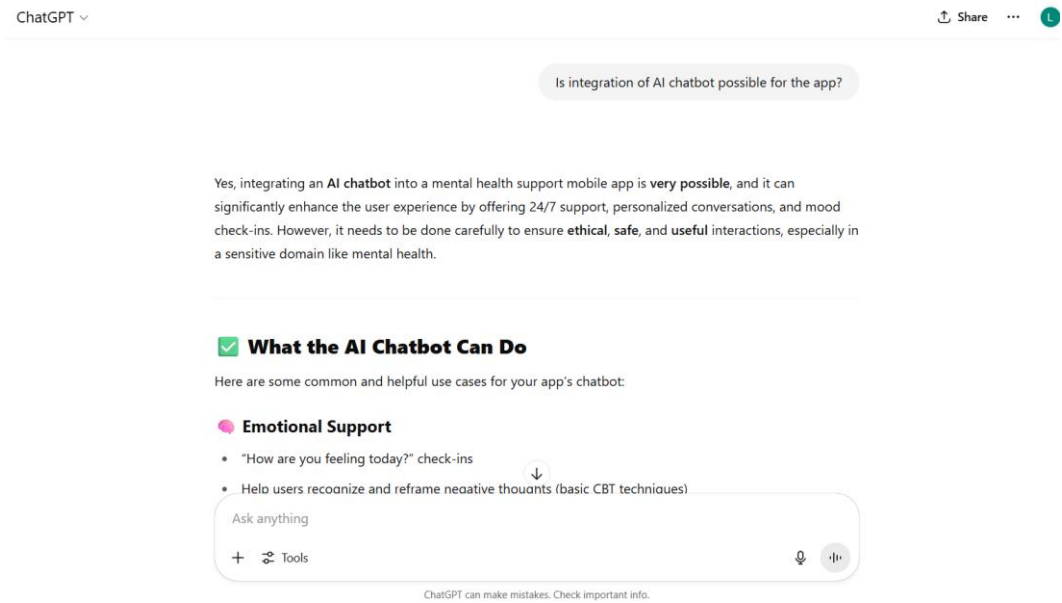
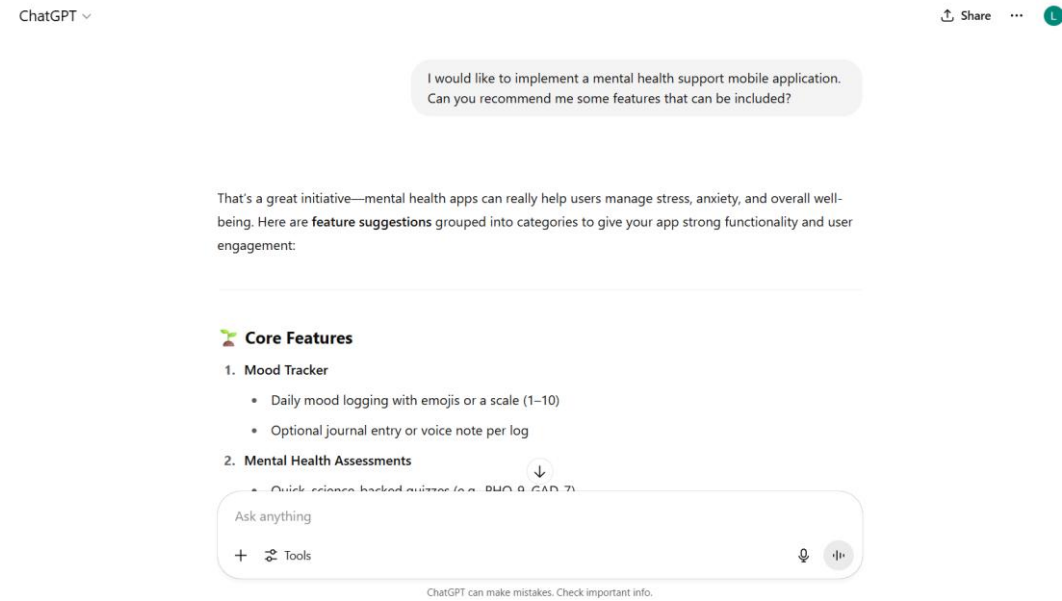
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Appendix A List of AI Prompts Used

1. <https://chatgpt.com/share/6857cc30-eb4-8003-a135-fe3d7d5db7bb>
(a)



ChatGPT

Share

How chatbot enhance user experience especially univwersoty student?

Great question. An AI chatbot can **significantly enhance the user experience** of a mental health app for **university students**, who often face academic stress, loneliness, time pressure, and emotional ups and downs. Here's how:

1. Always-Available Support

Problem: Students may not always have access to counselors or friends when they're struggling, especially late at night.

Chatbot Benefit:

- Available 24/7 to listen, offer calming exercises, or simply chat when they feel alone.
- No appointments or waiting lists.

Ask anything

+ Tools

ChatGPT can make mistakes. Check important info.

Appendix B Gantt Chart PSM

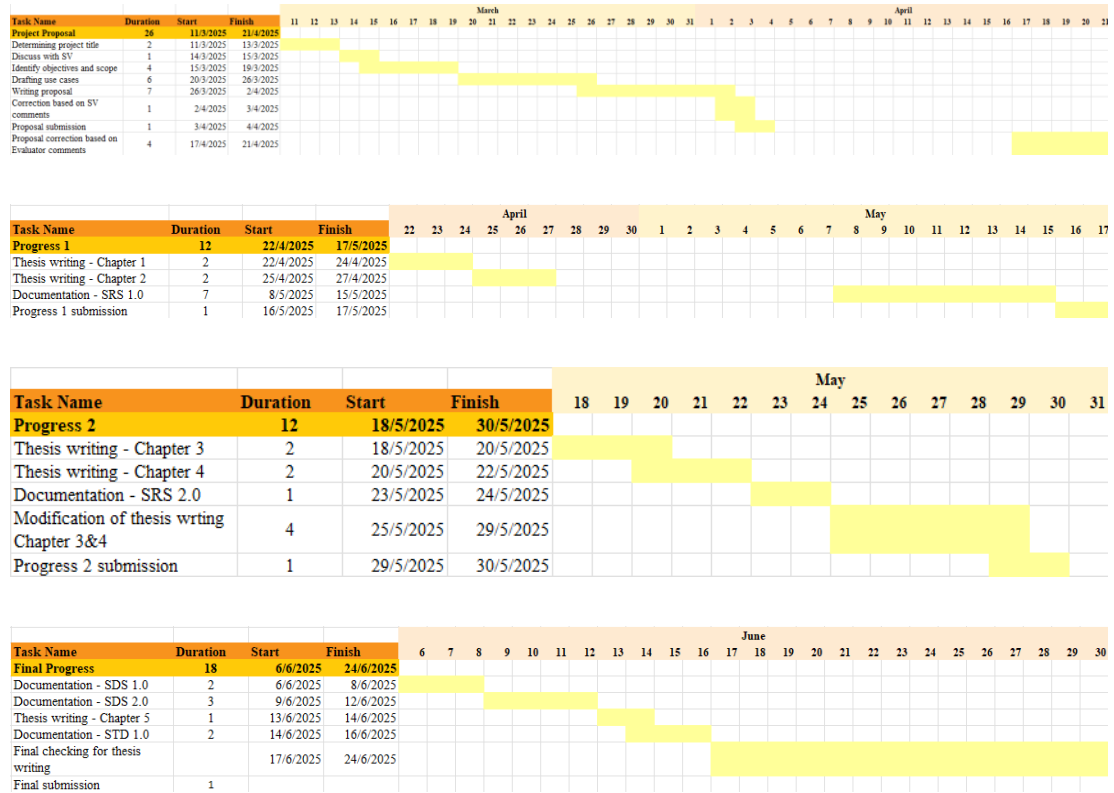


Figure A.1 Gantt Chart for PSM 1

Figure A.2 Gantt Chart for PSM 2

Appendix C Interview Transcripts & Summary

B.1 Interview Transcript – Y1S1 Student

Year/Semester: Year 1 Semester 1

Gender: Female

Age: 20

Date: 12 May 2025

Mode: Online (Google Meet)

Duration: 20 minutes

Q1: Can you describe how you have been feeling emotionally since starting university?

A: It's been a bit overwhelming. I'm from a different state, so I miss home a lot. There's also a lot of pressure to catch up with assignments and make friends.

Q2: Have you used any mental health apps before?

A: No, but I've seen some friends using journaling apps or mood trackers.

Q3: What kind of features would be helpful in an emotional support app?

A: I think having something simple to track my mood every day would help. Maybe a chatbot or something that gives positive quotes when I feel down.

Q4: Would you use a chatbot to talk about your feelings?

A: If it's private and supportive, yes. Sometimes I just want to type how I feel without being judged.

Q5: Do you find rewards or badges useful to stay motivated in using an app?

A: Yes, I like seeing progress or streaks. It makes me want to keep going.

B.2 Interview Transcript – Y4S2 Student

Year/Semester: Year 4 Semester 2

Gender: Male

Age: 23

Date: 13 May 2025

Mode: Online (Google Meet)

Duration: 20 minutes

Q1: How has your emotional experience been during your final year?

A: It's stressful, especially with the final year project and job hunting. Sometimes I feel mentally tired and don't really talk about it to anyone.

Q2: What kind of mental health support do you think students need?

A: Something that checks in regularly or reminds us to pause and reflect. I don't always notice how stressed I am until it gets bad.

Q3: Would you find an app helpful for tracking emotions or calming down?

A: Yes, especially if it can show a pattern or help me manage burnout. I also prefer something quick—not too many steps.

Q4: How do you feel about using a virtual assistant or AI chatbot?

A: I think it's a good idea. I wouldn't want a full therapy session, but if it gives short support messages or asks how I feel, that's great.

Q5: Would you continue using the app if it included badges or streaks?

A: Probably, especially if it feels rewarding or if it shows my progress visually.

B.3 Summary of Findings

Based on the interviews with a first-year and a final-year student, several key requirements for the MindMate app were identified:

Finding	Explanation

Simple Mood Tracking	Both students preferred an easy way to log their emotions daily using icons or short text.
AI Chat Support	Both participants liked the idea of a chatbot that offers quick replies, motivational messages, and journaling prompts.
Privacy	Users want a private space to express their feelings without fear of judgment.
Visual Progress	Mood charts and summaries were considered useful, especially for recognizing stress patterns.
Gamification	Badges, streaks, or rewards help motivate continued use, especially for younger users.
Short Interactions	Users prefer quick, non-intrusive features that don't require long input or effort.

Appendix D Software Requirement Specification (SRS)



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 3032: Software Engineering FYP1

Semester 02, 2024/2025

Software Requirements Specification (SRS)

**MindMate: A Flutter-Based Mental Health App for
University Students with Emotion Tracking and AI-
Powered Virtual Assistant**

Version 2.0

Prepared by: Lau Yun Xi

Revision Page

a. Overview

This document presents the introduction, requirements matrix, system architecture design, detailed description of each module, data design and user interface design.

b. Target Audience

The targeted audience are university students and admin.

c. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Lau Yun Xi	Initial Draft for SRS	08/05/2025
2.0	Lau Yun Xi	Completed	24/5/2025

Note:

This Software Requirements Specification (SRS) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998) that is simplified and customized to meet the need of SECJ3203 FYP1 SE course at Faculty of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly. **This template is tailored for SRS in a plan-driven software development approach.**

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the development of MindMate: A Flutter-Based Mental Health App for University Students with Emotion Tracking and AI-Powered Virtual Assistant. The purpose of this document is to serve as a guide for all stakeholders involved in the development, implementation, and evaluation of the system, including developers, testers, project supervisors, and evaluators. The domain diagram, sequence diagram and activity diagram can effectively capture the structures and relationships between components.

1.2 Scope

The software product to be built is MindMate, a mobile mental health application developed using the Flutter framework, aimed to support university students' emotional wellbeing. It allows users to log their moods, view mood trends, and receive personalized motivational content. The app integrates with Google Dialogflow via API to provide chatbot functionality for real-time virtual assistance, and Firebase is used to manage user authentication and store emotional data securely in the cloud. *MindMate* aims to help university students become more aware of their mental health through daily check-ins, simple reflections, and mood-related suggestions. It also includes gamification features such as streak counters and badges to promote consistent engagement. The app is developed as a prototype project and will focus on emotion tracking and mental health support rather than clinical diagnosis or therapy.

There are two types of users in the system:

- **End users (students):** Can interact with the app features such as logging moods, chatting with the virtual assistant, and reviewing mood history.
- **Admin (developer):** Has backend access to monitor data structure, manage application updates, and perform system testing.

The application will focus on supporting six core emotion states: happy, sad, stressed, anxious, frustrated, and focused. The system does not include professional counseling modules, emergency services, real-time human therapist chat, or any payment integration.

Additionally, the chatbot content is powered by external APIs and will not be developed manually from scratch.

1.3 Definitions, Acronyms and Abbreviation

API	Application Programming Interface – A set of rules that allows systems to communicate with each other.
Dialogflow	A Google cloud-based service that enables conversational AI, used for chatbot interaction.
Firebase	A cloud-based platform by Google that provides backend services like authentication and database management.
Flutter	An open-source UI toolkit developed by Google for creating natively compiled applications from a single codebase.
Gamification	The application of game-design elements such as rewards, streaks, or badges in non-game contexts to increase engagement.
OAuth	Open Authorization – A secure protocol for token-based user authentication (e.g., Google or Facebook login).
SRS	Software Requirements Specification – A document that outlines detailed software system requirements.
UI	User Interface – The layout and interactive elements through which users interact with an application.
UTM	Universiti Teknologi Malaysia – The institution under which this project is conducted.

1.4 References

1. Google Developers. (2024). *Flutter Documentation*. Retrieved from <https://flutter.dev>
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4. "IEEE Guide for Software Requirements Specifications," in IEEE Std 830-1984, vol., no., pp.1-26, 10 Feb. 1984, doi: 10.1109/IEEESTD.1984.119205.

1.5 Overview

The SRS report provides a clear perspective about the system, including the system interfaces, user, hardware and software interfaces. It also covers the product functions, user characteristics and constraints defined for the system. It also includes the external interface requirements to specify the interaction between external entities. Besides, this report outlines the system features based on the use case diagram. There are total of 12 use cases and the flow for each use case are explained more in-detailed through the activity diagram and sequence diagram. Moreover, this document discusses the functional and non-functional requirements of a system to ensure the system aligns to the software quality attributes during development.

2. Specific Requirements

2.1 User characteristics

2.1.1 University Students

- The users of this system are university students who have basic computer skills. Most of them are used to mobile apps and can do simple tasks like logging in, choosing options, and checking their mood history.
- While some students may be good with technology, others may not be as confident. Because of this, the app is designed to be easy to use, even for those who are not very tech-savvy.
- Some students may feel shy or uncomfortable talking to a counselor. Others might not have time or access to mental health services. That is why the chatbot in this app is helpful—it gives support in a private and simple way.

2.1.2 Admin

- The admin is usually the developer or system maintainer. In this project, it will be the project owner who uses Firebase and Dialogflow platforms to check the backend of the system.
- The admin will not use the mobile app like a normal user. Instead, they will access Firebase to manage data, user accounts, and fix any problems in the system if needed.
- The admin is expected to have technical knowledge in using tools like Flutter, Firebase, and Dialogflow, as they are responsible for maintaining and updating the system.

2.2 System Features

MindMate is a mobile application that runs on both Android and iOS platforms. The system is created to help university students understand and manage their emotional health better. It also gives students a way to get support in a simple and private way through a virtual assistant powered by chatbot technology. The app uses Firebase for storing data in real-time and Dialogflow for chatbot communication.

The system allows users to log their moods, talk to a chatbot, see their emotion history, and stay motivated through gamified features like streaks and badges. The app is made to be simple and easy to use by all types of students.

The system features are shown in Figure 2.2.1 below. The use case diagram displays how students and admin interact with different parts of the system.

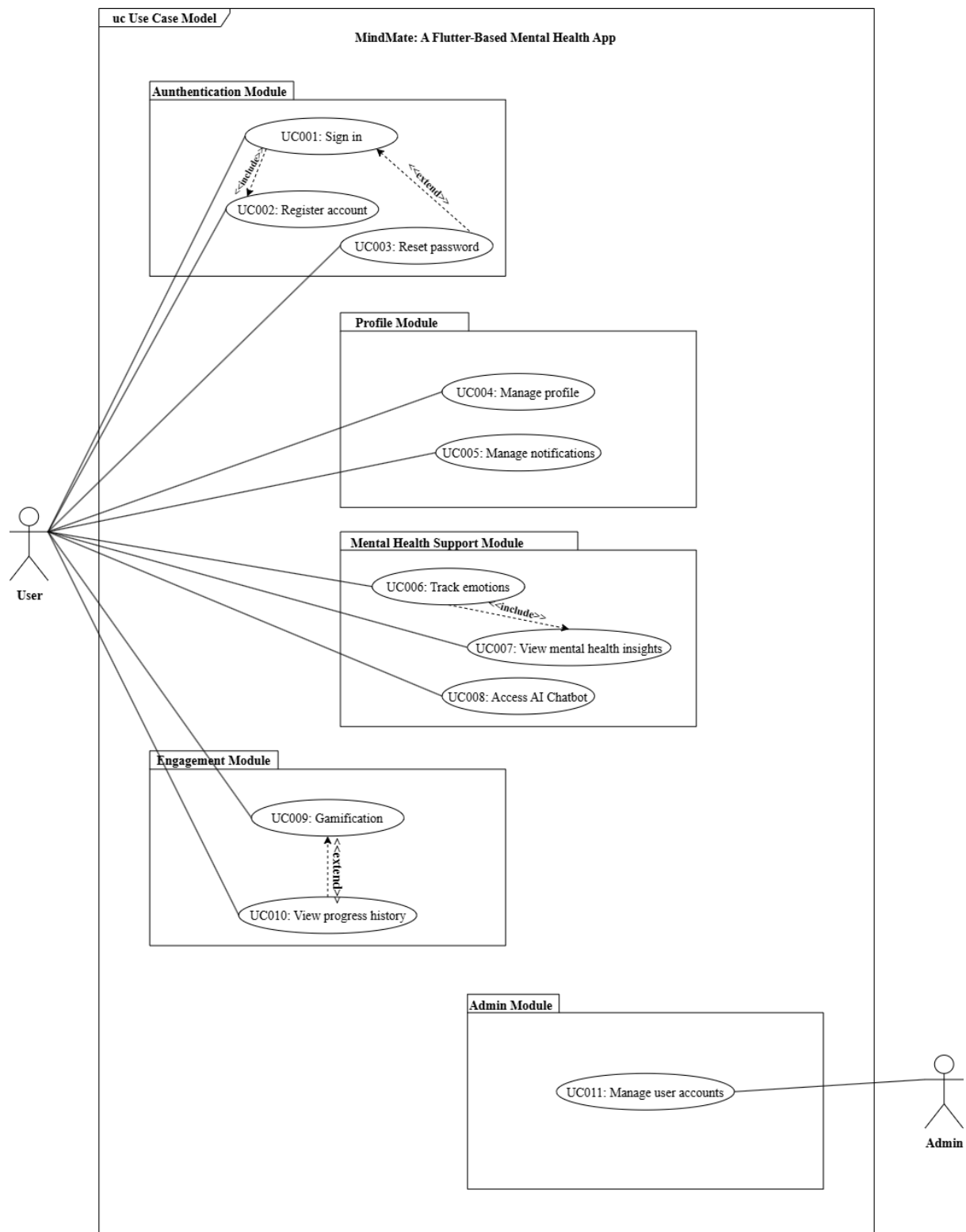


Figure 2.2.1: Use Case Diagram for MindMate

Table 2.2.1: Description of Module and Functions for MindMate

Module	Function	Description
Authentication Module	UC001 : Sign in	Allows users to log into the app using Google
	UC002 : Register account	New users can create an account to start using the app.
	UC003 : Forgot password	Helps users reset their password if they forget it.
Profile Module	UC004 – Manage profile	Users can update their name, age, or other personal details.
	UC005 – Manage notifications	Users can turn notifications on or off based on their preference
Mental Health Support Module	UC006 – Track emotions	Users log their daily mood using one of the emotion options.
	UC007 – View mental health insights	Shows users a summary of their emotion trends.
	UC008 – Access AI chatbot	Allows users to chat with the AI assistant for motivation or reflection.
Engagement Module	UC009 – Gamification	Gives users badges and streaks for using the app regularly.
	UC010 – View progress history	Shows users their past usage history and achievements.
Admin Module	UC011 – Manage user accounts	The admin can manage and monitor user data from the Firebase console.

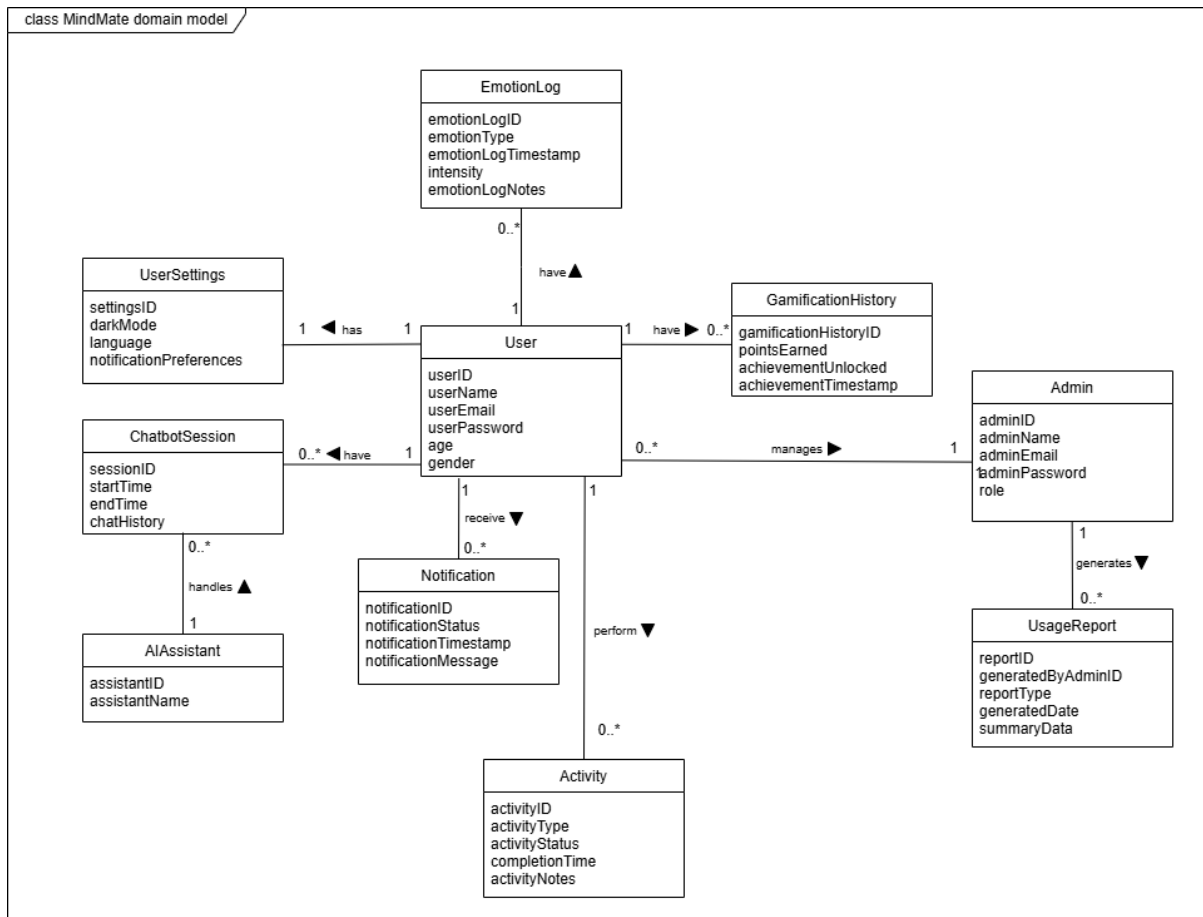


Figure 2.2.2: Domain Model for MindMate

Figure 2.2.2 shows user interactions, emotional tracking, chatbot sessions, gamification features, and administrative monitoring. The main part of the system is the **User** class. It stores basic user details like **userID**, **userName**, **userPassword**, **age**, and **gender**. Each user has their own **UserSettings**, which lets them choose preferences like **darkMode**, **language**, and **notificationPreferences**, allowing the system to customize the user experience based on individual settings.

Users can also record their emotions using the **EmotionLog** class. This keeps track of the **emotionType**, **emotionLogTimestamp**, **intensity**, and **emotionLogNotes**. Each emotion log is linked to **EmotionTracking**, which aggregates and evaluates emotion entries by recording a **trackingID**, **trackingDate**, and **trackingStatus**. One emotion tracking record can be connected to many emotion logs to help see patterns over time.

Users can chat with the system through chatbot sessions, which are saved in the **ChatbotSession** class. This includes the session's **sessionID**, **startTime**, **endTime**, and

the entire chatHistory. Each session is managed by an **AIAssistant**, identified by assistantID and assistantName, ensuring accountability and session management.

The system also sends **Notifications** to users. These include the notificationID, notificationStatus, notificationTimestamp, and notificationMessage. These notifications are generated and delivered to inform or alert users based on their system activity or administrative messages. Users can also engage in various **Activity** instances. The Activity class records events or tasks performed by users through attributes such as activityID, activityType, activityStatus, completionTime, and activityNotes.

To enhance user engagement, the system incorporates a GamificationHistory class. This tracks gamified progress, including pointsEarned, achievementUnlocked, and the achievementTimestamp. Admins are responsible for overseeing the system and are defined by adminID, adminName, adminEmail, adminPassword, and their role. They can create **UsageReports** which show summaries and stats, including the reportID, generatedByAdminID, reportType, generateDate, and a summaryData field that consolidates insights or statistics derived from system usage.

Overall, this class diagram demonstrates how MindMate supports emotional monitoring, interactive AI engagement, gamification, personalized settings, and administrative oversight, ensuring a robust and user-centric platform.

EmotionLog State Diagram

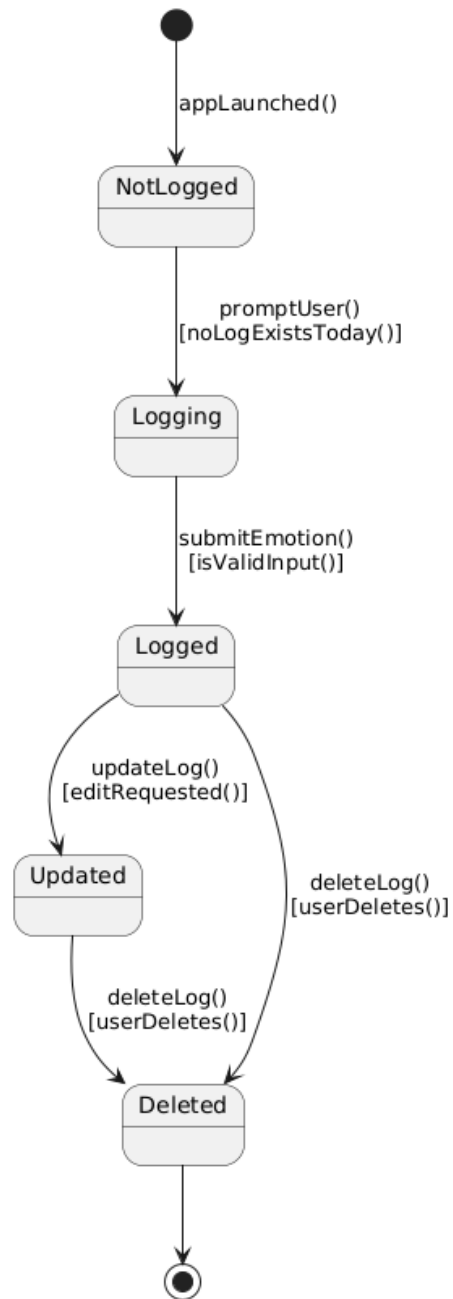


Figure 2.2.3: State Diagram for EmotionLog

ChatbotSession State Diagram

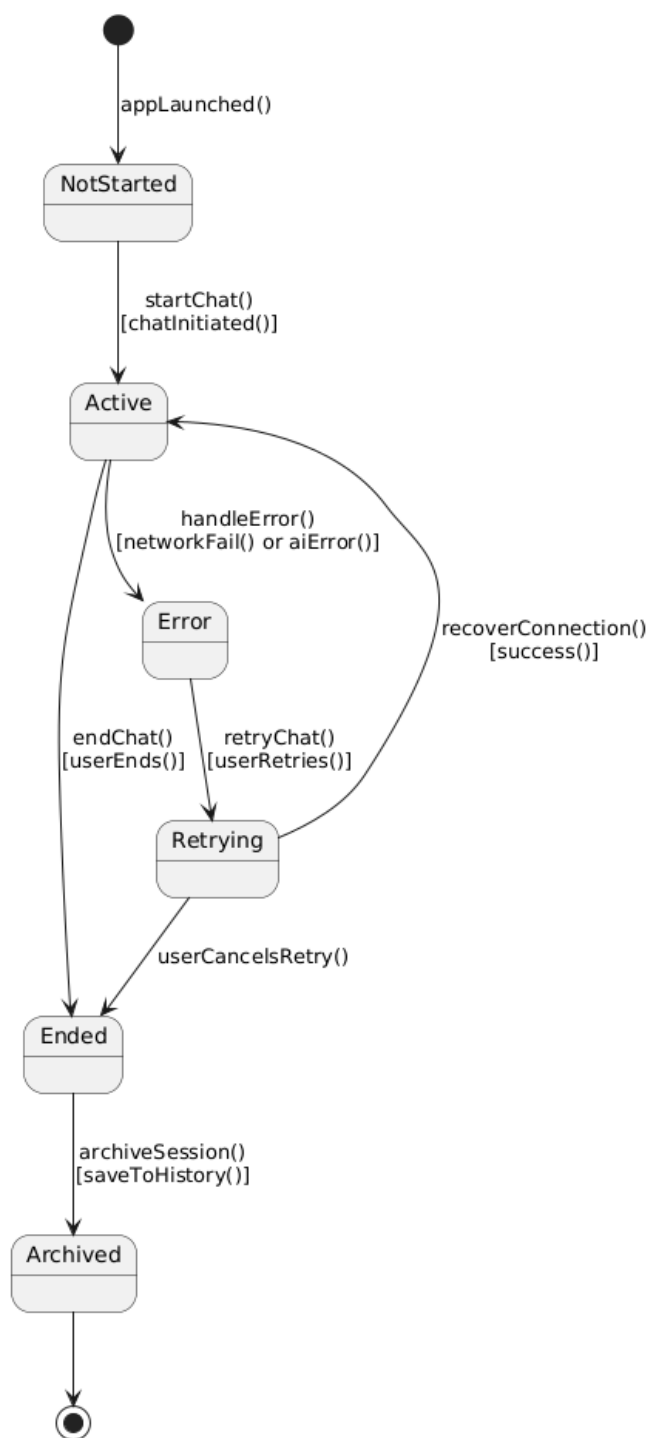


Figure 2.2.4: State Diagram for ChatbotSession

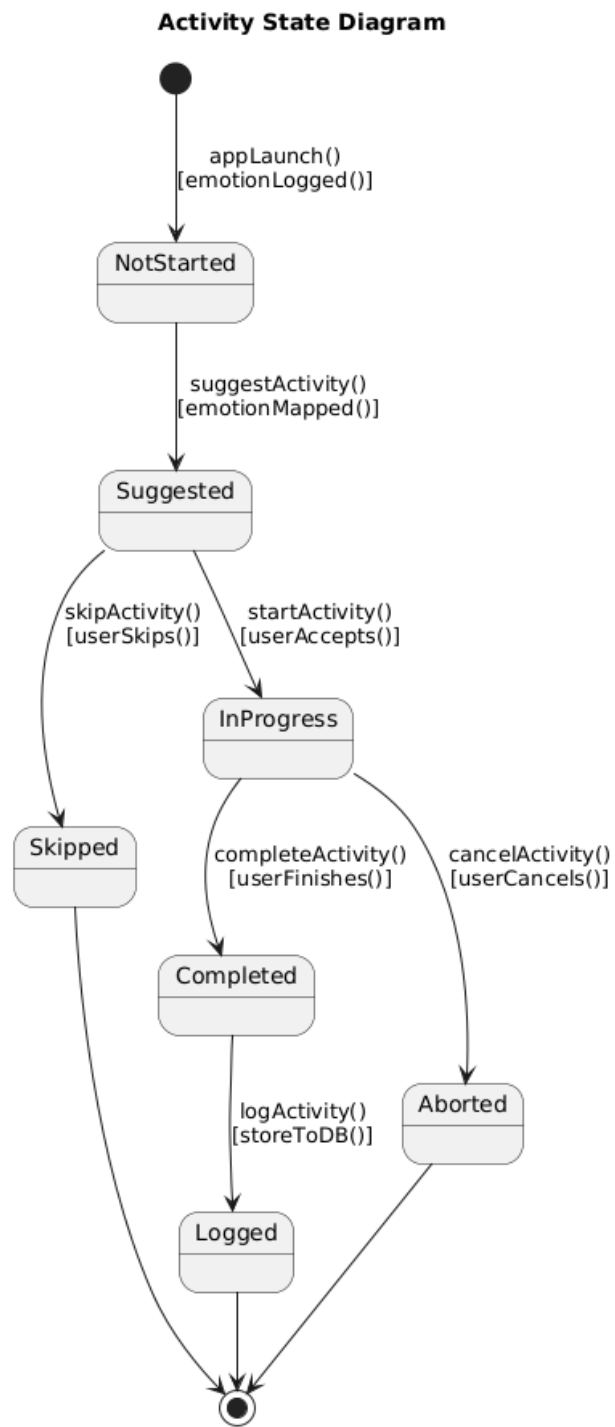


Figure 2.2.5: State Diagram for Activity

2.3 Use Case Details

2.3.1 UC001: Use Case <Sign in>

Table 2.3.1.1: Use Case Description for UC001 Sign in

Use Case ID	UC001
Use Case Name	Sign in
Description	Allows users or admins to log in using Google or email/password authentication.
Actor(s)	Primary: User, Admin Secondary: Google Authentication API, Firebase Authentication System
Pre-condition(s)	The user or admin must be registered with the system.
Normal Flow(s)-NF	Scenario 1 : User - Sign in via Google 1. The user clicks on the "Sign in with Google" button. 2. The system redirects the user to the Google authentication page. 3. The user enters their credentials and confirms. 4. Google API returns a success response. 5. The system logs in the user and navigates to the dashboard. Scenario 2 : User - Sign in via Email/Password 1. The user selects "Sign in with Email". 2. The user enters their email and password. 3. The system validates the credentials using Firebase Authentication. 4. When log in success, the user is navigated to the dashboard. Scenario 3 : Admin – Sign in via Email/Password 1. Admin accesses admin login panel. 2. Admin enters email and password. 3. System authenticates admin via Firebase. 4. Admin is taken to the admin console.

Alternative Flow(s) - AF	AF1. User or admin cancels login. AF2. User or admin chooses "Forgot Password" (redirects to UC003).
Exception Flow(s) - EF	EF1. Invalid credentials entered. EF2. Authentication system is unreachable or fails.
Post-condition(s)	1. Authenticated user is navigated to their dashboard. 2. Authenticated admin is directed to the admin panel.

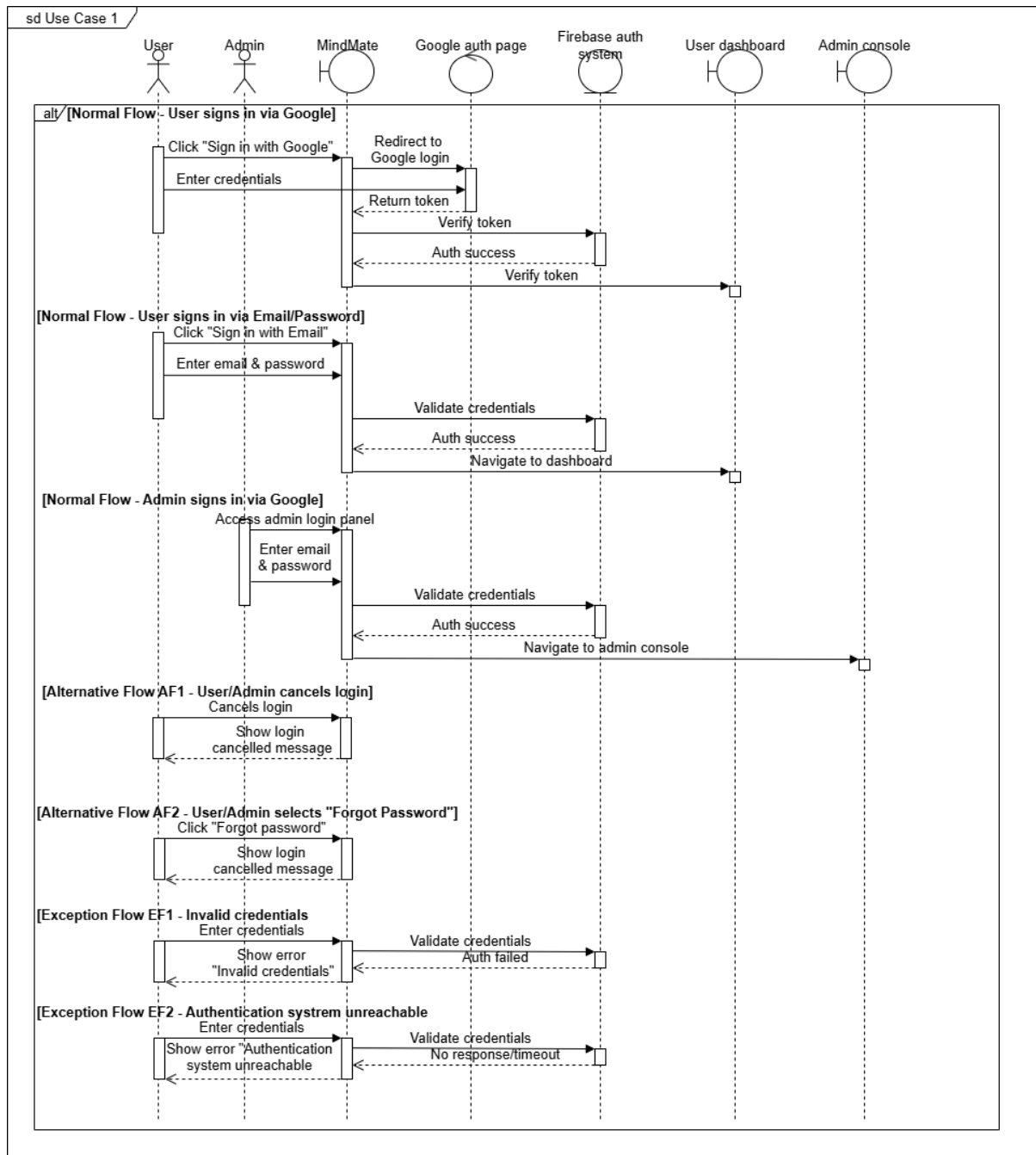


Figure 2.3.1.1: Sequence Diagram for UC001 Sign in

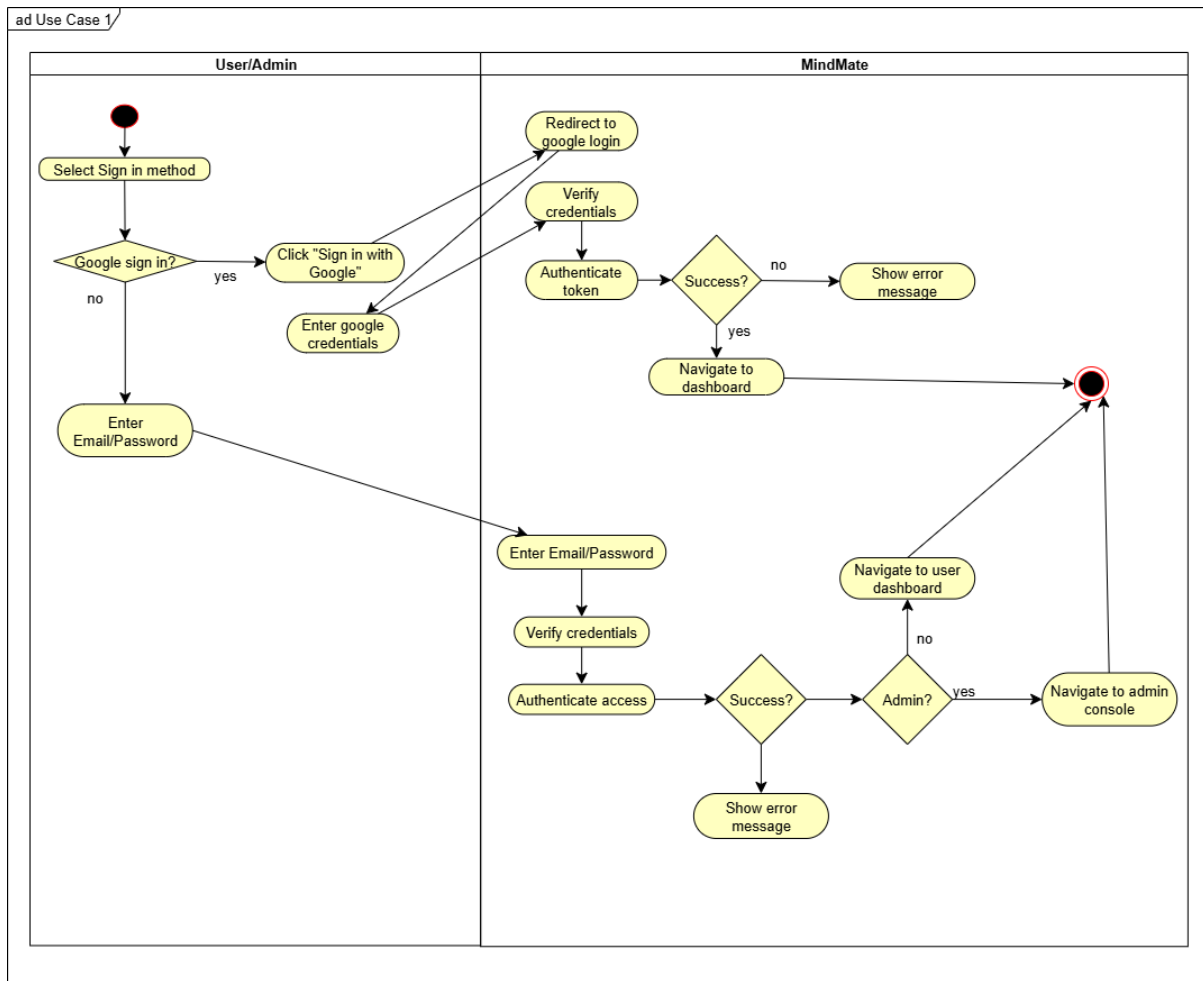


Figure 2.3.1.2: Activity Diagram for UC001 Sign in

2.3.2 UC002: Use Case <Register account>

Table 2.3.2.1: Use Case Description for UC002 Register account

Use Case ID	UC002
Use Case Name	Register account
Description	Allows new users to create an account using either Google Sign-Up or manual registration with email and password.
Actor(s)	Primary: User Secondary: Google Authentication API, Firebase Authentication System
Pre-condition(s)	The user must not already have an existing account in the system.
Normal Flow(s)- NF	Scenario 1 : User - Register via Google 1. The user selects "Register with Google". 2. The app redirects to Google Authentication. 3. The user signs in with Google. 4. Google API returns user info to the app. 5. The system creates a new user account using the info. 6. The user is redirected to the dashboard. Scenario 2 : User - Register via Email/Password 1. The user selects "Register with Email". 2. The user enters their name, email, password and age. 3. The system validates input and creates the user account. 4. The user is redirected to the dashboard.
Alternative Flow(s) - AF	AF1. User cancels the registration process. AF2. Email is already registered — the system prompts to log in instead.
Exception Flow(s) - EF	EF1. Google Authentication fails due to user denial or network error. EF2. Firebase fails to create a new user due to validation or backend error.

Post-condition(s)

A new user account is created and stored in the system. The user is logged in and redirected to the dashboard.

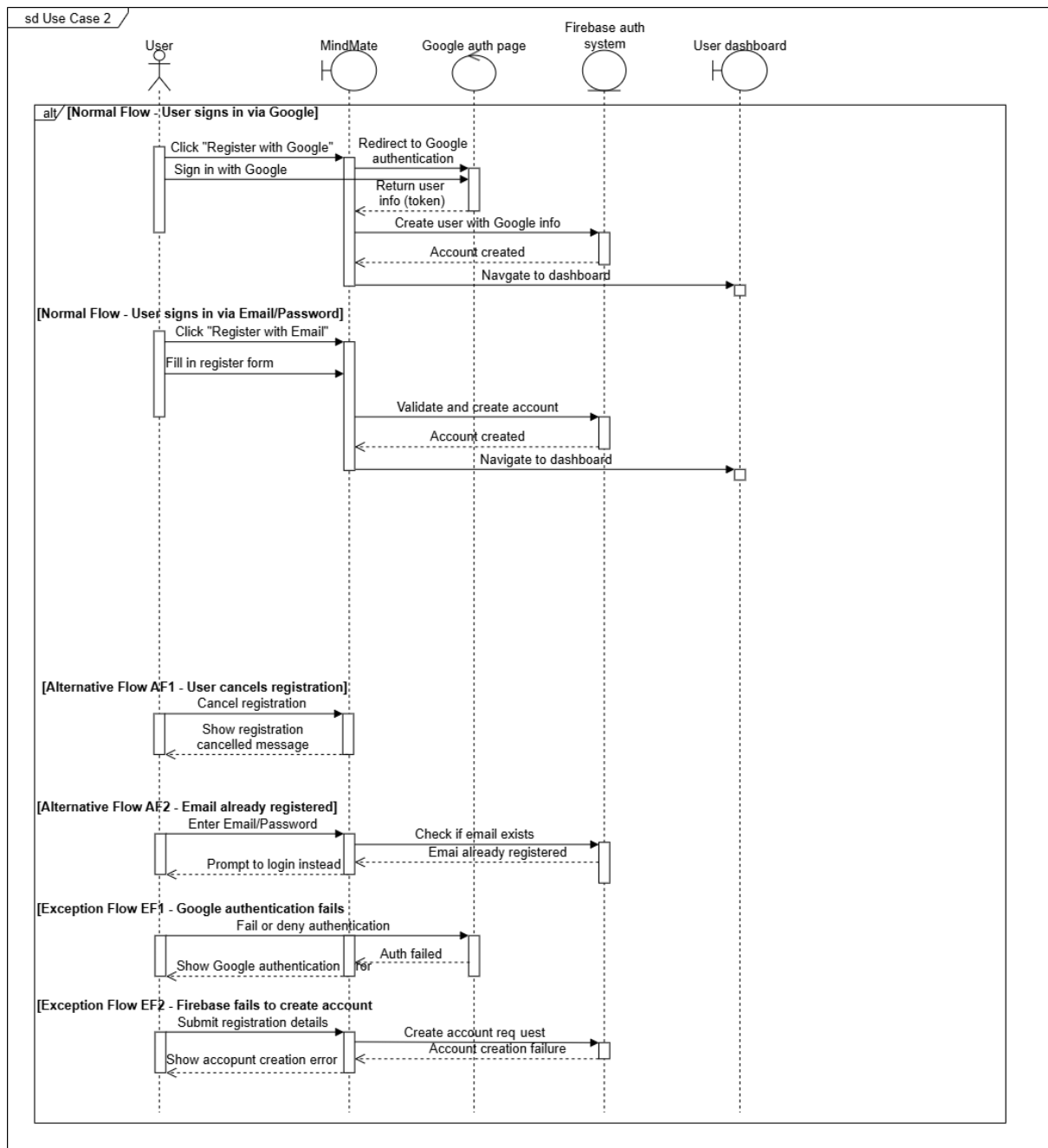


Figure 2.3.2.1: Sequence Diagram for UC002 Register account

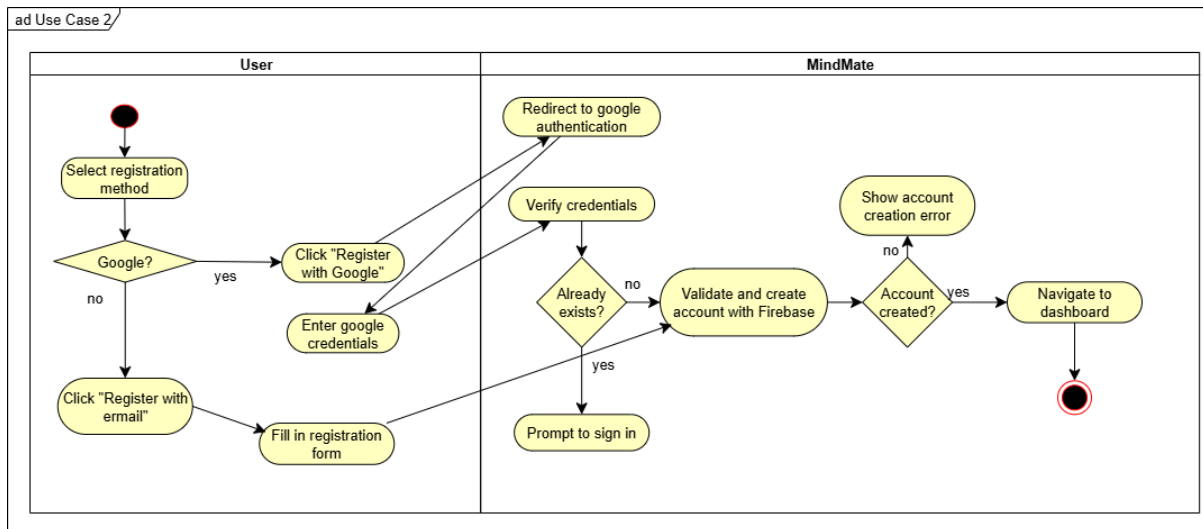


Figure 2.3.2.2: Activity Diagram for UC002 Register account

2.3.3 UC003: Use Case <Reset password>

Table 2.3.3.1: Use Case Description for UC003 Forgot pass Reset password

Use Case ID	UC003
Use Case Name	Reset password
Description	Allows users to reset their password via a password recovery email using Firebase Authentication.
Actor(s)	Primary: User Secondary: Firebase Authentication System
Pre-condition(s)	The user must have previously registered with an email/password account (manual registration).
Normal Flow(s)- NF	1. The user clicks on the “Forgot Password?” link on the login screen. 2. The system prompts the user to enter their registered email address. 3. The user enters the email address. 4. The system verifies if the email exists in the database. 5. The system sends a password reset email using Firebase. 6. The user follows the link in the email to reset their password. 7. Upon successful reset, the user can sign in with the new password.
Alternative Flow(s) - AF	AF1. User cancels the process and returns to the login screen.
Exception Flow(s) - EF	EF1. The email entered is not registered — the system shows an error message. EF2. Email delivery fails due to server or network issues.
Post-condition(s)	The user receives a password reset email and can recover access to their account upon resetting the password.

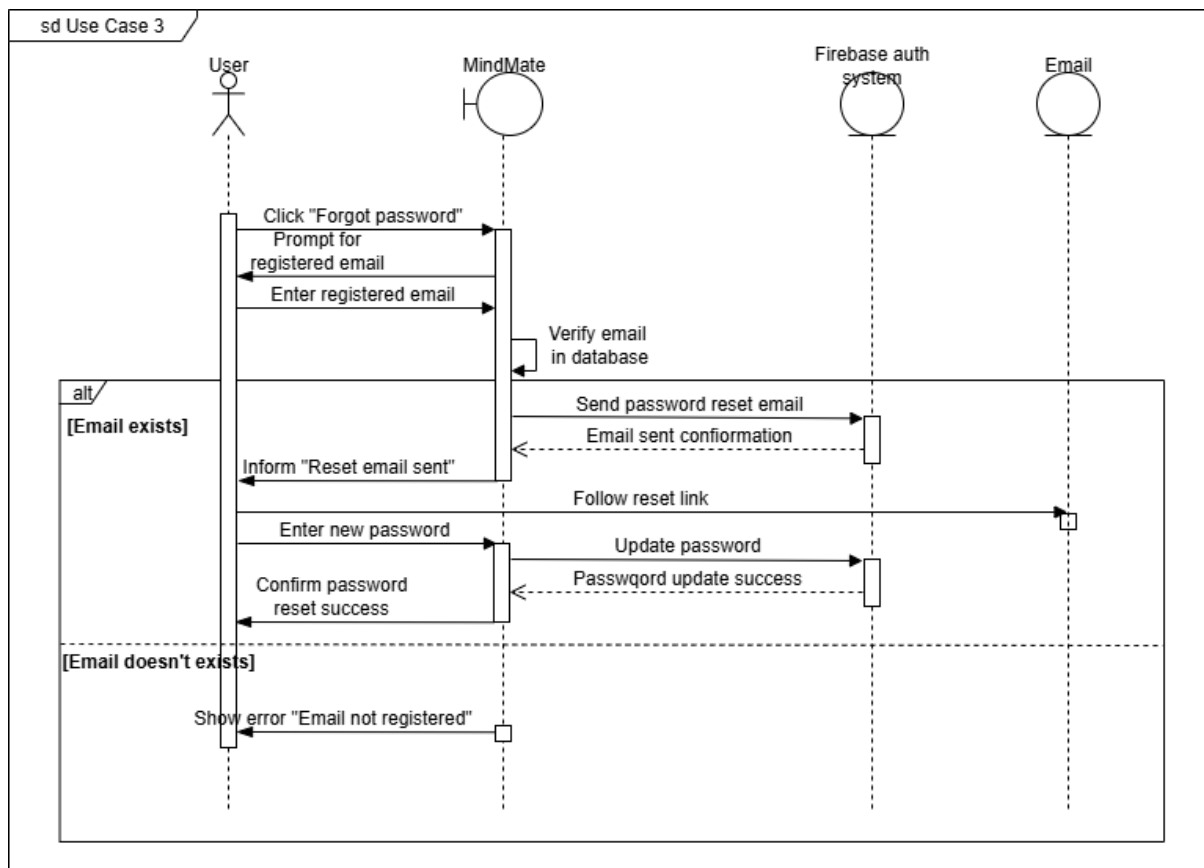


Figure 2.3.3.1: Sequence Diagram for UC003 Reset password

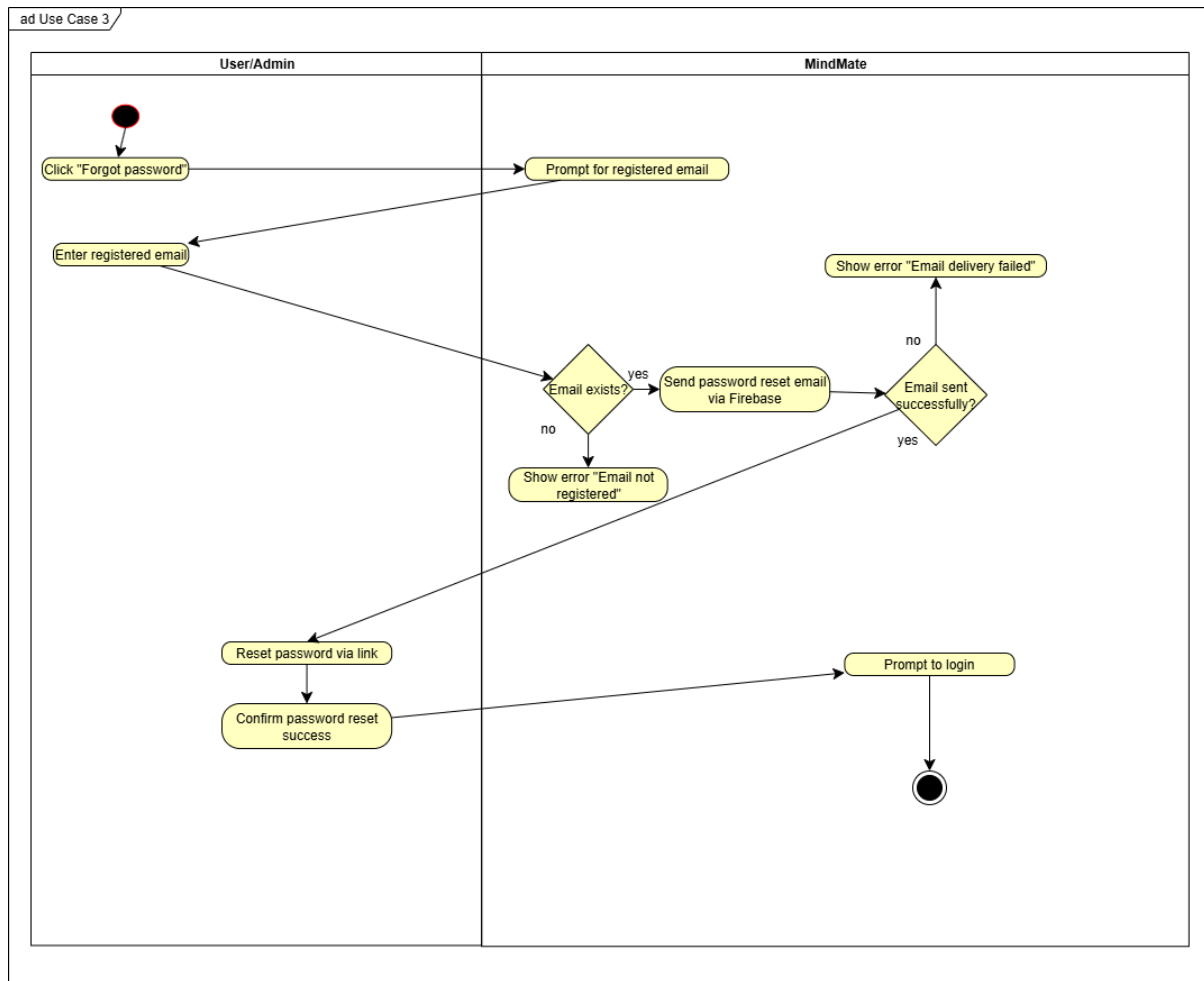


Figure 2.3.3.2: Activity Diagram for UC003 Reset password

2.3.4 UC004: Use Case <Manage profile>

Table 2.3.4.1: Use Case Description for UC004 Manage profile

Use Case ID	UC004
Manage profile	Manage profile
Description	Allows users to view and update their personal profile information such as name, age, email, and gender.
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged in to access their profile.
Normal Flow(s)- NF	1. The user navigates to the profile screen. 2. The system retrieves and displays current profile information. 3. The user clicks on the Edit button. 4. The user edits one or more fields (e.g., name, age, gender). 5. The user submits the updated profile. 6. The system validates and updates the user information in the database. 7. The user sees a confirmation message of successful update.
Alternative Flow(s) - AF	AF1. User decides not to make changes and returns to the previous screen.
Exception Flow(s) - EF	EF1. Invalid input — system displays an error and prompts correction. EF2. System fails to update data due to server/database issues.
Post-condition(s)	The user's updated information is saved and reflected in their profile.

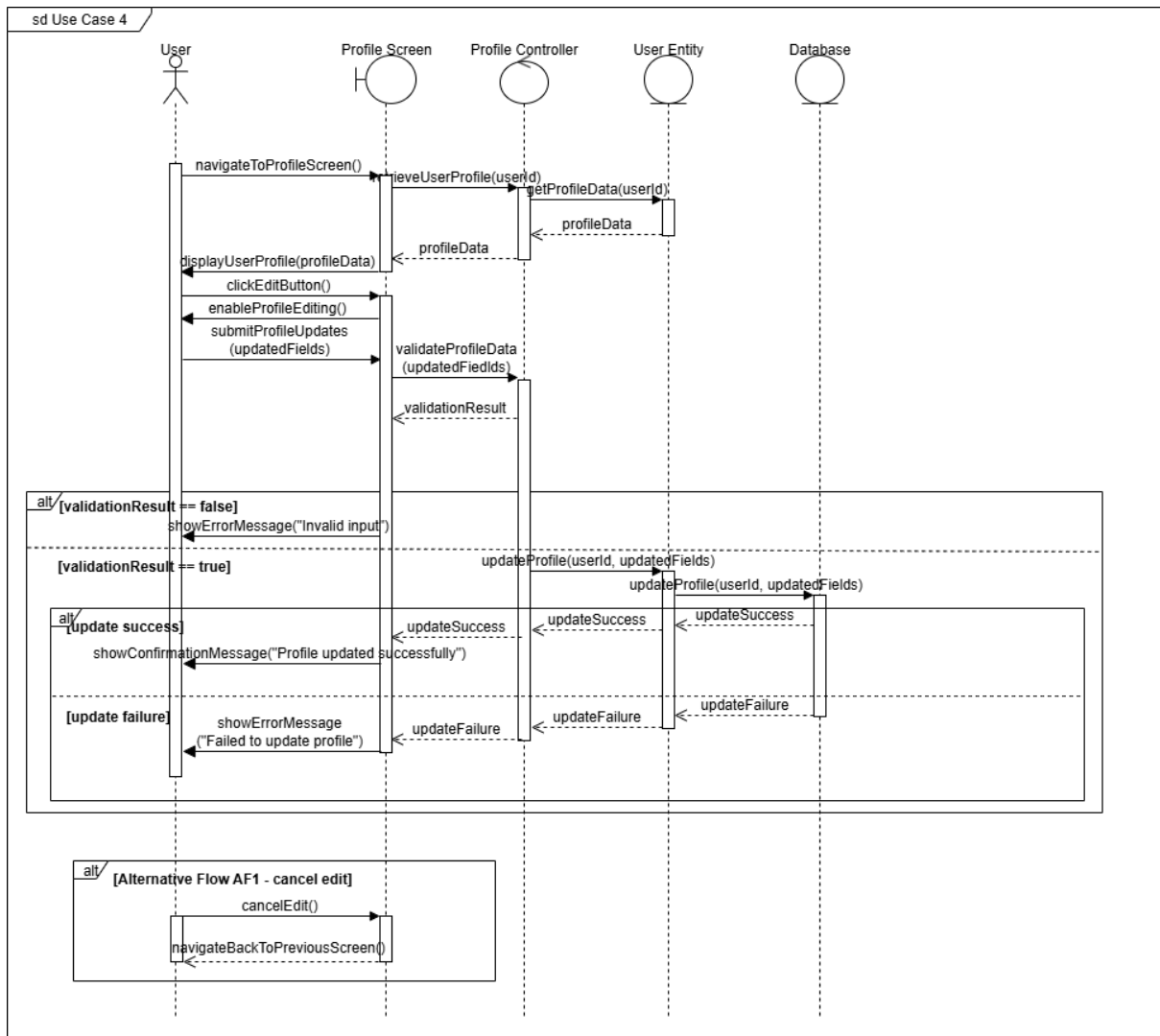


Figure 2.3.4.1: Sequence Diagram for UC004 Manage profile

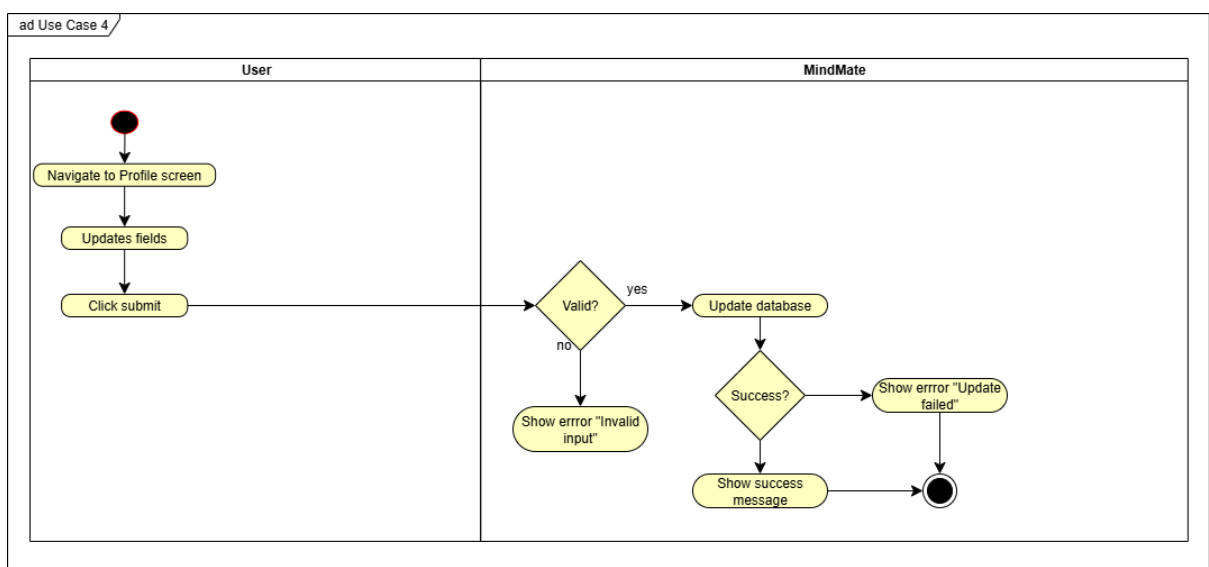


Figure 2.3.4.2: Activity Diagram for UC004 Manage profile

2.3.5 UC005: Use Case <Manage notifications>

Table 2.3.5.1: Use Case Description for UC005 Manage notifications

Use Case ID	UC005
Use Case Name	Manage notifications
Description	Allows users to configure their notification preferences such as enabling/disabling reminders or motivational messages.
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged into the app.
Normal Flow(s)- NF	1. The user navigates to the settings screen. 2. The system displays current notification preferences (e.g., reminders, motivational messages). 3. The user toggles switches or selects preferences (e.g., enable/disable types of notifications). 4. The user confirms or saves the settings. 5. The system updates and stores the new notification settings. 6. The user sees a confirmation that preferences have been saved.
Alternative Flow(s) - AF	AF1. The user exits the settings screen without saving — no changes are applied.
Exception Flow(s) - EF	EF1. Notification settings fail to save due to connectivity or database error — system displays an error message.
Post-condition(s)	The user's updated notification preferences are saved and applied by the system.

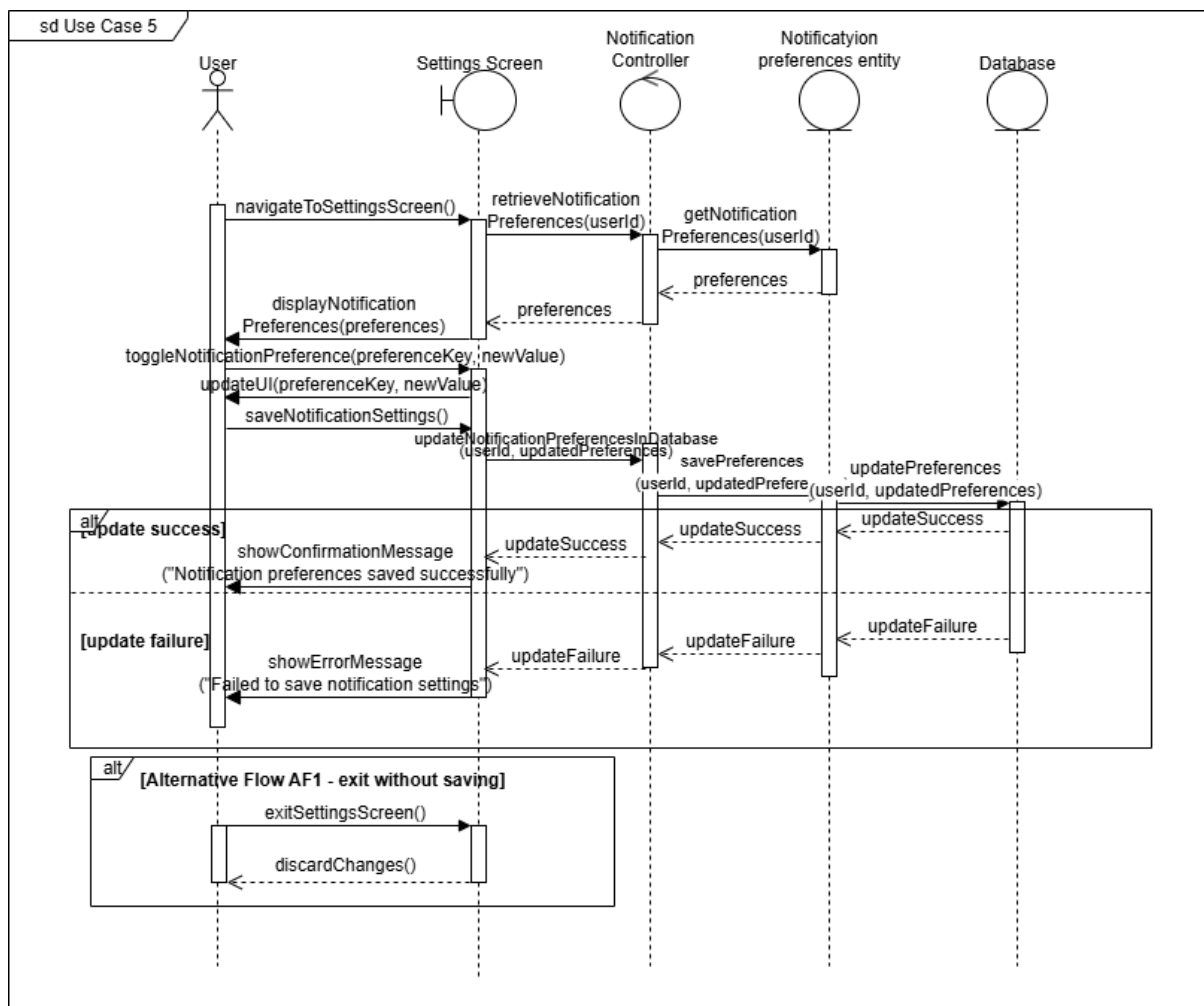


Figure 2.3.5.1: Sequence Diagram for UC005 Manage notifications

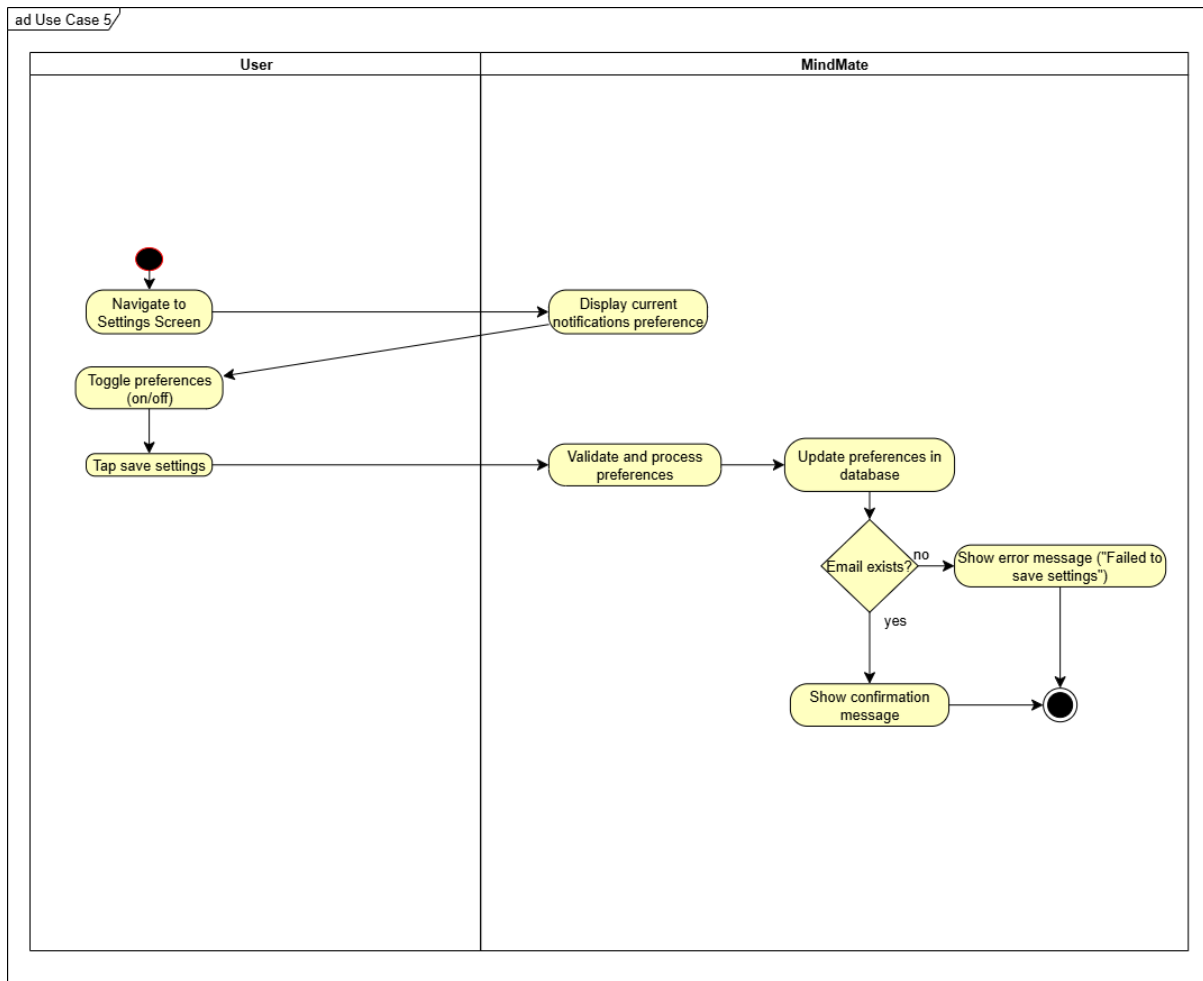


Figure 2.3.5.2: Activity Diagram for UC005 Manage notifications

2.3.6 UC006: Use Case <Track emotions>

Table 2.3.6.1: Use Case Description for UC006 Track emotions

Use Case ID	UC006
Use Case Name	Track emotions
Description	Allows users to log their emotional state for the day. Based on the emotion and recent trends, the system recommends a personalized self-help activity or offers to launch the AI Assistant.
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged into the app. If no emotion has been logged for the current date, the system prompts the user at app launch.
Normal Flow(s)- NF	1. System checks if an emotion has been logged for the current day 2. If not, system displays the emotion tracking screen 3. User selects an emotion 4. User optionally adds notes 5. User submits 6. System logs the emotion with timestamp 7. System checks for recent mood trend (e.g., multiple negative days) 8. System recommends a personalized activity (e.g., breathing, journaling) 9. User performs the activity 10. System logs the activity 11. If repeated negative emotions are detected, system prompts to open the AI chatbot
Alternative Flow(s) - AF	AF1. The user chooses to skip the prompt or close the check-in popup. AF2. The user accesses the emotion tracking manually later through the navigation tab. AF3. The user skips the suggested activity. AF4. System detects recurring negative mood:

	AF4.1. System prompts: “Would you like to talk to our AI Assistant?” AF4.2. If Yes → UC008 triggered AF4.3. If No → return to home
Exception Flow(s) - EF	EF1. Emotion log fails to save due to connectivity or database error — system displays an error message.
Post-condition(s)	A new emotion log entry is created and stored with a timestamp and any optional notes. An activity is logged optionally to support the emotion. If pattern of distress detected, chatbot is prompted. The system will not prompt again for that day.

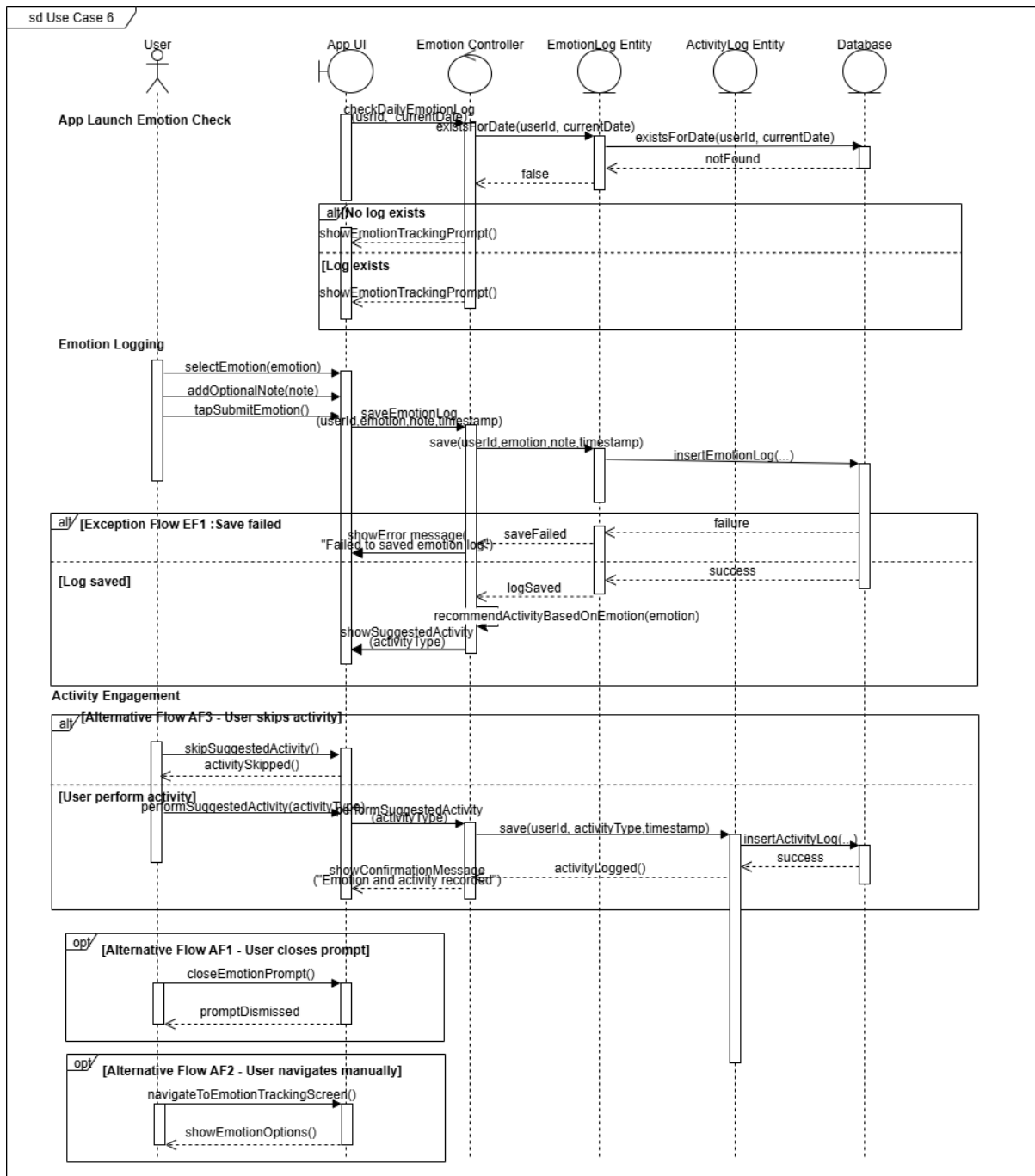


Figure 2.3.6.1: Sequence Diagram for UC006 Track emotions

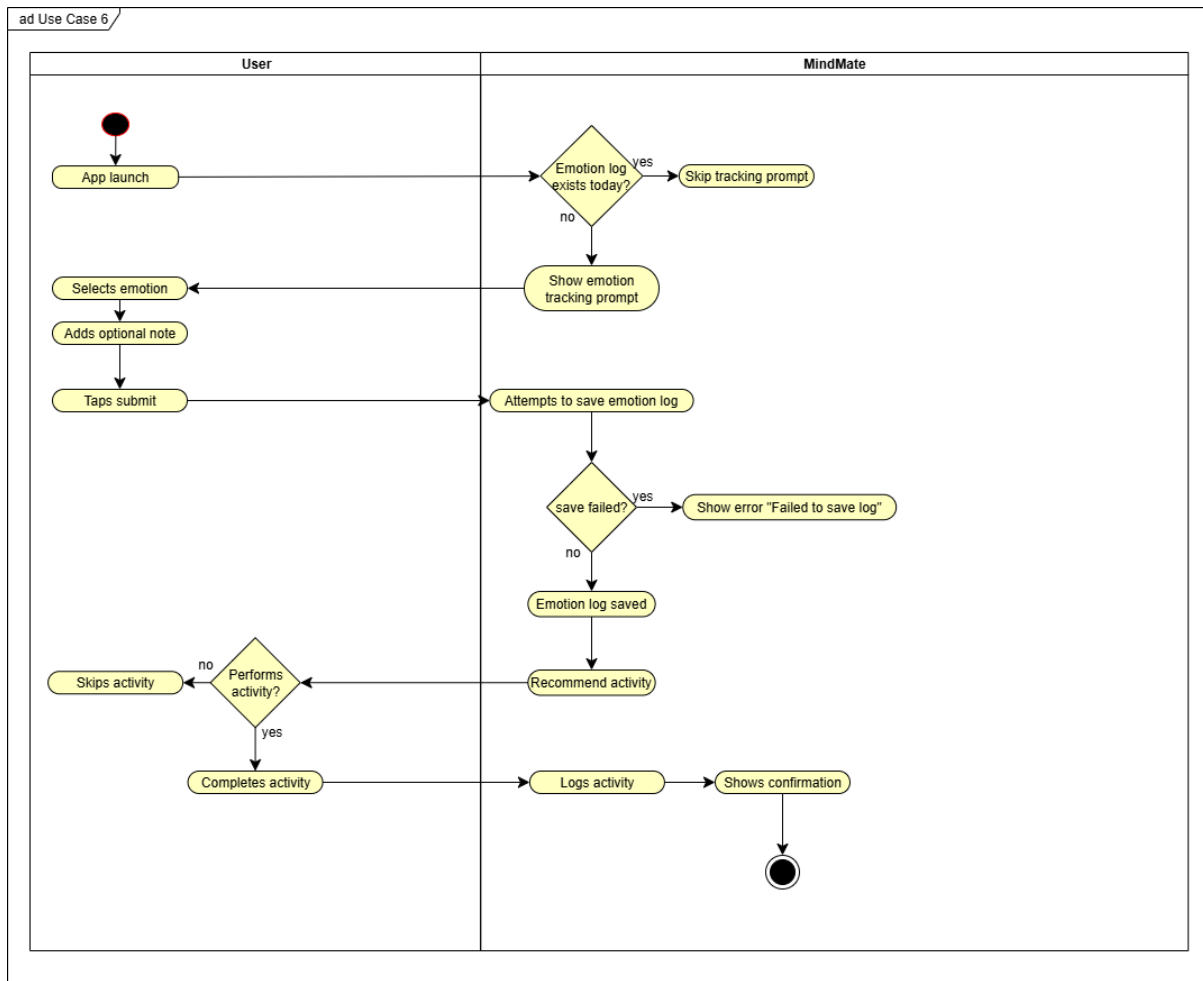


Figure 2.3.6.2: Activity Diagram for UC006 Track emotions

2.3.7 UC007: Use Case <View mental health insights>

Table 2.3.7.1: Use Case Description for UC007 View mental health insights

Use Case ID	UC007
Use Case Name	View mental health insights
Description	Allows users to view a visual summary of their emotional trends and mental health patterns over time based on their daily emotion logs.
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged into the app and have at least one emotion log entry in the system.
Normal Flow(s)- NF	1. The user navigates to the Mental Health Insights screen. 2. The system retrieves the user's historical emotion logs. 3. The system analyzes and summarizes trends (e.g., most frequent emotion). 4. The system displays visualizations (e.g., charts, mood history timeline, word clouds). 5. The user views and interprets their emotional patterns over days/weeks. 6. The system suggests actions (e.g., chatbot, breathing) based on trend
Alternative Flow(s) - AF	AF1. The user filters insights by date range (e.g., past week, month, custom). AF2. The user switches chart types or toggles between summary and detailed views. AF3. The user ignores suggestions
Exception Flow(s) - EF	EF1. No emotion data available — system notifies the user with a message like “Start tracking your emotions to see insights.” EF2. Data retrieval failure due to connectivity issues — system displays an error.
Post-condition(s)	The user is informed of their emotional trends and patterns visually, supporting better self-awareness and reflection. Personalized, pattern-

based suggestions are presented. System may recommend chatbot session if prolonged negative trend detected.

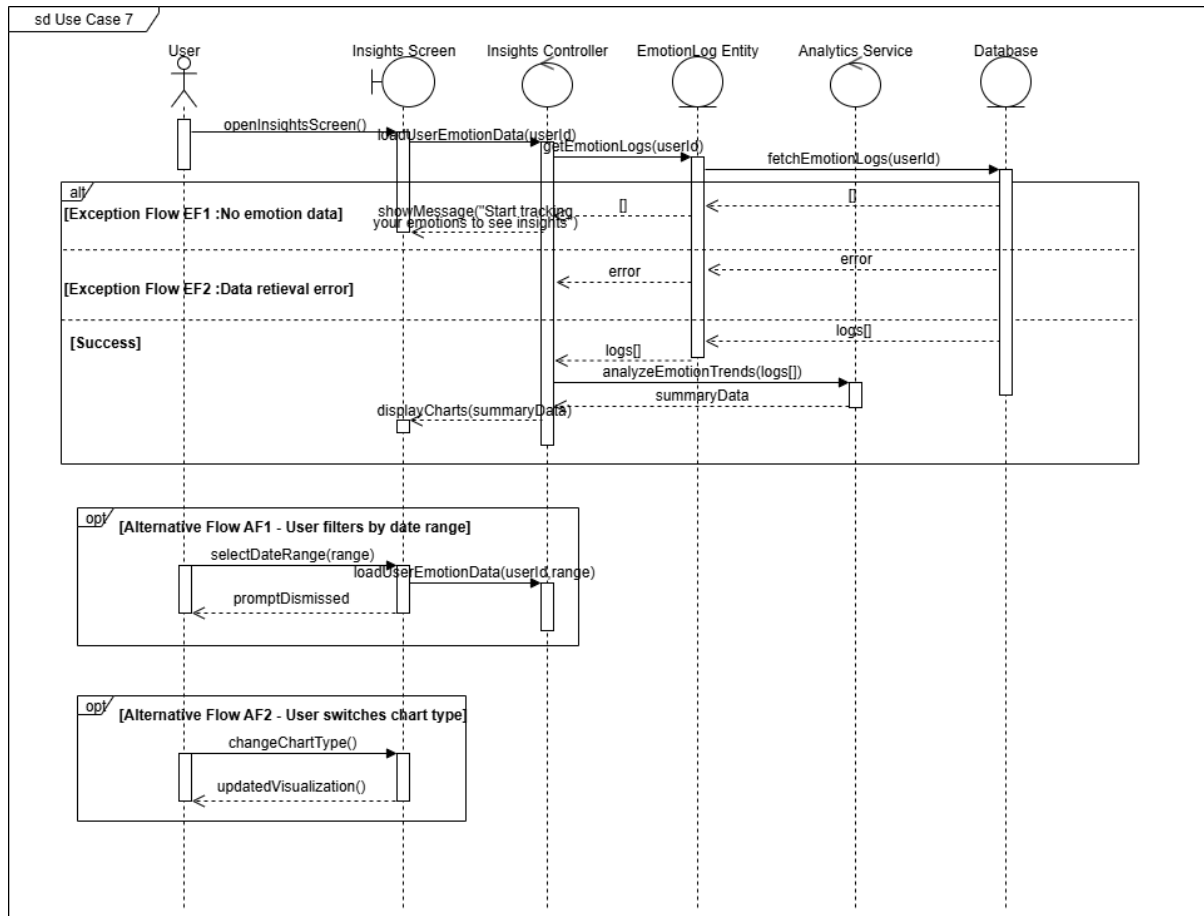


Figure 2.3.7.1: Sequence Diagram for UC007 View mental health insights

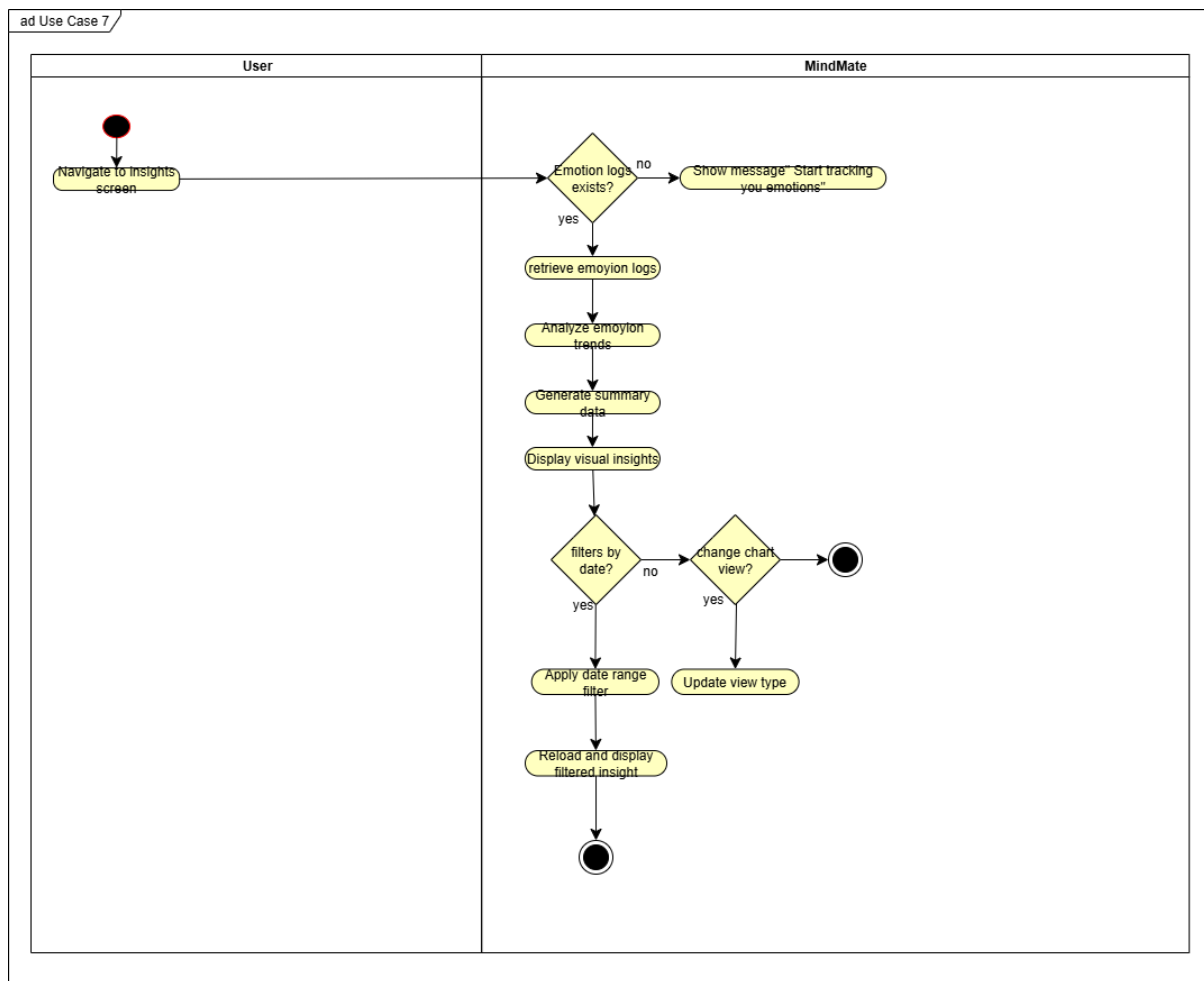


Figure 2.3.7.2: Activity Diagram for UC007 View mental health insights

2.3.8 UC008: Use Case <Access AI chatbot>

Table 2.3.8.1: Use Case Description for UC008 Access AI chatbot

Use Case ID	UC008
Use Case Name	Access AI chatbot
Description	Allows students to interact with the AI-powered virtual assistant for emotional support, motivational prompts, and self-awareness recommendations. The chatbot simulates basic counselling responses based on the student's emotional state.
Actor(s)	Student, System (AI Assistant)
Pre-condition(s)	• Student is successfully logged into the MindMate app • Emotion logs are available in the system (for system-triggered cases)
Normal Flow(s)- NF	1. The user navigates to the Chatbot screen. 2. The system displays a chat interface and greets the user. 3. The user types a message or selects a suggested prompt (e.g., "I'm feeling stressed"). 4. The system sends the message to the AI Assistant. 5. The AI Assistant processes the input and returns a thoughtful response. 6. The system displays the AI's reply in the chat interface. 7. The user continues the conversation as needed. 8. The chat session is saved in the user's chat history.
Alternative Flow(s) - AF	AF1. Student completes mood logging via UC006 AF2. System analyzes logged emotion AF3. If negative emotion detected (e.g., sadness, stress, anxiety): AF3.1. System displays a prompt: "We've noticed you're feeling [emotion]. Would you like to talk to our AI Assistant for support?" AF4. Student selects "Yes": AF4.1. System automatically launches AI Assistant (steps continue from Basic Flow step 3) AF5. Student selects "No":

	AF5.1. System returns to Home screen
Exception Flow(s) - EF	EF1. Network or AI service error — system displays an error message and suggests retrying.
Post-condition(s)	The chat session is recorded and can be accessed later. Responses tailored using logged data and trend recognition. May include suggestions for reflection, activities, or real-life counselling

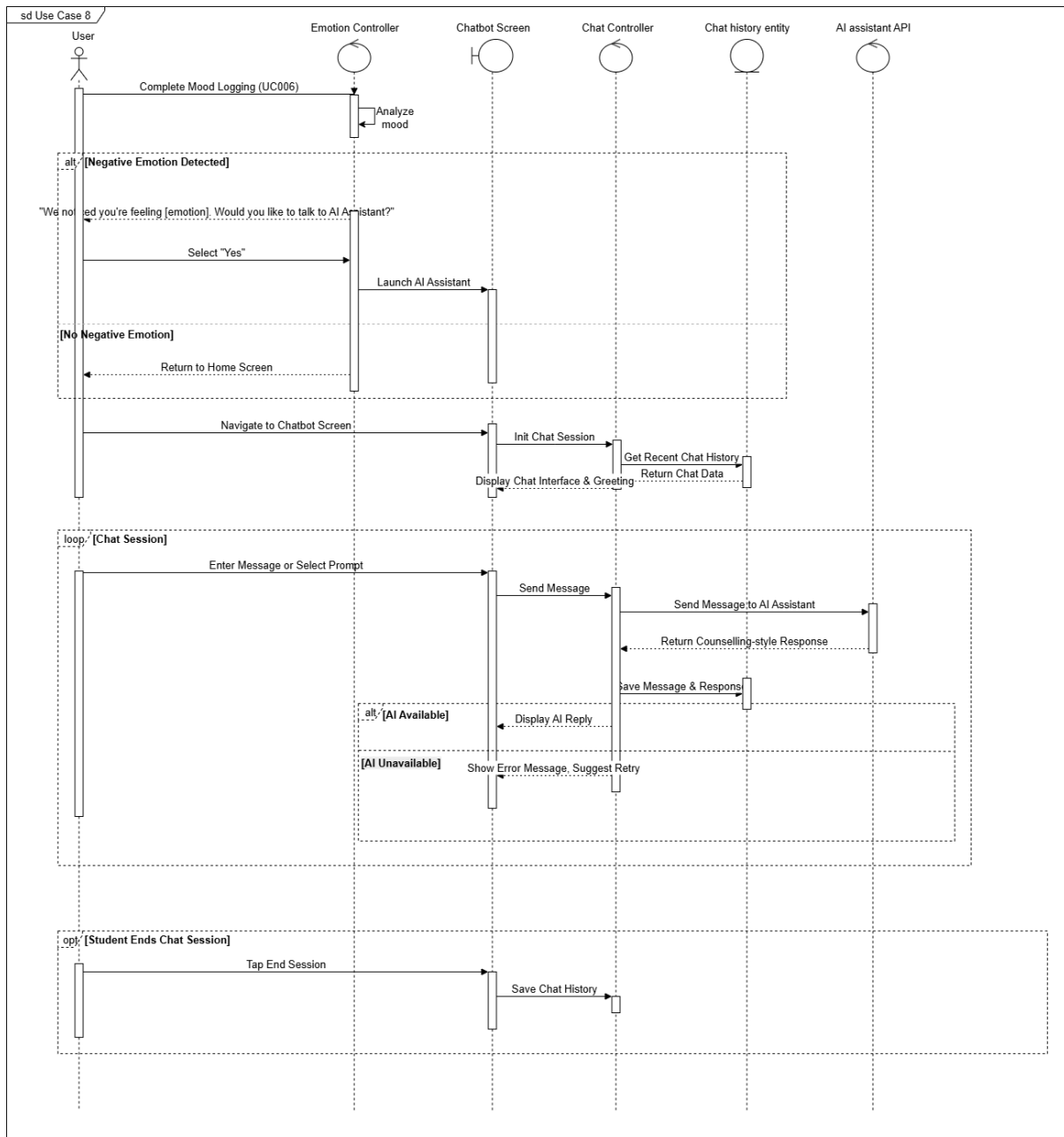


Figure 2.3.8.1: Sequence Diagram for UC008 Access Ai chatbot

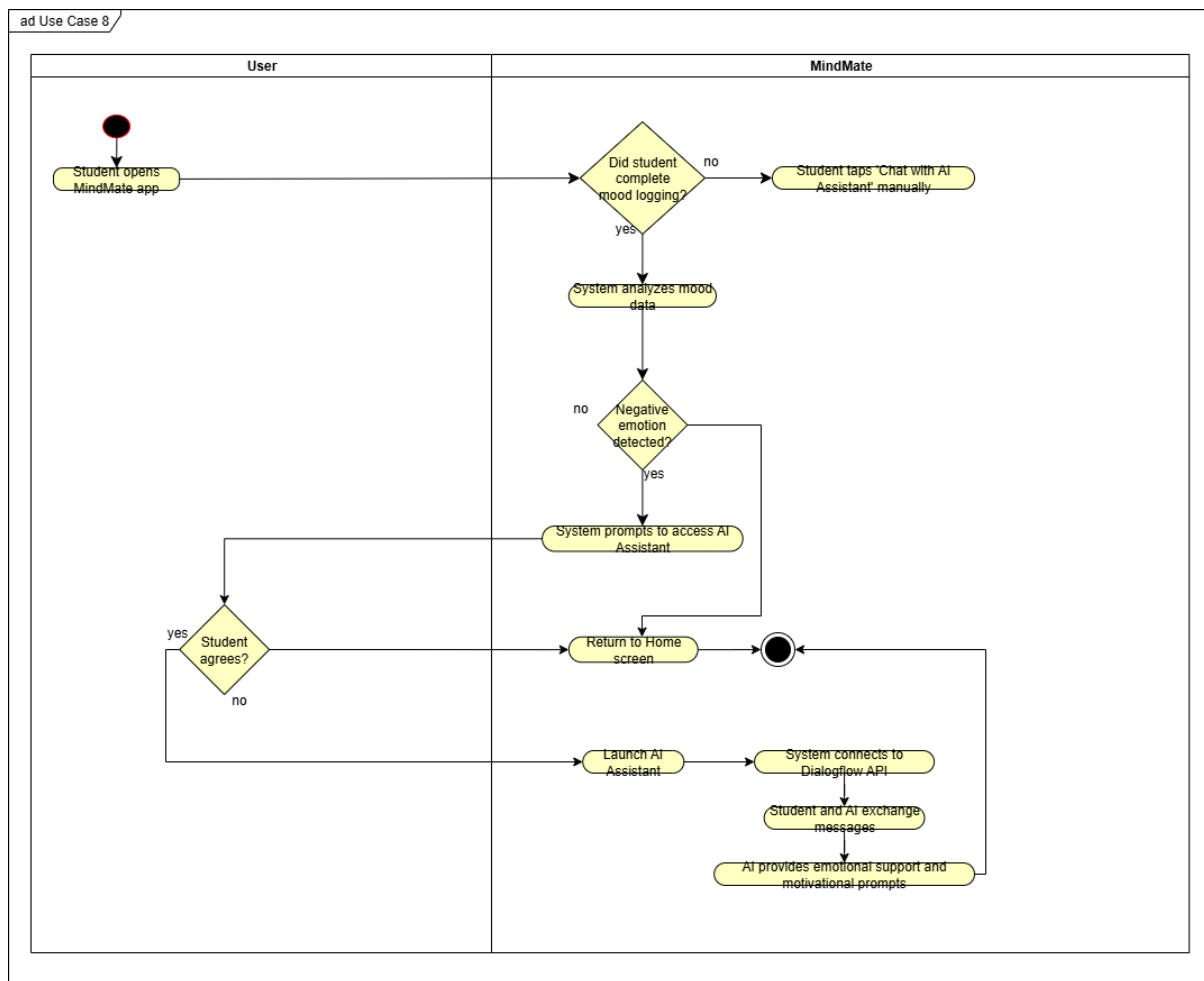


Figure 2.3.8.2: Activity Diagram for UC008 Access Ai chatbot

2.3.9 UC009: Use Case <Gamification>

Table 2.3.9.1: Use Case Description for UC009 Gamification

Use Case ID	UC009
Use Case Name	Gamification
Description	Encourages consistent app usage by rewarding users with achievements, badges, and points based on their activity (e.g., daily check-ins, streaks).
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged into the app. The system must have access to the user's activity and emotion log history.
Normal Flow(s)- NF	1. The user performs activities such as emotion logging, accessing the chatbot, or completing mindfulness exercises. 2. The system evaluates if any achievements criteria are met (e.g., 7-day streak, first mood entry). 3. If criteria are met, the system awards a badge or increases the user's point total. 4. The system notifies the user of their new achievement or progress milestone. 5. The achievement is saved in the user's gamification history.
Alternative Flow(s) - AF	AF1. The user views a list of available achievements and their progress toward them.
Exception Flow(s) - EF	EF1. Achievement update fails due to database error — system displays a temporary error message.
Post-condition(s)	The user earns badges or points, which are reflected in their gamification history and visible in their profile. Motivation and engagement are reinforced.

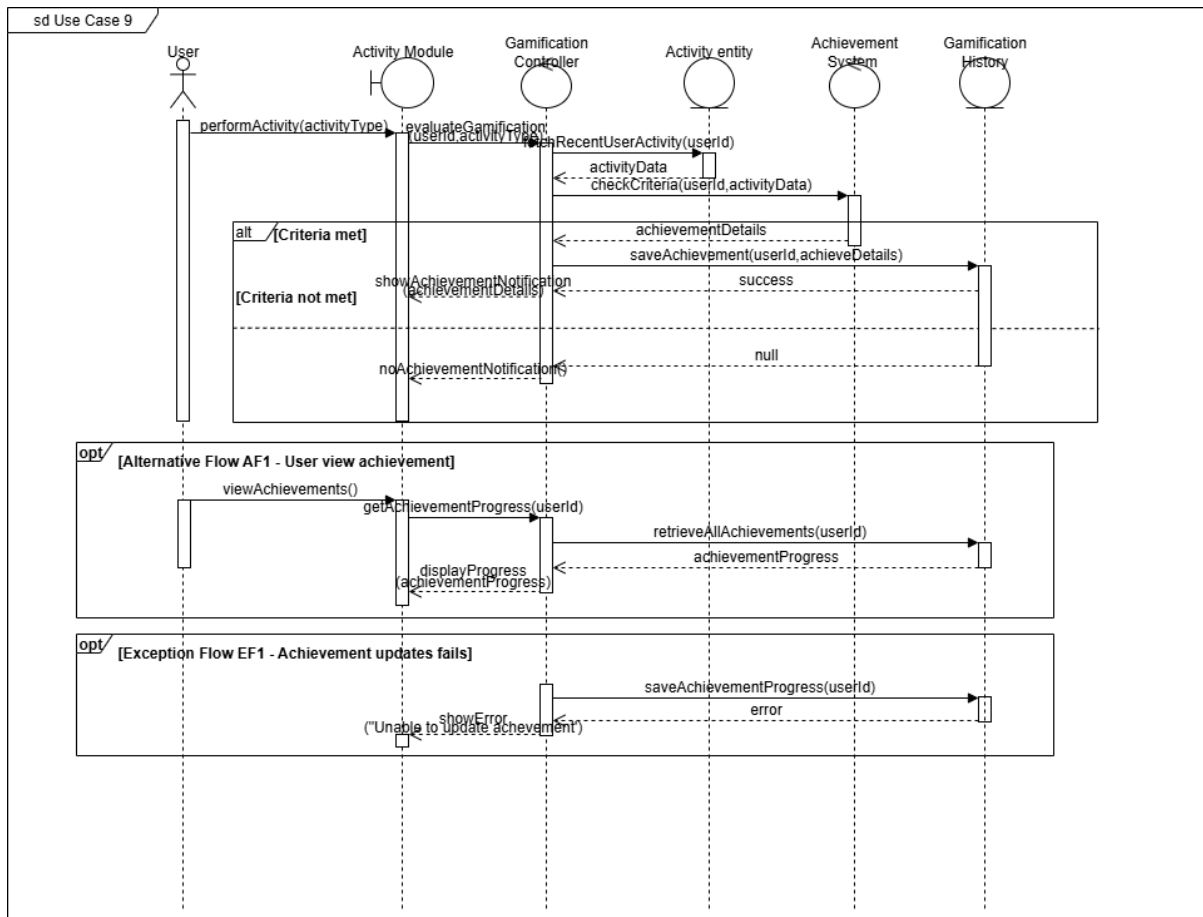


Figure 2.3.9.1: Sequence Diagram for UC009 Gamification

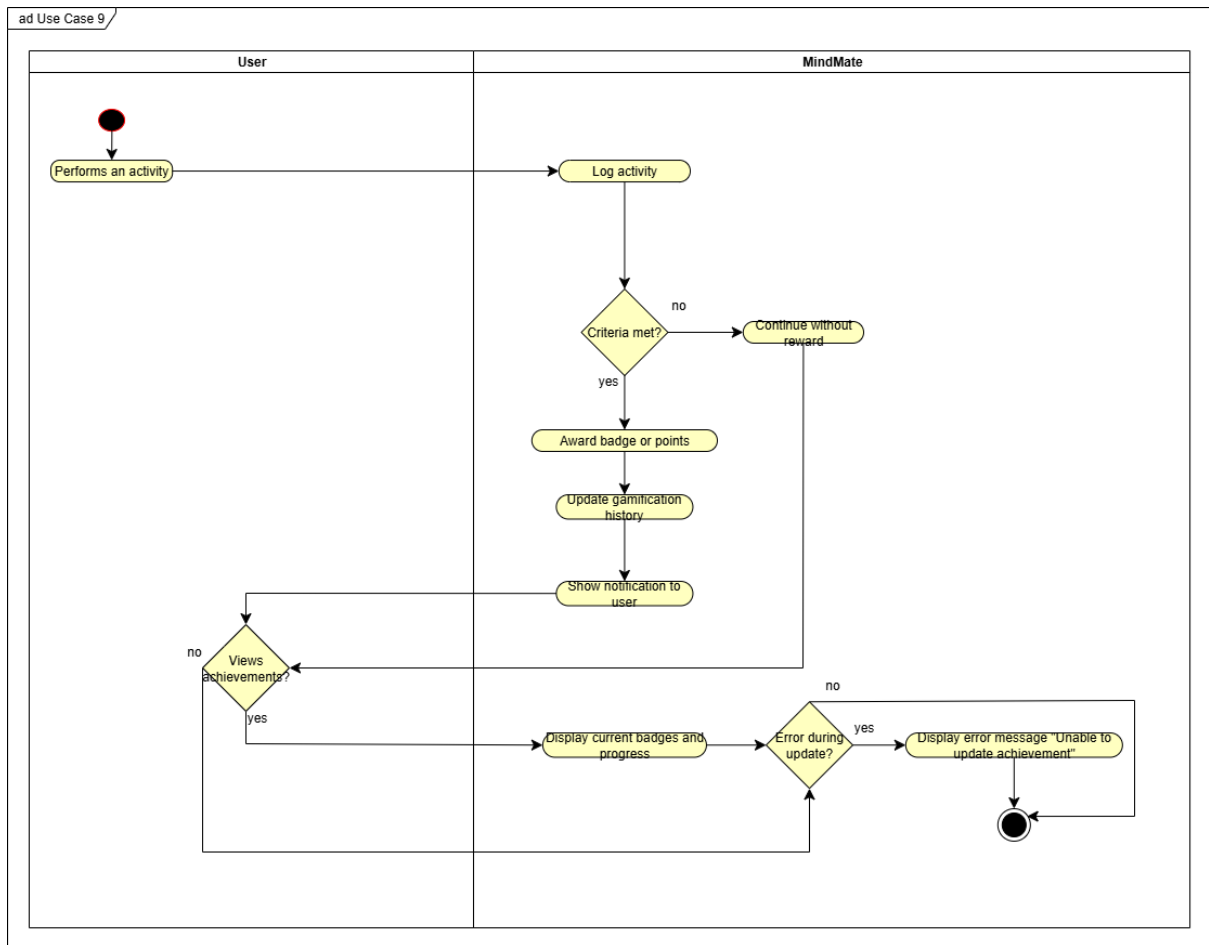


Figure 2.3.9.2: Activity Diagram for UC009 Gamification

2.3.10 UC010: Use Case <View progress history>

Table 2.3.10.1: Use Case Description for UC010 View progress history

Use Case ID	UC010
Use Case Name	View progress history
Description	Enables users to review their historical app usage, logged emotions, chatbot sessions, and earned achievements over time.
Actor(s)	Primary: User
Pre-condition(s)	The user must be logged into the app. The system must have relevant historical data available (e.g., emotion logs, chat sessions, achievements).
Normal Flow(s)- NF	1. The user navigates to the Progress History screen. 2. The system fetches the user's historical records from the database: a) Emotion logs b) Chatbot session history c) Achievements and streaks 3. The system displays a categorized list of entries. 4. The user browses through the timeline or filters the history by type or date range. 5. The user selects an entry (e.g., a chat session or achievement) to view details.
Alternative Flow(s) - AF	AF1. The user filters by category (e.g., only achievements or only mood entries).
Exception Flow(s) - EF	EF1. No data available — system shows a message like “No records found yet.” EF2. Data retrieval error — system notifies the user and suggests retrying.
Post-condition(s)	The user views detailed information on their app engagement and can reflect on their growth and patterns.

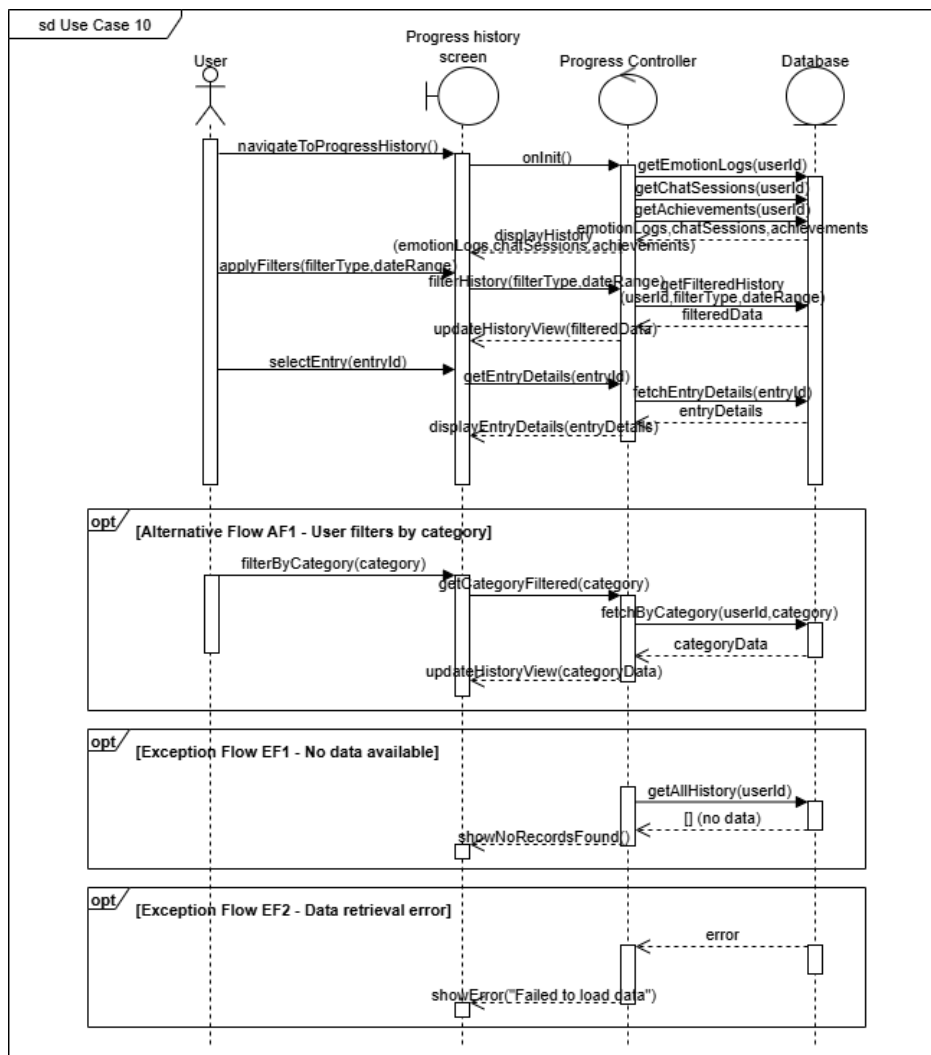


Figure 2.3.10.1: Sequence Diagram for UC010 View progress history

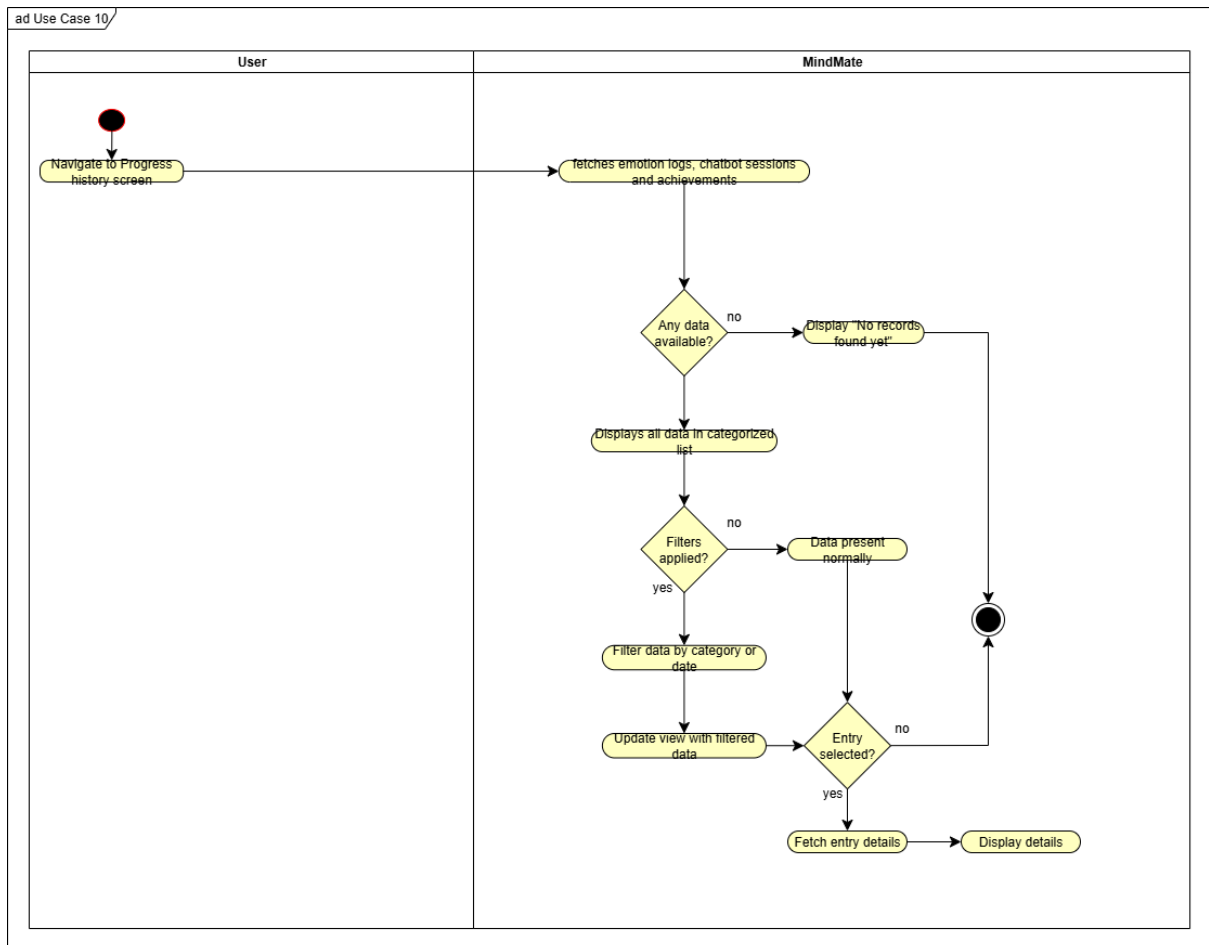


Figure 2.3.10.2: Activity Diagram for UC010 View progress history

2.3.11 UC011: Use Case <Manage user accounts>

Table 2.3.11.1: Use Case Description for UC011 Manage user accounts

Use Case ID	UC011
Use Case Name	Manage user accounts
Description	Allows an admin to monitor, view, and manage registered user accounts via Firebase Console.
Actor(s)	Primary: Admin
Pre-condition(s)	The admin must be logged into the Firebase Console with appropriate privileges.
Normal Flow(s)- NF	1. The admin logs into the Firebase Console. 2. The admin navigates to the Users section. 3. The system displays the list of all registered users with basic details (e.g., email, last login). 4. The admin selects a user account to view more details. 5. The admin performs management actions such as: a) Resetting a user password b) Disabling or enabling an account c) View anonymized usage statistics (e.g., number of mood logs, last active date)
Alternative Flow(s) - AF	AF1. The admin searches for a user by name, email, or UID.
Exception Flow(s) - EF	EF1. Connectivity issue — system fails to fetch user data and shows an error message.
Post-condition(s)	Admin successfully views or modifies user account data as needed. All changes are logged by the system.

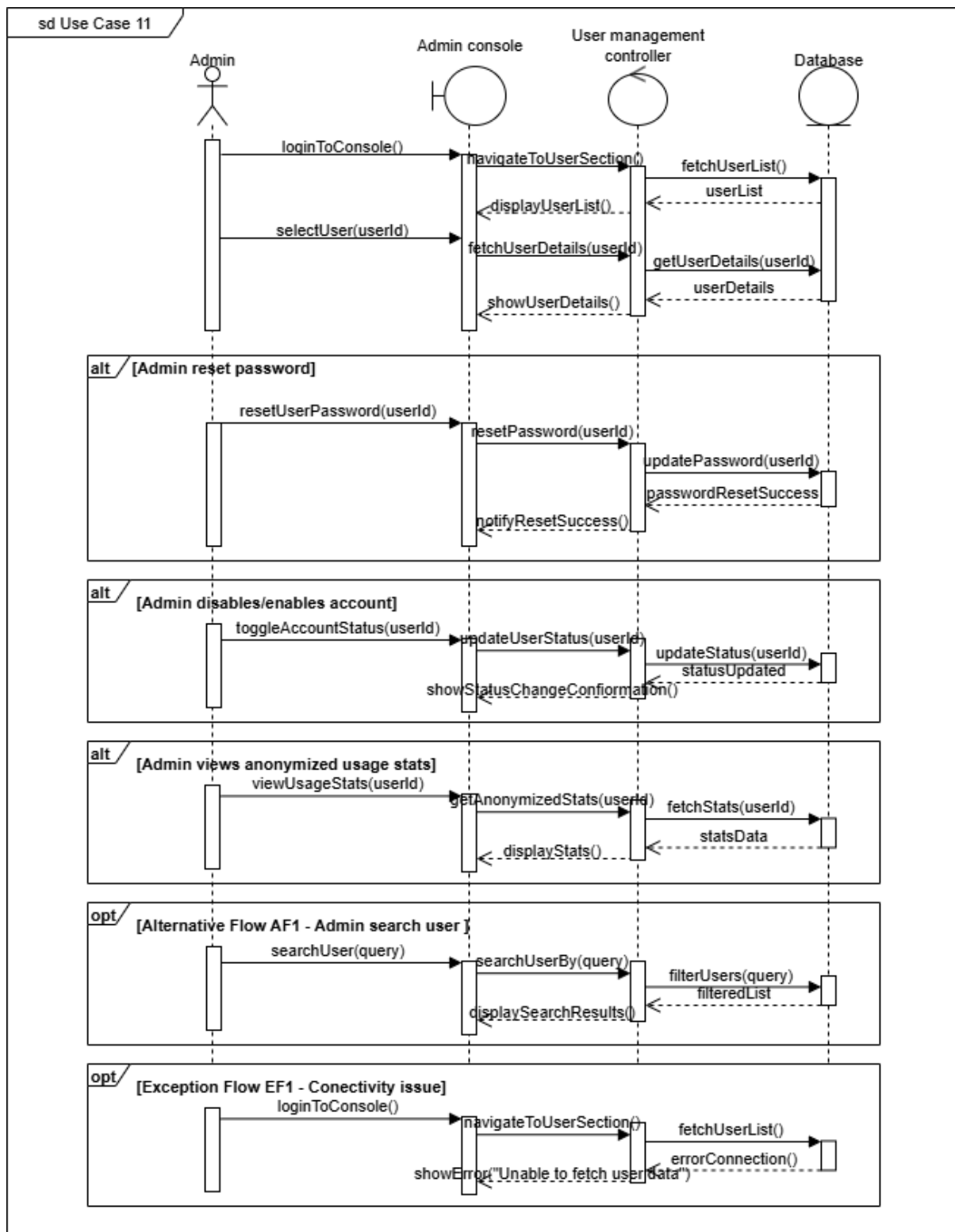


Figure 2.3.11.1: Sequence Diagram for UC011 Manage user accounts

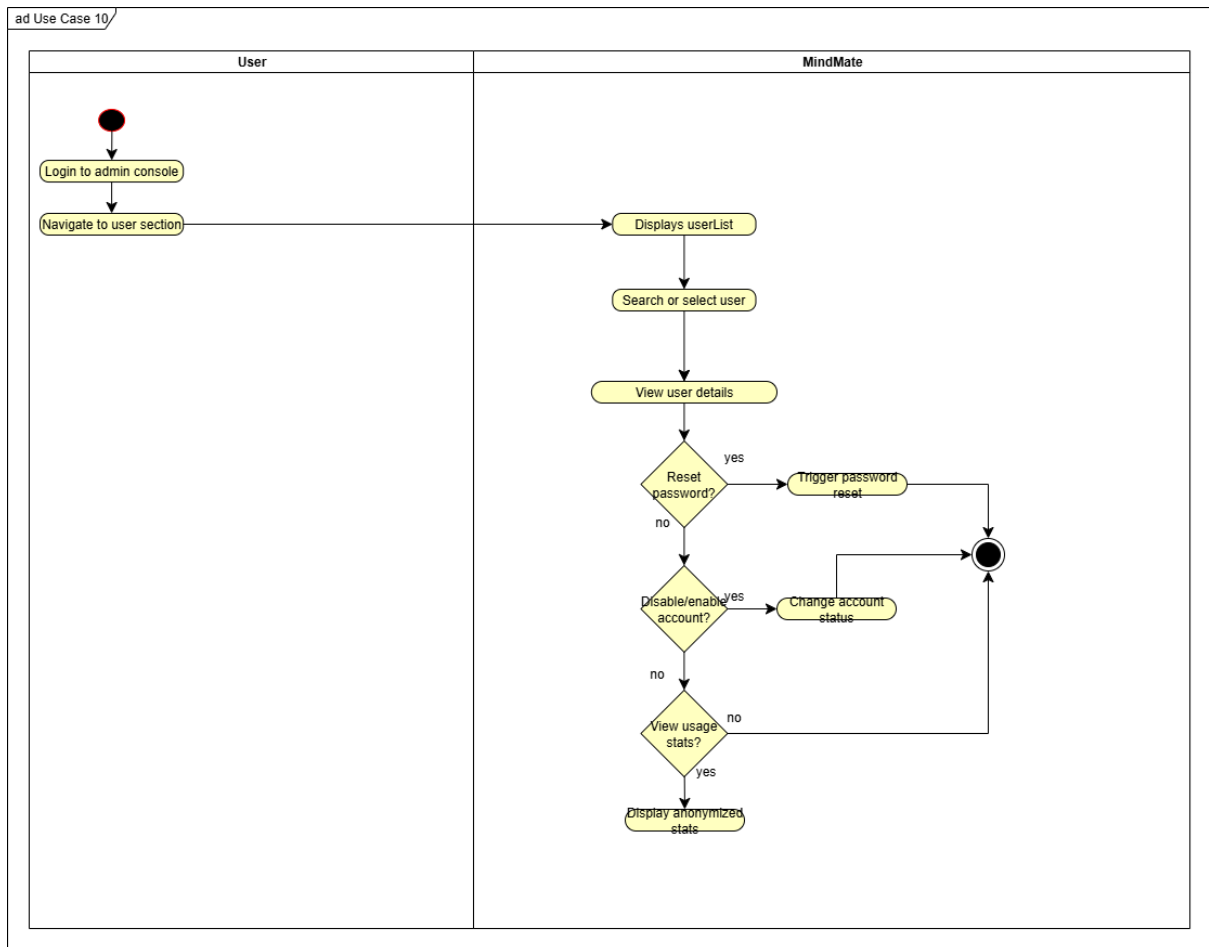


Figure 2.3.11.2: Activity Diagram for UC011 Manage user accounts

2.4 Performance and Other Requirements

This section describes The non-functional requirement of MindMate mobile application. It focuses on the system attributes, performance and qualities to ensure usability, efficiency and robustness of the system.

2.4.1 Software System Attributes

- **Usability:**

The application must offer an intuitive and user-friendly interface, ensuring that users with minimal technical experience can easily navigate and utilize core features such as emotion logging and chatbot interaction.

- **Reliability:**

The application must maintain 99% uptime, particularly for core features like authentication, emotion logging and chatbot interaction. The app must function consistently without crashing under typical use conditions.

- **Maintainability:**

The application must be structured using modular Flutter components to allow easy updates, bug fixes, and integration of additional features in future iterations.

- **Portability:**

The application must run effectively on both Android and iOS platforms.

- **Compatibility:**

The system must integrate seamlessly with Firebase and Dialogflow services. All APIs and libraries used should be compatible with the latest Flutter SDK version.

2.4.2 Performance Requirements

- **Response time:**

The application should respond to user interactions such as daily emotion logging or chatbot queries within 2 seconds under normal network conditions.

- **Throughput:**

The application must be capable of supporting at least 100 concurrent users during peak hours without significant performance degradation.

- **Capacity:**

Firebase should be able to store up to 10,000 user accounts and maintain logs for at least one year of usage data per user without affecting access speed.

- **Availability:**

The system should be available 24/7, except during planned updates or maintenance.

2.4.3 Other Requirements

- **Security:**

All user data must be encrypted using Firebase's built-in security features. User authentication should be managed through secure OAuth mechanisms such as Google Sign-In.

- **Legal and Regulatory:**

The system must comply with Malaysia's Personal Data Protection Act (PDPA) and any relevant university policies regarding student data privacy.

- **Environmental:**

The system is cloud-hosted and mobile-based, ensuring minimal environmental impact in terms of energy consumption and physical infrastructure.

2.5 Design Constraints

This section explains the limitations that affect the design and development of the MindMate application.

- **Environmental Constraints:**

The application is designed for indoor use on mobile devices under normal environmental conditions (0°C to 40°C). No special environmental protections are required.

- **Hardware Constraints:**

The application must be optimized to run on devices with at least 2GB of RAM and 100MB of available storage. Performance may degrade on lower-spec devices.

- **Security Constraints:**

All sensitive data such as user profiles, emotion log, activity log and chat history must be encrypted using AES-256. Firebase rules must enforce secure access control to database entries and authentication services.

- **Compatibility Constraints:**

The application must maintain compatibility with the latest stable version of the Flutter SDK and integrate with Google Firebase and Dialogflow APIs without third-party middleware.

- **Performance Constraints:**

The application must maintain a response time of less than 2 seconds per user interaction and support up to 500 simultaneous users during stress testing scenarios.

Appendix E Software Design Specification (SDS)



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 3032: Software Engineering FYP1

Semester 02, 2024/2025

Software Design Specification (SDS)

MindMate: A Flutter-Based Mental Health App for
University Students with Emotion Tracking and AI
Powered Virtual Assistant

Version 2.0

Prepared by: Lau Yun Xi

1. Revision Page

a. Overview

This document presents the Software Design Specification (SDS) for the MindMate mobile application. It describes the software architecture, module breakdown, data flow, system components, and design viewpoints to support the implementation of the MindMate application.

b. Issuing Organization / Team

Faculty of Computing, Universiti Teknologi Malaysia (UTM)

Bachelor Of Computer Science (Software Engineering)

SECJ3032-01 Final Year Project 1

c. Authorship / Project Team Members

Name	Role	Assigned tasks	Status
Lau Yun Xi	Sole Developer	Documentation of SDS	Completed

d. Revision History

REVISION NO.	ISSUE DATE	DETAILS OF REVISION
1.0	6/6/2025	Initial Draft for SDS
2.0	8/6/2025	Final Daft for SDS

Note:

This Software Design Descriptions (SDD template is adapted from IEEE Recommended Practice based on Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), that are simplified and customized to meet the need of SECJ3203 FYP1 SE course at Faculty of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

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2. Introduction

2.1 Purpose

This SDS describes the architecture, components, interfaces, and other characteristics of the MindMate application. It is intended for developers, project supervisors, testers, and stakeholders involved in building and evaluating the MindMate mobile application. The SDS provides a detailed blueprint to guide the implementation of the application.

2.2 Scope

The software product is MindMate, a Flutter-based mobile application designed for university students to monitor their mental well-being. It allows users to log their moods, view mood trends, and receive personalized motivational content. The app integrates with Google Dialogflow via API to provide chatbot functionality for real-time virtual assistance, and Firebase is used to manage user authentication and store emotional data securely in the cloud. MindMate aims to help university students become more aware of their mental health through daily check-ins, simple reflections, and mood-related suggestions. It also includes gamification features such as streak counters and badges to promote consistent engagement. The app is developed as a prototype project and will focus on emotion tracking and mental health support rather than clinical diagnosis or therapy.

There are two types of users in the system:

- End users (students): Can interact with the app features such as logging moods, chatting with the virtual assistant, and reviewing mood history.
- Admin (developer): Has backend access to monitor data structure, manage application updates, and perform system testing.

The application will focus on supporting six core emotion states: happy, sad, stressed, anxious, frustrated, and focused. The system does not include professional counseling modules, emergency services, real-time human therapist chat, or any payment

integration. Additionally, the chatbot content is powered by external APIs and will not be developed manually from scratch.

2.3 Context

This SDS is developed in line with the software engineering practices recommended for academic projects at Universiti Teknologi Malaysia. It is based on IEEE software design documentation standards and tailored to the requirements of the course. The document is intended for technical stakeholders, including software designers and evaluators, to provide a consistent and well-organized representation of the system's architecture and design choices. It ensures that the development process remains well-documented, traceable, and ready for future enhancement.

2.4 Summary

This SDS includes a detailed breakdown of the system's design viewpoints. These cover the context, composition, logical, interface, structure, interaction, state, and algorithmic perspectives. Each viewpoint is supported with relevant diagrams such as use case, class, activity, and state machine diagrams. The document concludes with user interface design elements and supporting appendices to complete the overall design documentation.

3. References

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4. Glossary

Terms / Acronyms / Abbreviation	Definitions
AI	Artificial Intelligence. In MindMate, used to support the chatbot virtual assistant for mental health conversations.
API	Application Programming Interface. Allows communication between software modules or with backend services like Firebase.
Chatbot	A conversational interface powered by AI, designed to interact with users and provide emotional support.
ERD	Entity Relationship Diagram. A data modeling technique that shows the relationships between different data entities.
SDS / SDD	Software Design Specification / Software Design Description. A document that outlines how software will be designed and structured.
UML	Unified Modeling Language. A standardized modeling language used for visualizing system design (e.g., use case, class, and activity diagrams).

5. Design Body

5.1 Design Stakeholders and Their Concerns

In the context of the MindMate mobile application, several stakeholders are identified based on their involvement in the design, development, evaluation, and future use of the system. Each stakeholder has specific concerns that are addressed through the system's architecture, functionality, and usability.

Stakeholder	Role	Design Concerns
Developer	Designer, implementer, and document author	Maintain a clear modular design; ensure manageable codebase and documentation.
FYP Supervisor	Evaluator and academic guide	Ensure adherence to academic guidelines; support traceability of design decisions.
FYP Evaluator	Assessor of project design and implementation quality	Review system completeness, functionality coverage, and technical feasibility.
University Students (Users)	Primary users of the mobile app	Require an easy-to-use, responsive, and supportive mental health tool.

5.2 Context Viewpoint

5.2.1 Design Concerns

The purpose of the Context viewpoint is to identify the boundary between the system and its environment, including how external users and systems interact with MindMate. The key concern is to define what services the system offers to its users and how those actors are expected to interact with the system. This helps determine the external functional scope of the application without knowing the internal implementation.

In the MindMate application, the primary actors are university student, the main user who interacts with the system to monitor their emotional well-being and admin, the developer who is responsible for managing user accounts and the maintenance of the application.

The application offers several features, such as authentication, emotion tracking, AI chatbot interaction, gamified engagement, notification management, and account administration. These functionalities are grouped into modules to reflect their logical and functional roles in the system.

5.2.2 Design View

The system features are shown in Figure 5.2.2.1 below. The use case diagram displays how students and admin interact with different parts of the system. Additionally, Table 5.2.2.1 describes the modules and functions for MindMate application in detailed.

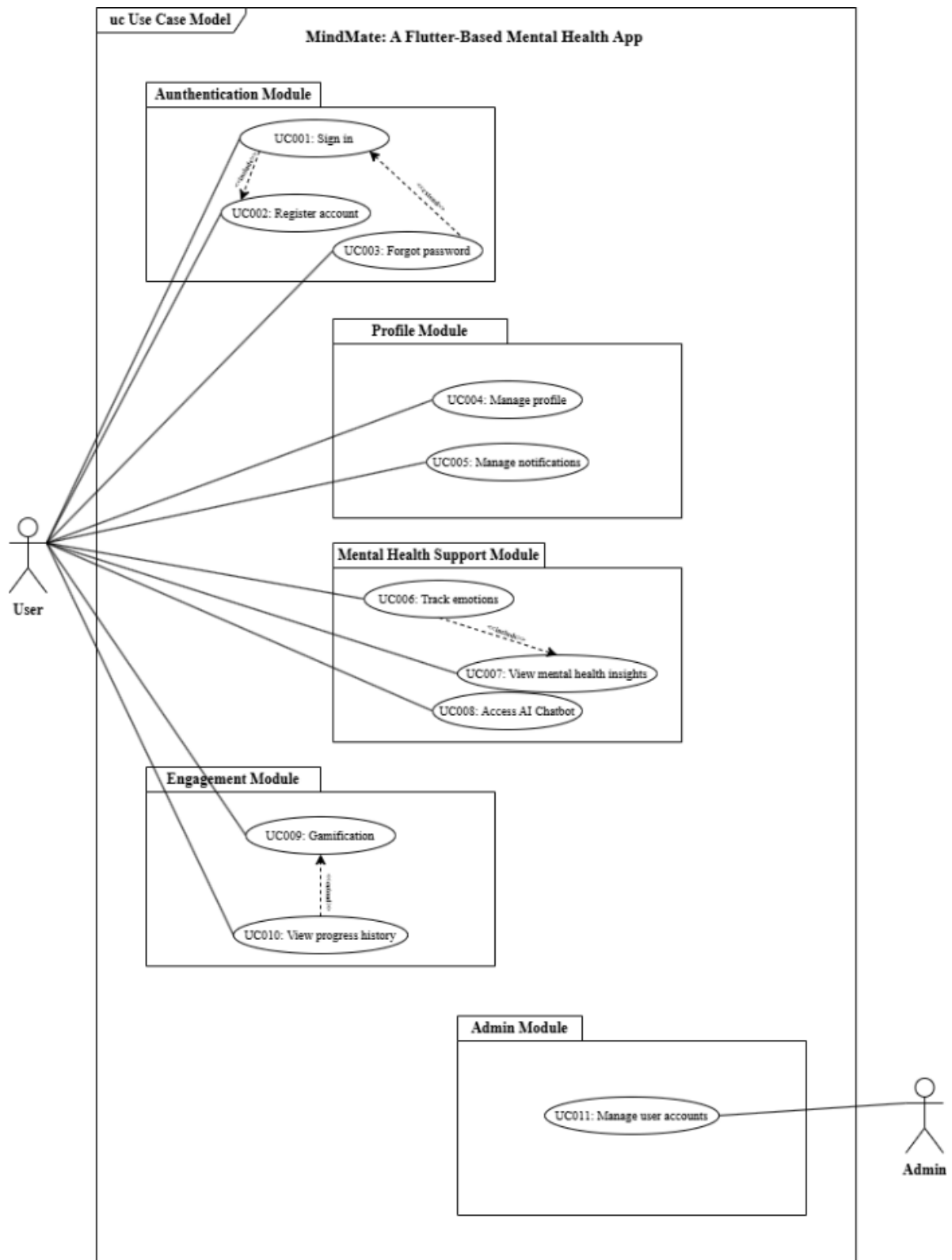


Figure 5.2.2.1: Use Case Diagram for MindMate

Table 5.2.2.1: Description of Module and Functions for MindMate

Module	Function	Description
Authentication Module	UC001: Sign in	Allows users to log into the app using Google
	UC002: Register account	New users can create an account to start using the app.
	UC003: Forgot password	Helps users reset their password if they forget it.
Profile Module	UC004: Manage profile	Users can update their name, age, or other personal details.
	UC005: Manage notifications	Users can turn notifications on or off based on their preference
Mental Health Support Module	UC006: Track emotions	Users log their daily mood using one of the emotion options.
	UC007: View mental health insights	Shows users a summary of their emotion trends.
	UC008: Access AI chatbot	Allows users to chat with the AI assistant for motivation or reflection.
Engagement Module	UC009: Gamification	Gives users badges and streaks for using the app regularly.
	UC010: View progress history	Shows users their past usage history and achievements.
Admin Module	UC011: Manage user accounts	The admin can manage and monitor user data from the Firebase console.

5.3 Composition Viewpoint

5.3.1 Design Concerns

The purpose of the composition viewpoint is to define the subsystems composed. The system is formed by 5 modules: Authentication, Profile, Mental health support, engagement and Admin. Each module is designed to interact with a central backend (Firebase) and ensures separation of concerns for better scalability and maintainability.

The system is organized into the following major layers:

- **Presentation Layer (View):** Renders the UI using Flutter and reflects changes received from the ViewModel.
- **Business Logic Layer (ViewModel):** Manages UI state, validates inputs, and handles asynchronous operations with Firebase and Dialogflow.
- **Data Layer (Model):** Handles data operations such as reading and writing to Firebase Firestore, including user data, emotion logs, chatbot history, and gamification data.

5.3.2 Design View

MindMate is designed following the MVVM pattern, integrated within a 3-tier layered structure. This design presents a clear separation of concerns between presentation, business logic, and data handling layers.

The Model layer represents the data access layer, which interacts with Firebase services such as Firestore and Firebase Authentication to manage persistent data like user profiles, emotion logs, chatbot conversations, and progress tracking.

The ViewModel layer, serving as the business logic layer, functions as an intermediary between the user interface and the underlying data. It is responsible for transforming raw data into a format suitable for presentation, managing application state, handling user input, and performing asynchronous operations such as fetching or storing data through Firebase or interacting with the AI chatbot powered by Dialogflow.

The View layer, implemented using Flutter widgets, consists of user-facing screens including emotion tracking, chatbot interaction, profile management, and progress visualization. This layer reflects updates from the ViewModel and facilitates an interactive experience for users.

The structure is visually represented in Figure 5.3.2.1, which illustrates the MVVM architecture applied in the MindMate application.

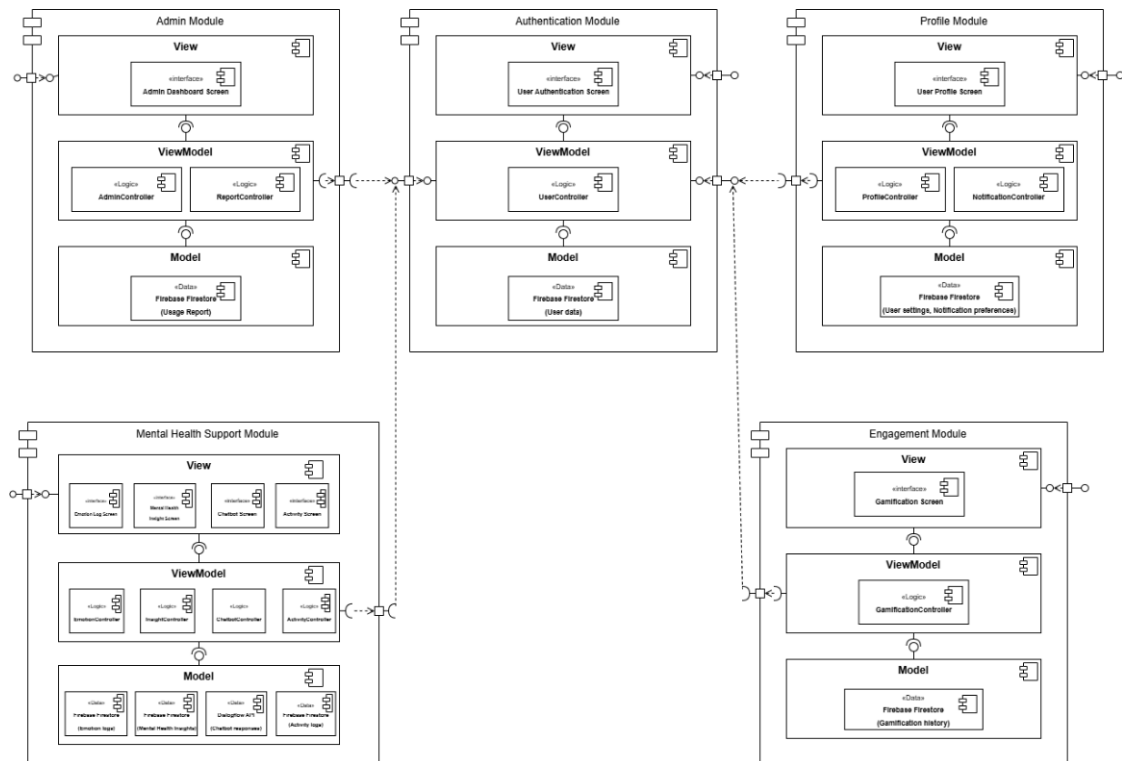


Figure 5.3.2.1: Component Diagram of MiindMate

5.4 Logical Viewpoint

5.4.1 Design Concerns

The purpose of the logical viewpoint is to elaborate the internal object-oriented structure of the system, particularly the relationships between classes and how responsibilities are distributed. This viewpoint focuses on the static structure of the application through the identification of key classes, their attributes, and associations. For MindMate, the domain model represents the core entities of the system such as users, chatbot sessions, emotion logs, and gamification history, along with their relationships and data flows.

5.4.2 Design View

The following includes all the class diagram for each modules, which are Authentication Module, Profile Module, Mental Health Support Module, Engagement Module and Admin Module.

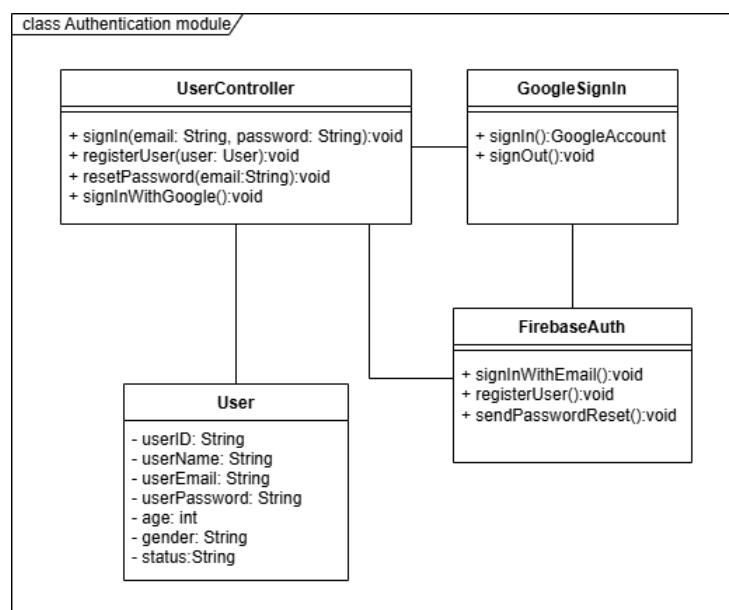


Figure 5.4.2.1: Class diagram for Authentication Subsystem

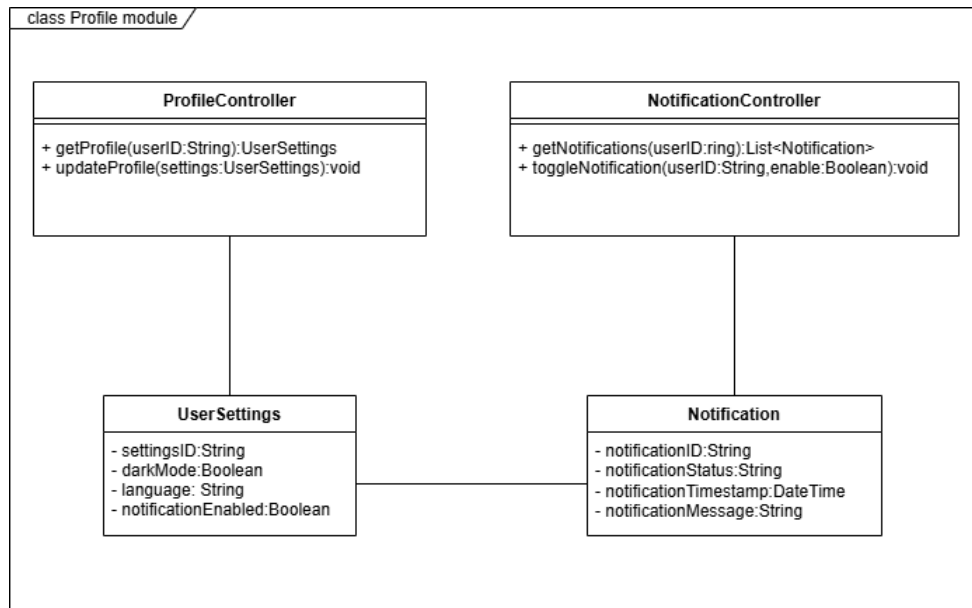


Figure 5.4.2.2: Class diagram for Profile Subsystem

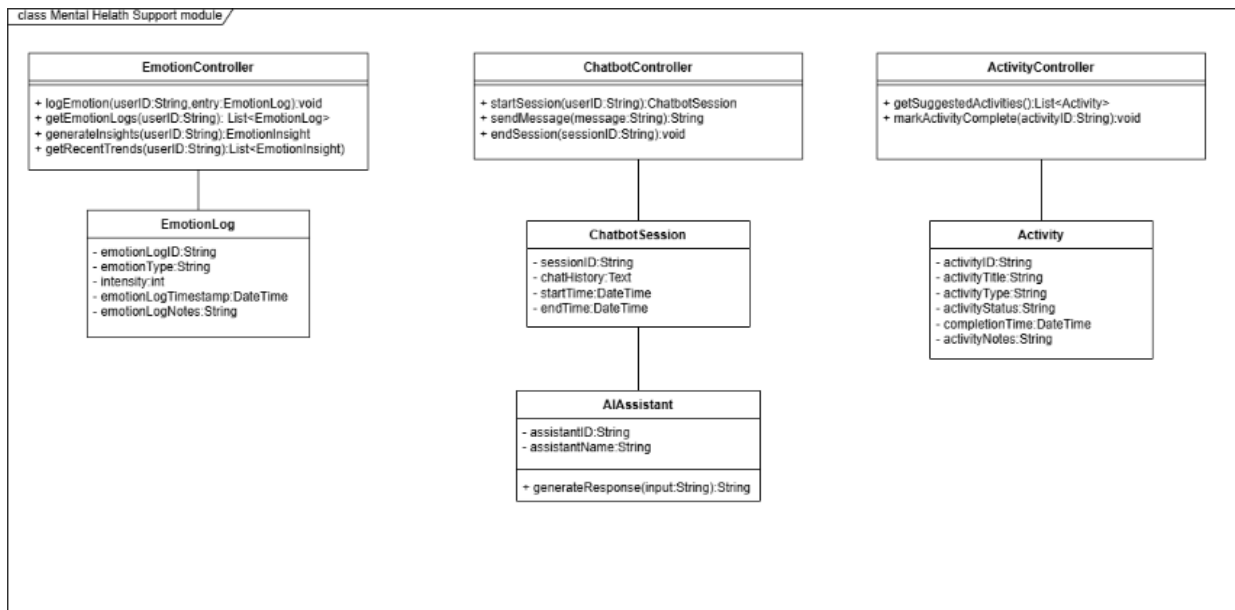


Figure 5.4.2.3: Class diagram for Mental Health Support Subsystem

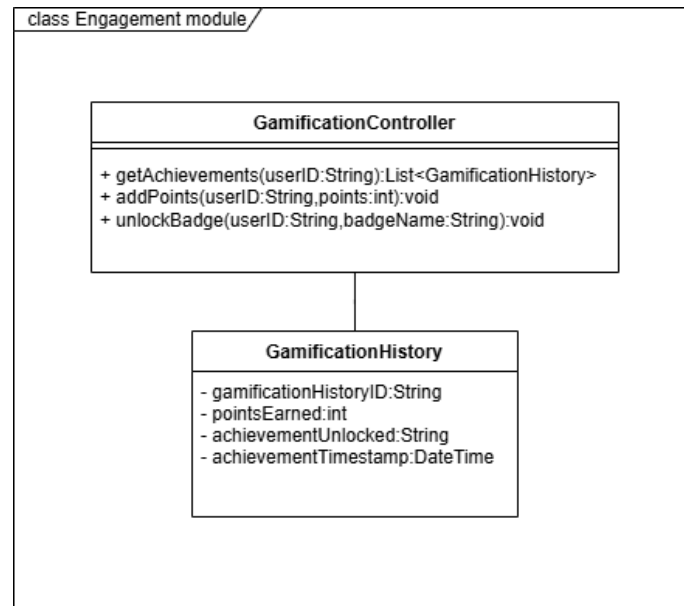


Figure 5.4.2.4: Class diagram for Engagement Subsystem

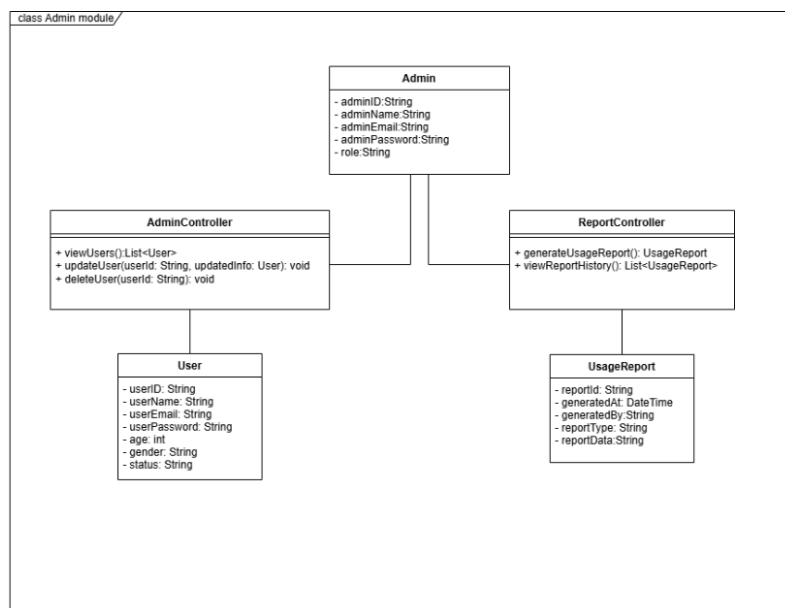


Figure 5.4.2.5: Class diagram for Admin Subsystem

5.5 Information Viewpoint

5.5.1 Design Concerns

The Information Viewpoint is relevant for systems like MindMate where large volumes of user-generated content such as emotion logs, chatbot sessions, and gamification data must be persistently stored and retrieved efficiently. It focuses on how real-world data is modeled and transformed into system-level entities, as well as how the relationships between those entities are maintained.

In the MindMate application, the information domain is modeled using object-oriented classes, which are mapped directly into collections in Firebase Firestore, a scalable NoSQL document-based database. Each major domain class corresponds to a Firestore collection where each document stores one record (e.g., a user, a log, or a session).

The major data or systems entities are stored into a relational database named as MindMateDB comprises 9 entities, listed below

Entity Name	Description
User	Core user account details
UserSettings	User personal preferences
Notification	Stores in-app messages and reminders
EmotionLog	Tracks user emotional states over time
ChatbotSession	Stores user-AI conversations per session
AIAssistant	Represents chatbot logic and session handling
Activity	Records user activities like journaling or breathing exercises
GamificationHistory	Tracks points and badges earned through engagement
Admin	Stores system administrator details and roles
UsageReport	Stores usage summaries

5.5.2 Design View

This section lists and describes the attributes of each domain-layer class in alphabetical order. Controllers and handler classes are omitted.

Data Dictionary

Entity: Activity

Attribute Name	Type	Description
activityID	String	Unique ID for the activity
activityTitle	String	Activity name
activityType	String	Activity type (e.g., journaling, breathing)
activityStatus	String	Completed, pending, skipped
completionTime	DateTime	Timestamp when activity was finished
activityNotes	String	Notes written by user during activity
userID (FK)	String	Reference to user

Entity: Admin

Attribute Name	Type	Description
adminID	String	ID for chatbot instance
adminName	String	Name for the AI assistant
adminEmail	String	Admin login email address
adminPassword	String	Hashed password or token
role	String	admin

Entity: AIAssistant

Attribute Name	Type	Description
assistantId	String	ID for chatbot instance

assistantName	String	Name for the AI assistant
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Entity: ChatbotSession

Attribute Name	Type	Description
sessionId	String	ID for chatbot instance
startTime	String	Start of session
endTime	DateTime	End of session
chatHistory	Text	Full conversation text
userID (FK)	String	Reference to user

Entity: EmotionLog

Attribute Name	Type	Description
emotionLogID	String	Unique ID for the emotion record
emotionType	String	Type of emotion (e.g., happy, sad)
intensity	int	Intensity scale (e.g., 1–5)
emotionLogTimestamp	DateTime	Time of log creation
emotionLogNotes	String	Optional user reflection
userID (FK)	String	Reference to user

Entity: GamificationHistory

Attribute Name	Type	Description
gamificationHistoryID	String	ID for chatbot instance
pointsEarned	int	Points accumulated
achievementUnlocked	String	Badge or milestone earned
achievementTimestamp	DateTime	Date and time badge was unlocked

userID (FK)	String	Reference to user
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Entity: Notification

Attribute Name	Type	Description
notificationID	String	Unique identifier for the notification
notificationStatus	String	Read/unread
notificationTimestamp	DateTime	Time the notification was issued
notificationMessage	String	Notification content
userID (FK)	String	Reference to user

Entity: UsageReport

Attribute Name	Type	Description
reportID	String	Unique identifier for report
generatedDate	DateTime	Date report was created
generatedBy	String	Admin who created the report
reportType	String	Type (summary, engagement, usage, etc.)
reportData	String	Structured or textual report content

Entity: User

Attribute Name	Type	Description
userID	String	Unique user identifier
userName	String	Full name
userEmail	String	Email address
userPassword	String	Hashed password or token

age	int	Gender identity
gender	String	Age in years
status	String	User account status

Entity: UserSettings

Attribute Name	Type	Description
settingsId	String	Unique settings ID
darkMode	Boolean	UI preference
language	String	Preferred language
notificationEnabled	Boolean	Enable/disable system notifications
userID (FK)	String	Reference to User

Entity Relationship Diagram (ERD)

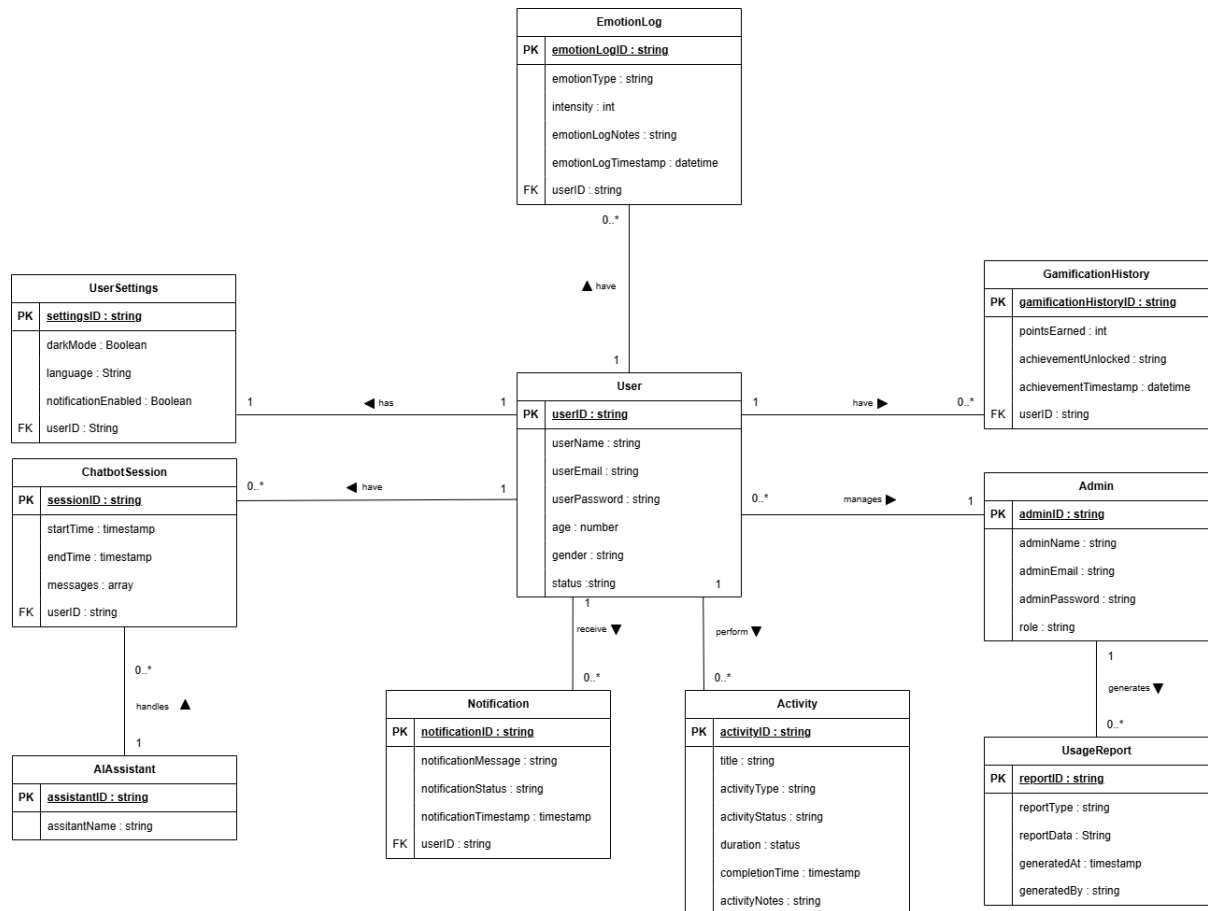


Figure 5.5.2.1: Entity-Relationship-Diagram of MindMate Database Design

5.6 Interface Viewpoint

5.6.1 Design Concerns

The interface viewpoint covers user interfaces, hardware interfaces and software interfaces. It defines how the MindMate application interacts with users and other systems. The primary goal is to ensure that all external and internal interfaces are clearly specified to support consistent user experience and reliable data exchange. The MindMate application primarily supports interaction through a mobile application UI built with Flutter. It also interacts with cloud services such as Firebase Firestore, Firebase Authentication, and Dialogflow API for storage, user authentication, and AI chatbot services respectively.

From the user's point of view, MindMate is a mobile app that helps students manage their mental health. When users open the app, they can log in using their email and password or sign in with their Google account. If they forget their password, they can reset it through email. After logging in, users will see a simple home screen where they can access all the main features.

One of the main features is emotion tracking. Users can choose how they feel by selecting a mood icon, adjusting the intensity using a slider, and writing notes if they want. When they submit their emotion, the app shows a confirmation message and saves the entry. These logs are later used to create emotion trends, which are shown in the “Insights” screen using graphs. Users can filter the data by week or month to better understand their feelings over time.

The app also includes a chatbot that users can talk to. This virtual assistant replies with helpful and friendly messages. Users type their messages in a chat box and get responses that feel like a real conversation. All chat sessions are saved so users can review them later.

In the profile section, users can change their personal preferences such as switching to dark mode, choosing their language, or turning notifications on or off. These settings are saved immediately and applied throughout the app.

To keep users motivated, MindMate uses a gamification system. Users earn points and badges when they log emotions or complete activities. These rewards are shown on a special screen with progress bars and icons.

For admin users, there is a admin panel where they can view user lists, reset accounts, and generate reports. This section is protected and only accessible by admins.

5.6.2 Design View

User Interfaces

The user interface is developed by using Flutter and Dart language, focusing on simplicity, responsiveness and role-specific access. Each interface has clearly defined layouts to support users with varying levels of technical knowledge.

The logical characteristics of each interface between MindMate and its users:

Screen	Logical Characteristics
Login / Registration	Simple forms with email/password fields, Google Sign-In button, and password reset link.
Emotion Log	Mood selection icons, rating slider (1–5), optional notes field, submit button.
Chatbot	Message input box, chat history area, speech bubble layout for bot/user messages
Mental Health Insights	Graphs showing emotion trends over time, filter by date range.
Profile	User name, age, gender, theme toggle, language selector, notification switch.
Gamification	Badges display area, points tally, progress bar for achievements.
Admin Panel	Accessible only to admins. User list for account management, report generation button.

Interface Optimization Guidelines for MindMate application:

Do's	Don'ts
Use consistent iconography and colors	Avoid crowded UI or overlapping elements
Support dark/light mode for comfort	Don't hard-code labels without localization
Provide form validation with helpful messages	Don't allow silent submission failures
Display loading indicators during processing	Avoid long response times without feedback
Minimal training required for user	Don't hide buttons on smaller screen sizes

Hardware Interfaces

The MindMate application uses the following hardware interfaces:

Aspect	Description
Platform	Android 8.0+ compatible mobile devices
Touch Input	Required for UI interaction (tap, swipe)
Storage	Minimum 100 MB free space for local cache
Memory	Minimum 2 GB RAM
Network	Required for login, chatbot, data sync

Software Interfaces

The MindMate application uses the following software interfaces:

Component	Name	Mnemonic	Version	Source
Mobile Framework	Flutter SDK	flutter	3.x (stable)	https://flutter.dev
Authentication	Firebase Auth	firebase_auth	Latest	https://firebase.google.com

Database	Firebase Firestore	firestore	Latest	Google Cloud
AI Chatbot	Dialogflow	dialogflow	v2	https://dialogflow.cloud.google.com
IDE (Development)	Visual Studio Code	VSCode	Latest	https://code.visualstudio.com
IDE (Emulator)	Android Studio Emulator	AVD	2022+	Google Developer Tools

Communication Interfaces

The MindMate application uses the following communication interfaces:

Interface Type	Protocol / Format	Purpose
Firebase API	HTTPS + JSON	Auth, read/write logs, gamification data
Dialogflow API	HTTPS + JSON	Chatbot session initiation and response
Local Preferences	Key-Value	Store minimal offline settings
Authentication Token	JWT (Firebase ID Token)	Secure user session management

5.7 Structure Viewpoint

5.7.1 Design Concerns

The Structure Viewpoint describes how the system is organized in terms of its internal structure and package-level relationships.

In MindMate, the software system is structured using the package/module approach, where each core feature is divided into its own self-contained module. This viewpoint illustrates how components are logically grouped into packages according to functionality, with defined dependencies between them.

By visualizing these packages and their relationships, the system promotes modularity, reusability, and maintainability. Developers can work on individual modules independently while understanding their roles and dependencies in the overall architecture.

5.7.2 Design View

MindMate is divided into five packages, which are Authentication module, Profile module, Mental health support module, Engagement module and Admin module. Each package contains its own domain classes and controllers, and interacts with shared backend services like Firebase and Dialogflow.

1. Authentication module: Manages login, registration, Google sign-in, and password recovery.
2. Profile module: Handles user preferences, language, and notification settings.
3. Mental Health Support module: Responsible for emotion tracking, mental health insights, and chatbot services.
4. Engagement module: Manages gamification including points, achievements, and progress milestones.
5. Admin module: Used by administrators to manage users and generate usage reports.

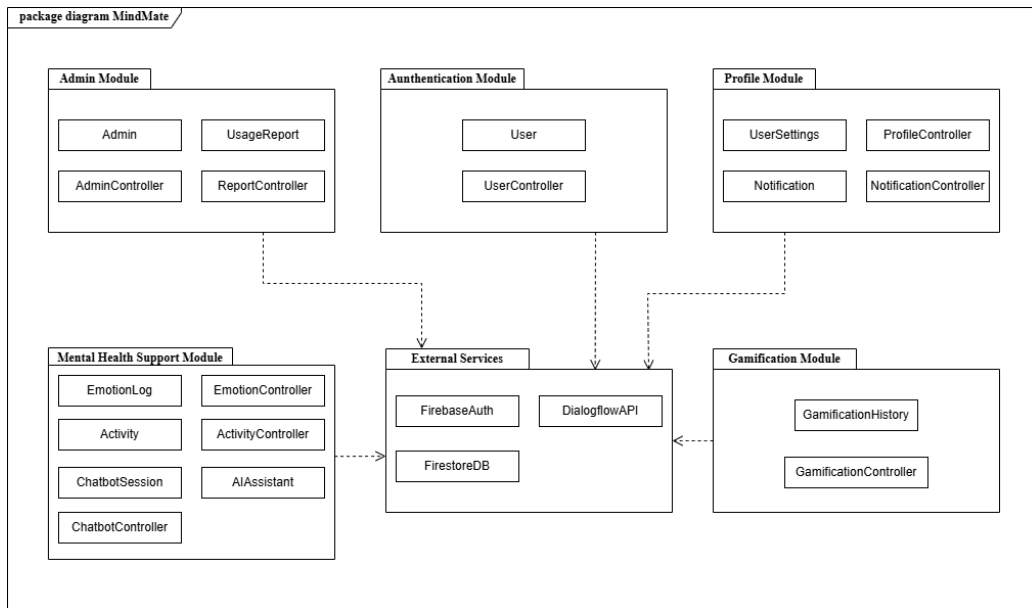


Figure 5.7.2.1: Package Diagram for MindMate

5.8 Interaction Viewpoint

5.8.1 Design Concerns

The Interaction Viewpoint defines how different system components communicate with each other to fulfill functional requirements. It shows the sequence and actions that occur during user interaction, and helps developers understand the logic flow at both the UI and backend levels. It also highlights how each controller communicates with entities and external services such as Firebase Authentication, Firestore, and Dialogflow.

5.8.2 Design View

a) SD001: Sequence diagram for Sign In scenario

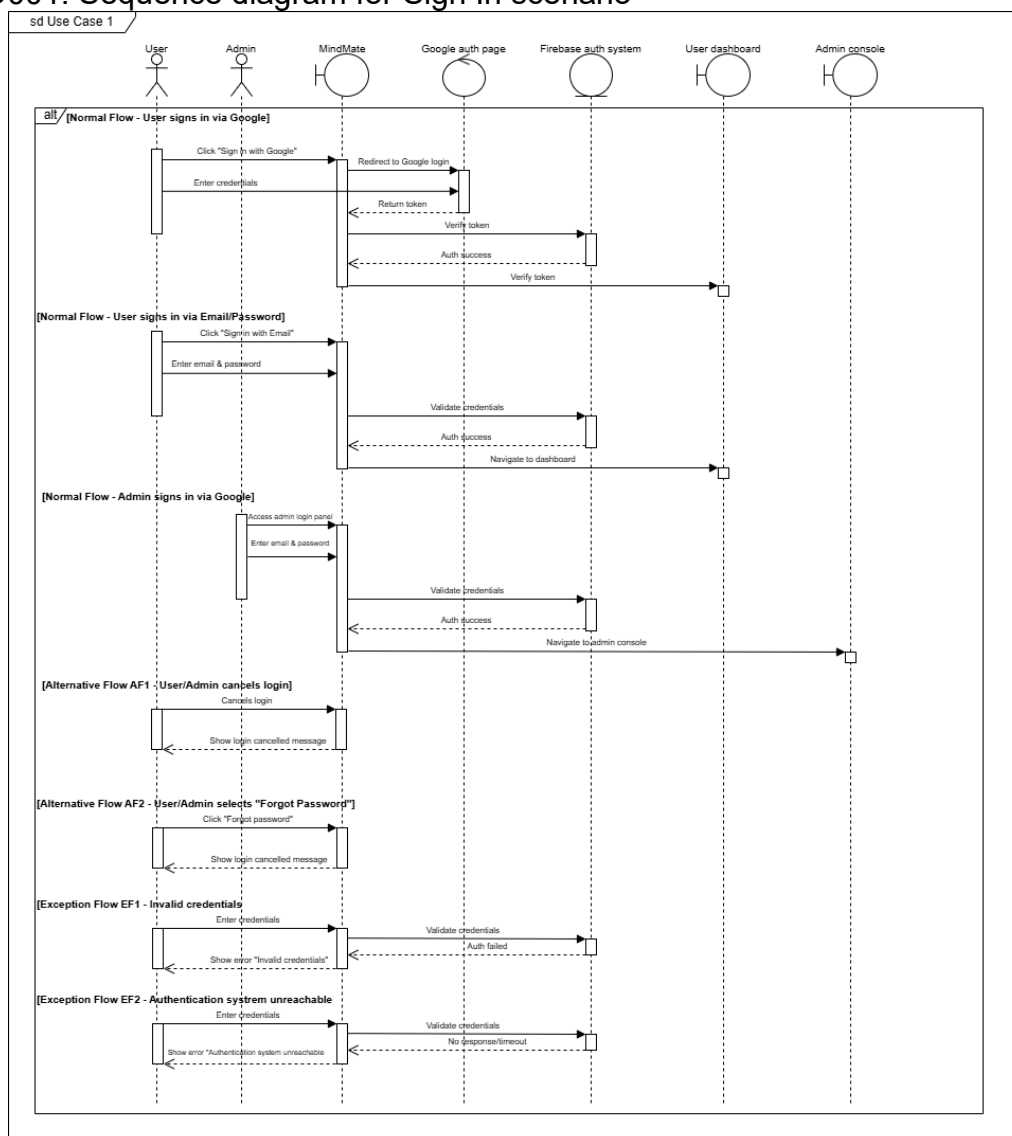


Figure 5.8.2.1: Sequence Diagram for Sign In scenario

b) SD002: Sequence diagram for Register account scenario

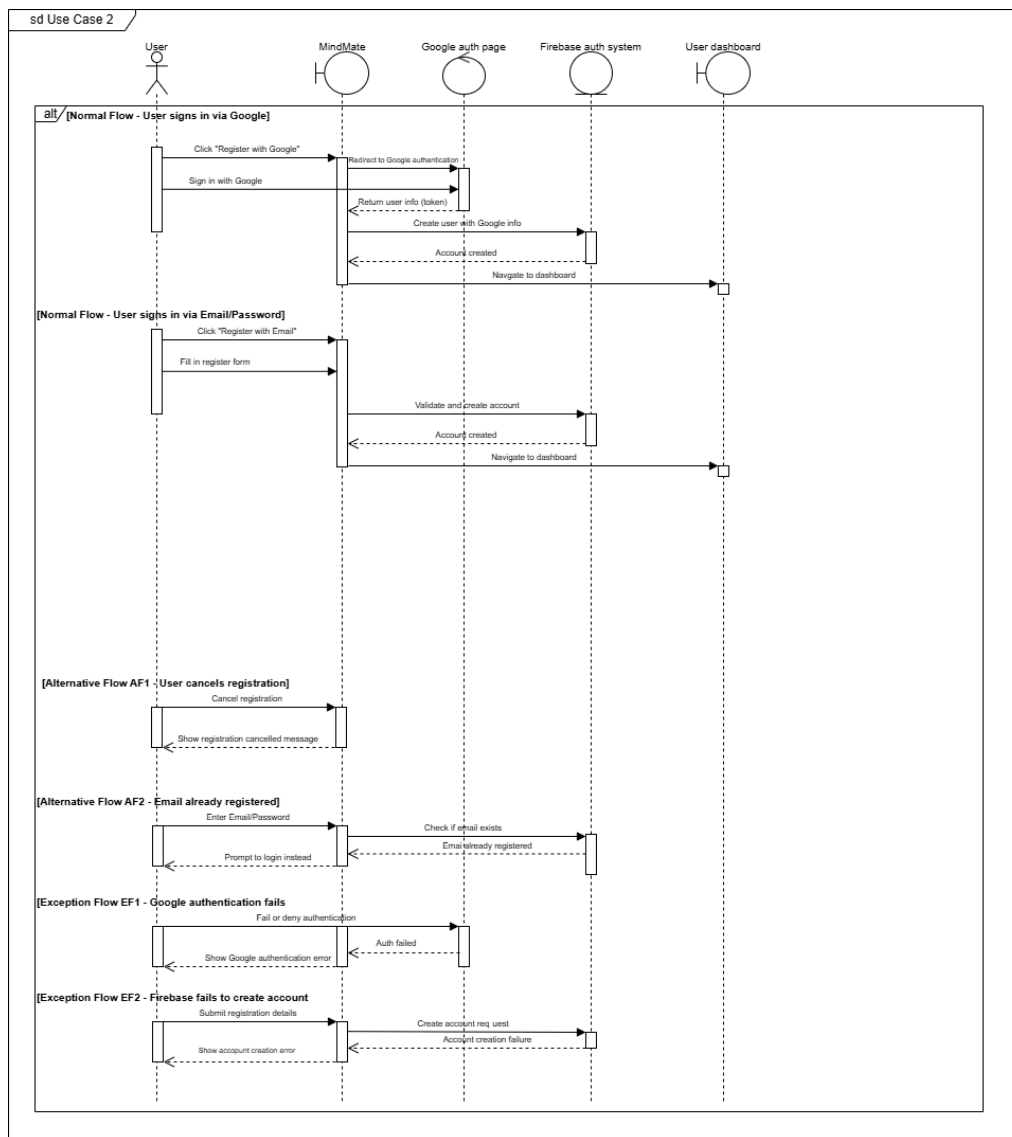


Figure 5.8.2.2: Sequence Diagram for Register account scenario

c) SD003: Sequence diagram for Reset password scenario

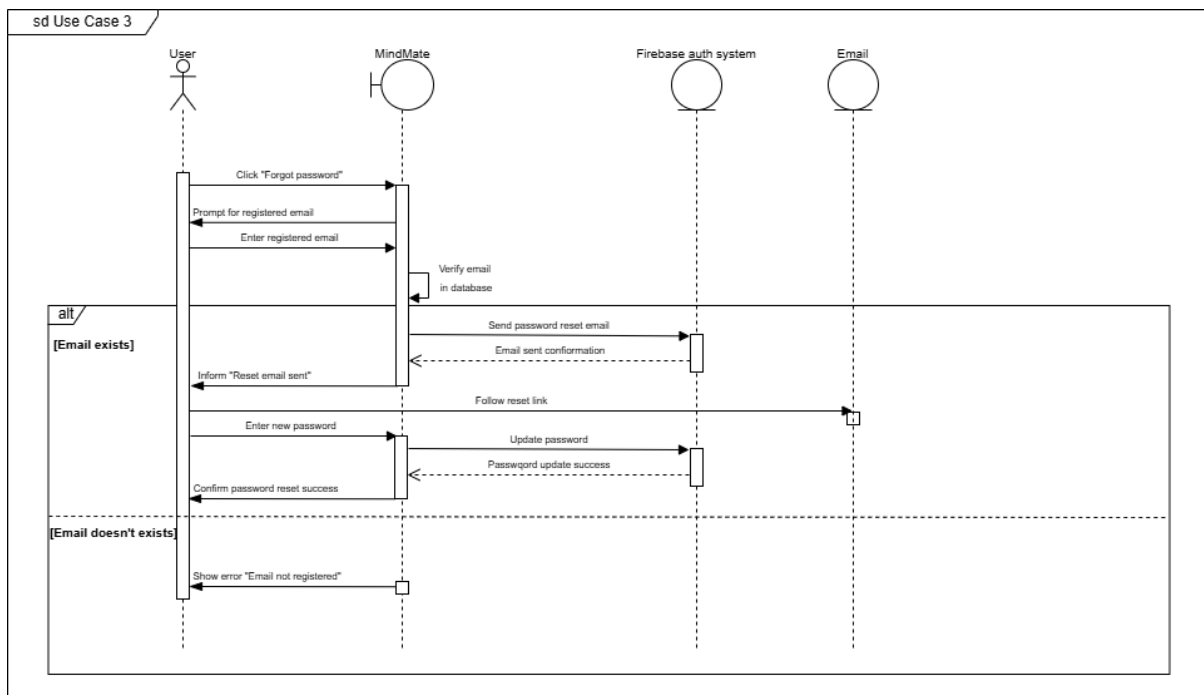


Figure 5.8.2.3: Sequence Diagram for Reset password scenario

d) SD004: Sequence diagram for Manage profile scenario

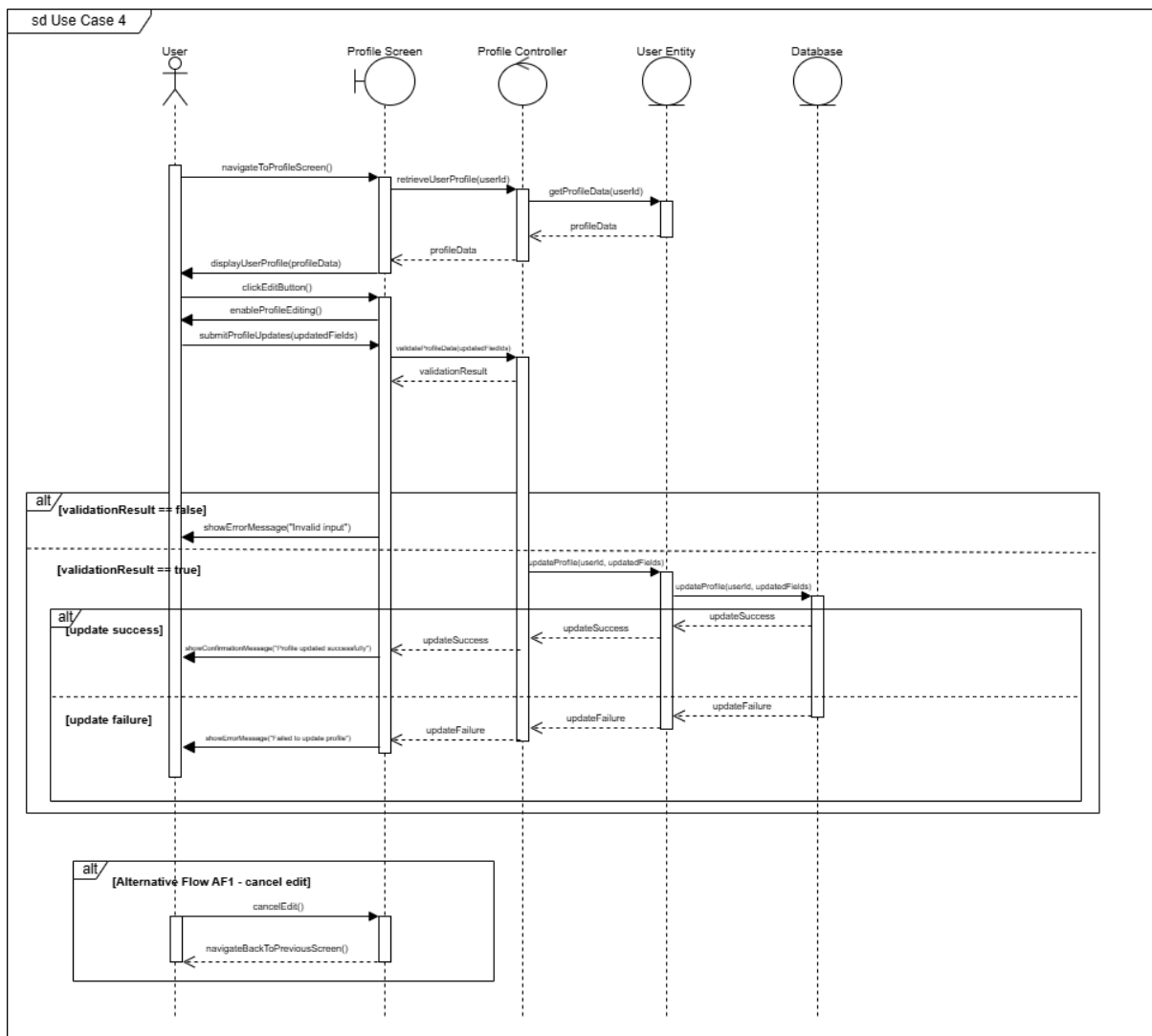


Figure 5.8.2.4: Sequence Diagram for Manage profile scenario

e) SD005: Sequence diagram for Manage notifications scenario

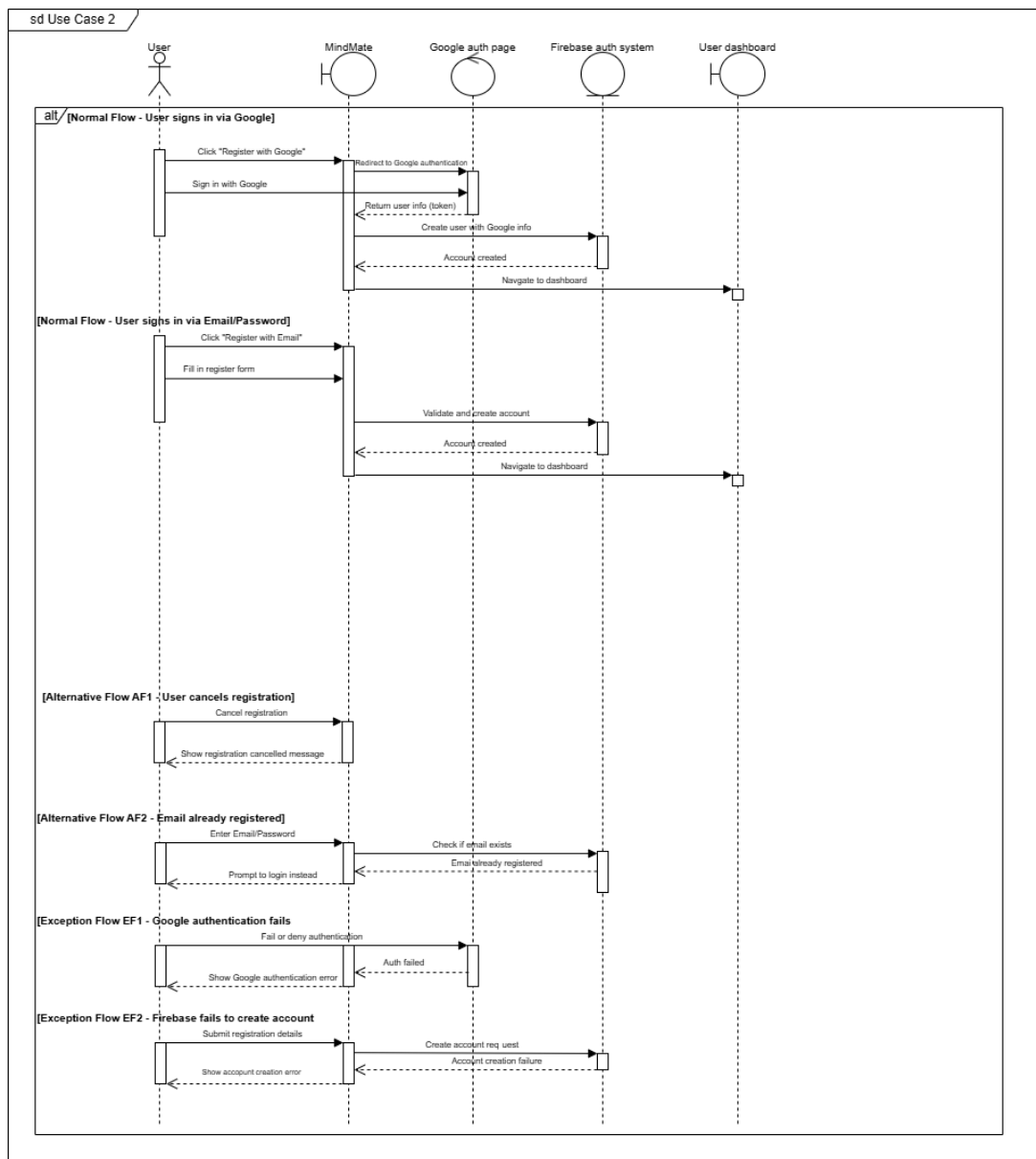


Figure 5.8.2.5: Sequence Diagram for Manage notifications scenario

f) SD006: Sequence diagram for Track emotions scenario

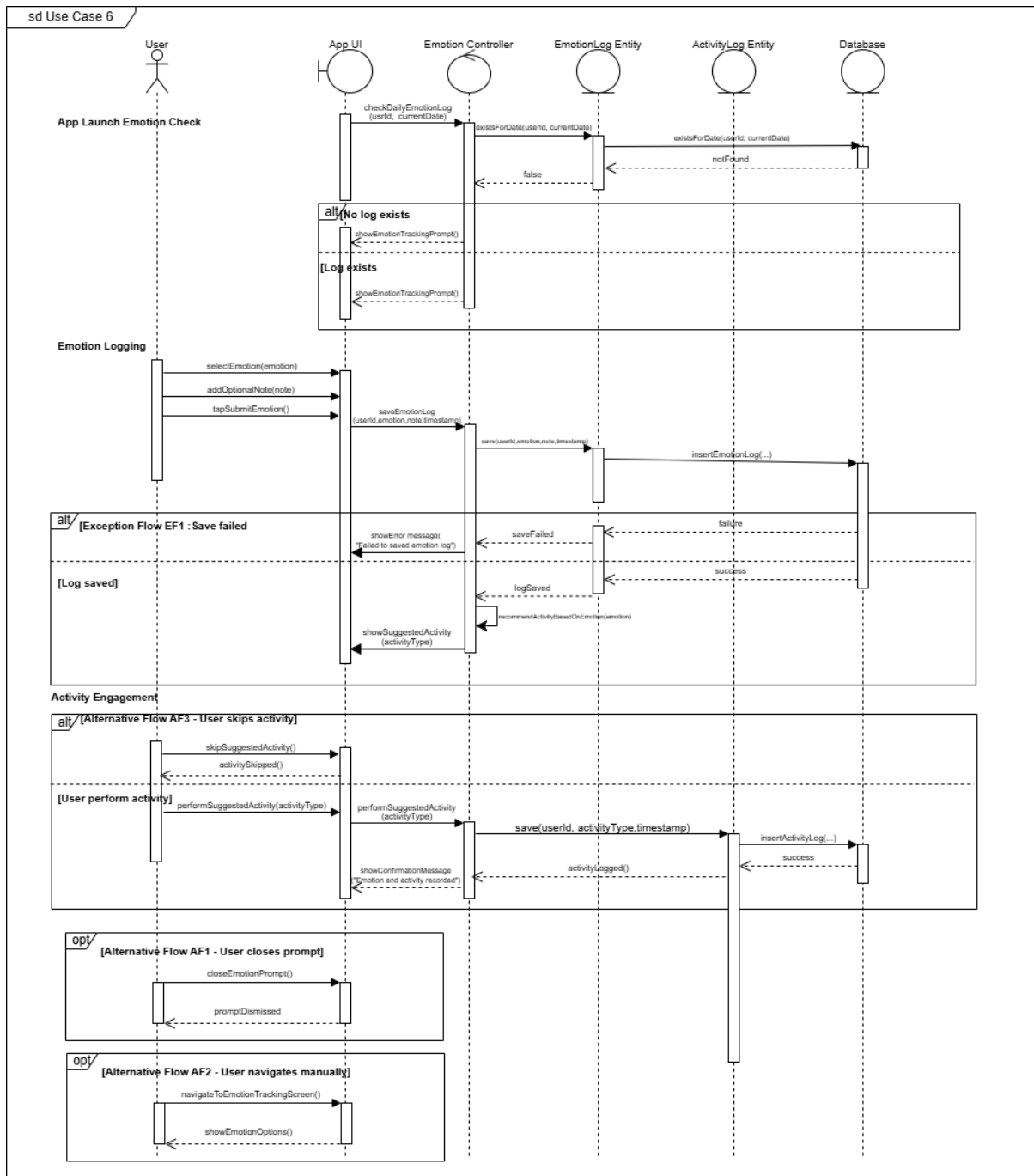


Figure 5.8.2.6: Sequence Diagram for Track emotions scenario

g) SD007: Sequence diagram for View mental health insights scenario

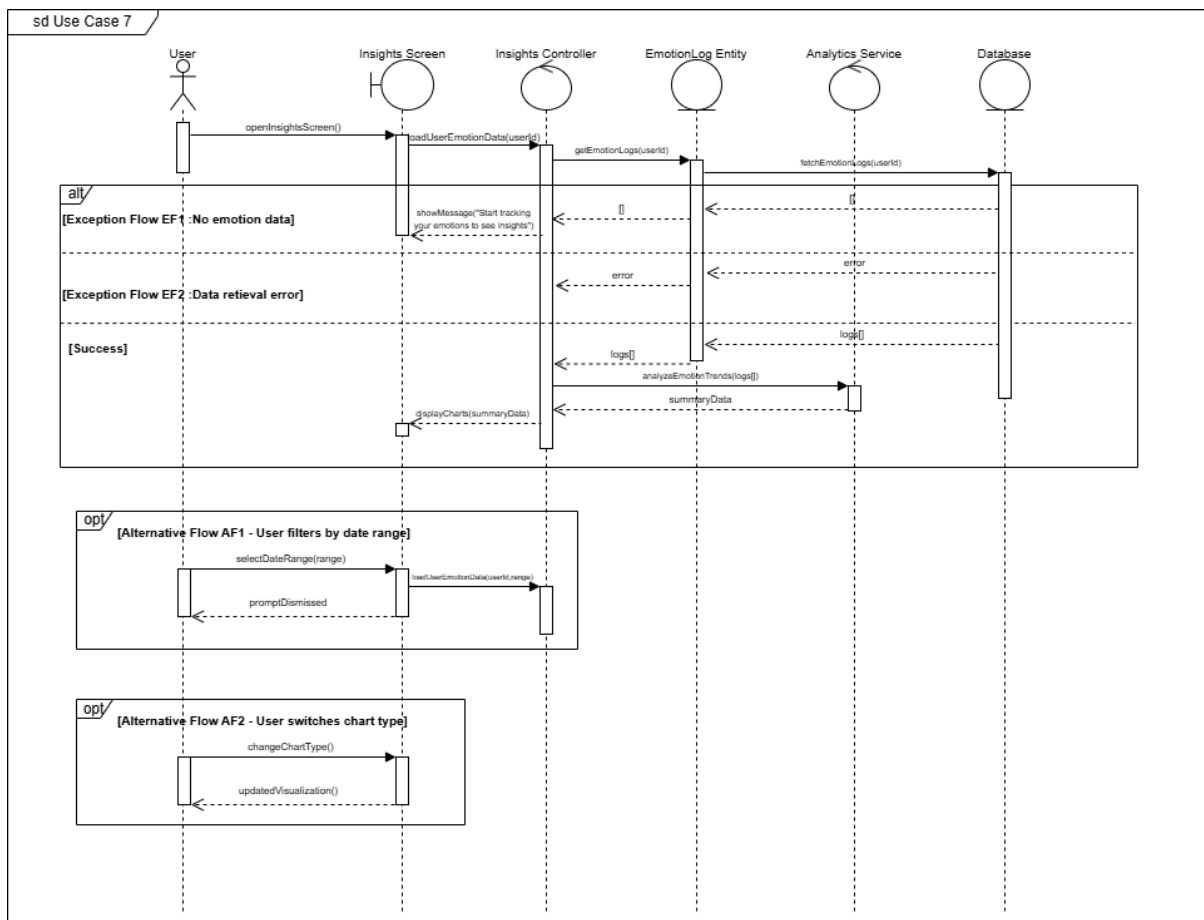


Figure 5.8.2.7: Sequence Diagram for View mental health insights scenario

h) SD008: Sequence diagram for Access AI chatbot scenario

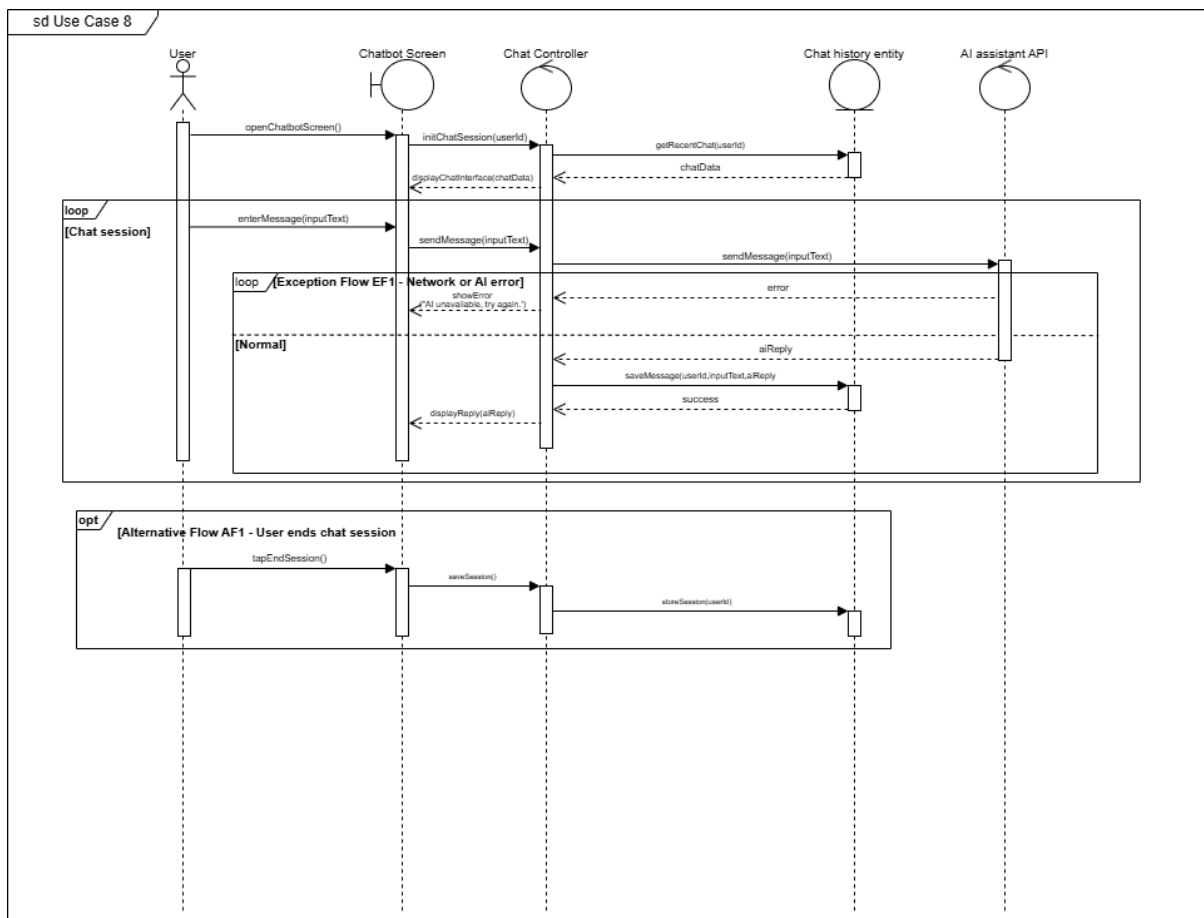


Figure 5.8.2.8: Sequence Diagram for Access AI chatbot scenario

i) SD009: Sequence diagram for Gamification scenario

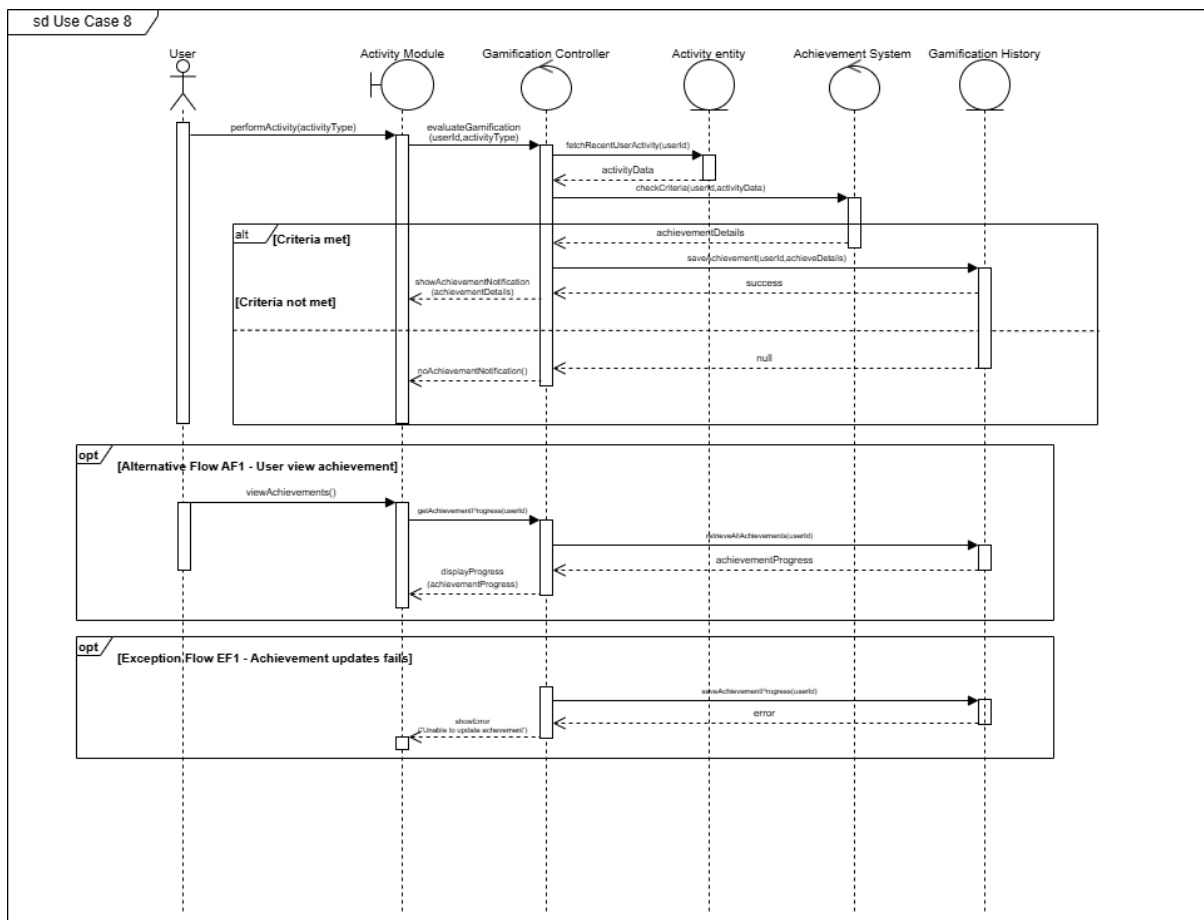


Figure 5.8.2.9: Sequence Diagram for Gamification scenario

j) SD010: Sequence diagram for View progress history scenario

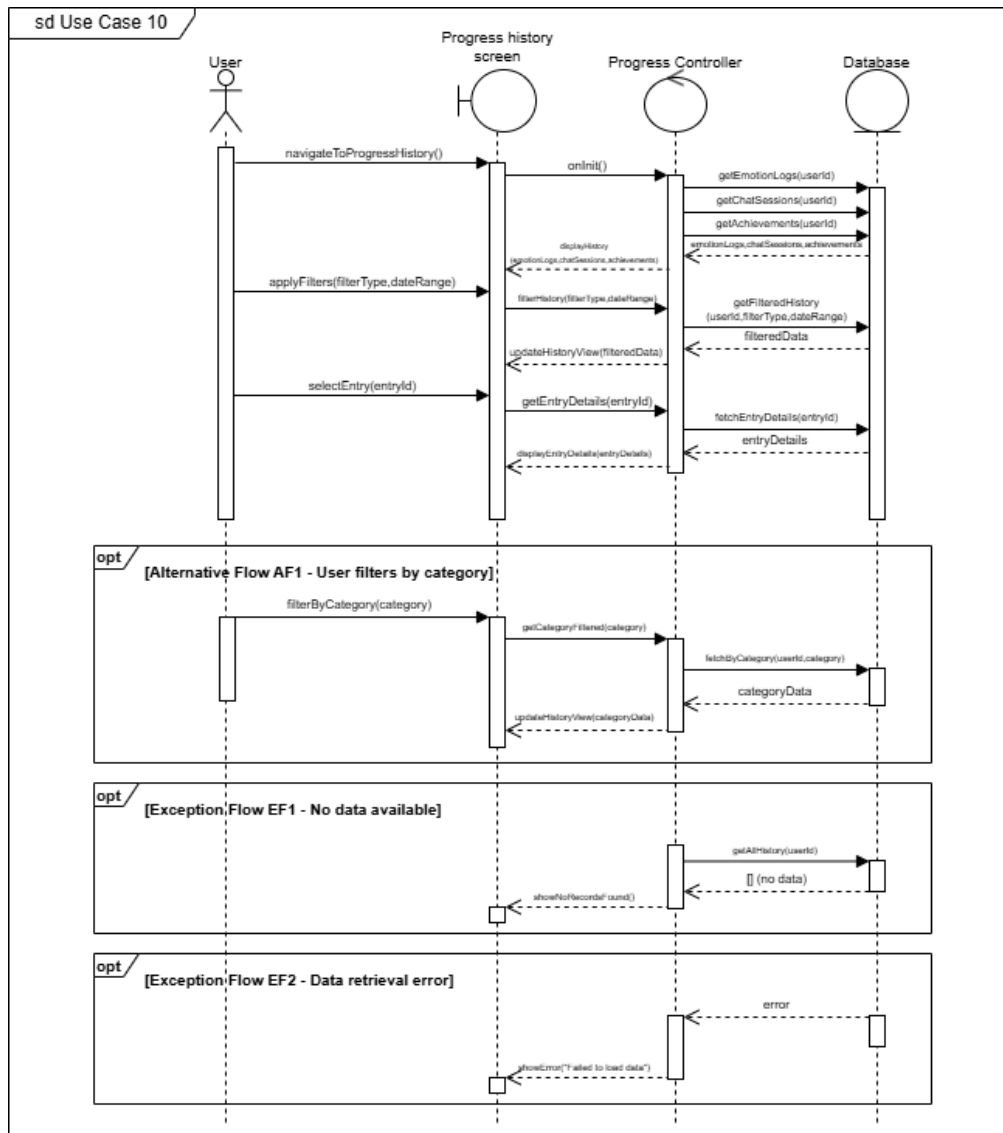


Figure 5.8.2.10: Sequence Diagram for View progress history scenario

k) SD011: Sequence diagram for Manage user accounts scenario

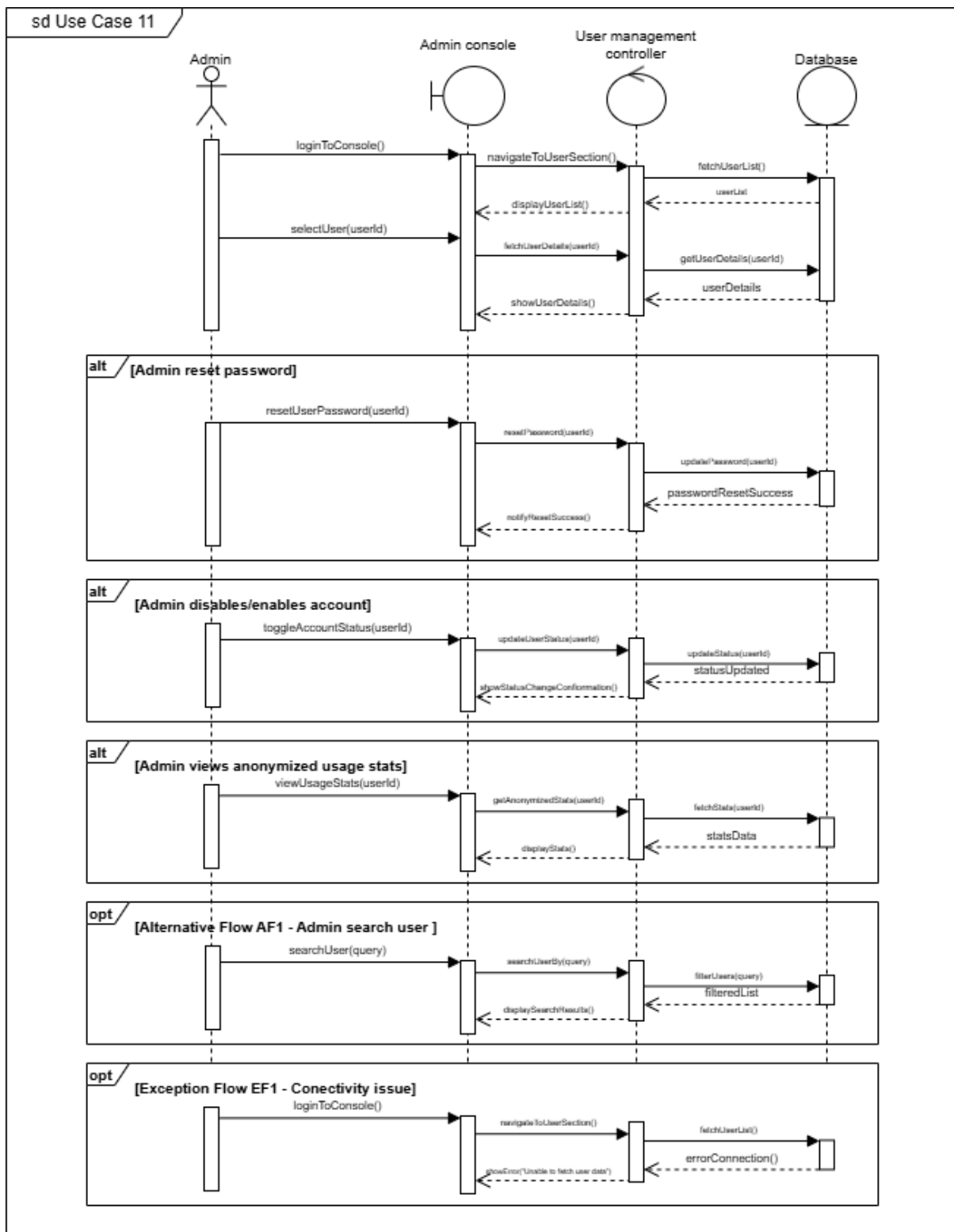


Figure 5.8.2.11: Sequence Diagram for Manage user accounts scenario

5.9 State Dynamic Viewpoint

5.9.1 Design Concerns

The State Dynamic Viewpoint describes the internal behavior and transitions between states of key entities within the system. It is especially useful for reactive systems where the flow of states depends on user actions and external events. It provides insight into the possible states, transitions, and lifecycle behaviors of core entities like emotion logs, chatbot sessions, and activities.

In MindMate, several entities operate in multiple states depending on user interactions and validation rules. For example, an emotion log can transition from “NotLogged” to “Logged”, then to “Updated” or “Deleted” depending on actions taken. Similarly, chatbot sessions respond dynamically to user events such as errors or retries, while activities move from being suggested to completed, skipped, or aborted..

5.9.2 Design View

Below are the state transition diagrams for key entities in the system. These diagrams reflect how each object transitions between states based on specific events or triggers.

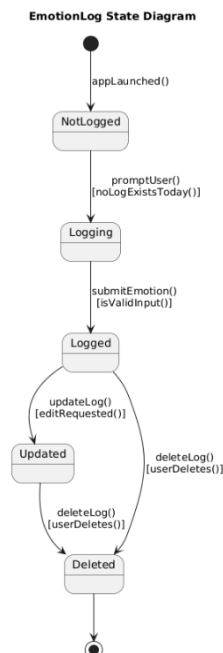


Figure 5.9.2.1: State Machine Diagram for EmotionLogs

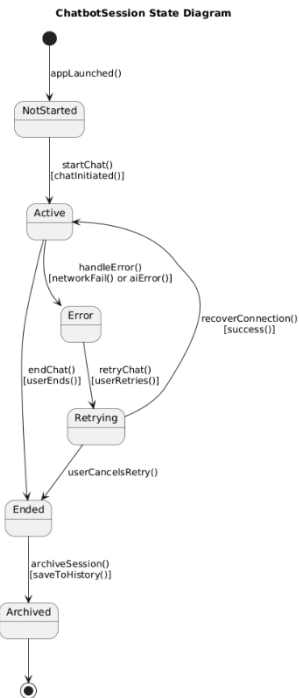


Figure 5.9.2.2: State Machine Diagram for ChatbotSession

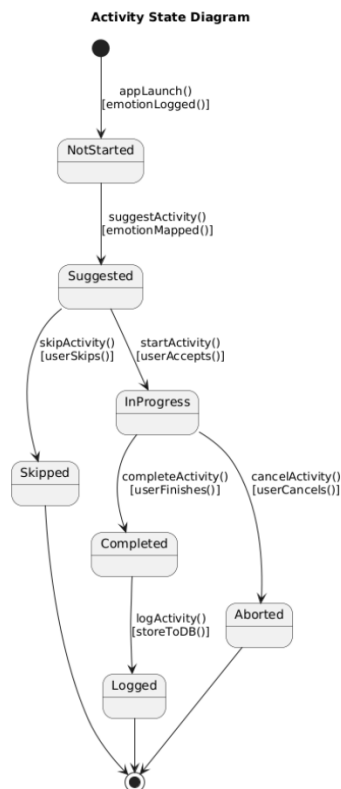


Figure 5.9.2.3: State Machine Diagram for Activity

5.10 Algorithm Viewpoint

5.10.1 Design Concerns

The Algorithm Viewpoint provides a detailed description of the internal logic and operations behind the MindMate application's main functions. This includes the computational steps performed within core methods such as user authentication, emotion tracking, chatbot processing, gamification progress, and admin reporting. It describes how inputs are processed, conditions are checked, and outputs are generated.

5.10.2 Design View

This section is organized according to the five major system modules:

a) Authentication module

Method	Input	Processing Logic	Output
signIn()	Email, password	Validate input → call FirebaseAuth → verify credentials → return login result	Success or error message
registerUser()	Name, email, password	Validate input → check uniqueness → hash password → store in Firestore	Registration result
resetPassword()	Email	Validate email → call Firebase reset API	Reset link email sent
signInWithGoogle()	OAuth token from Google	Fetch user details → check existing user → register	Authenticated session

		or sign in via Firebase	
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b) Profile module

Method	Input	Processing Logic	Output
getProfile()	User ID	Fetch user settings from Firestore	UserSettings object
updateProfile()	New profile settings	Validate changes → overwrite settings in Firestore	Confirmation or error
getNotifications()	User ID	Query Firestore → retrieve all notifications ordered by time	List of notifications
toggleNotification()	User ID, enable/disable flag	Update notification preference field in Firestore	Confirmation message

c) Mental Health Support module

Method	Input	Processing Logic	Output
logEmotion()	Emotion type, intensity, notes	Validate input → assign timestamp → save log to Firestore	Save confirmation
generateInsights()	User ID	Retrieve emotion logs → calculate	Summary insight object

		trends & averages → return EmotionInsight object	
startSession()	User ID	Initialize session → create session ID → log start time	ChatbotSession object
sendMessage()	Message string	Sanitize input → send to Dialogflow → receive response → save exchange	Chatbot reply
endSession()	Session ID	Mark session as ended → update end timestamp → archive if needed	Archived session or confirmation

d) Engagement module

Method	Input	Processing Logic	Output
getAchievements()	User ID	Retrieve user's achievement logs from Firestore	List of badges and points
addPoints()	User ID, points	Fetch current score → add points → update entry → check badge unlock	Updated score or badge popup

unlockBadge()	User ID, badge name	Add badge to user's achievements list if not already unlocked	Confirmation message
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e) Admin module

Method	Input	Processing Logic	Output
viewUsers()	N/A	Query all user documents from Firestore	List of users
updateUser()	User ID, updated user info	Validate updates → apply changes in Firestore	Update confirmation
deleteUser()	User ID	Check permissions → remove user record → cascade delete related data if needed	Deletion confirmation
generateUsageReport()	N/A	Aggregate data (emotion logs, sessions, activities) → compile summary → save report	UsageReport object

viewReportHistory()	Admin ID	Fetch all reports generated by admin	List of UsageReport items
---------------------	----------	--------------------------------------	---------------------------

6. User Interface Design

6.1 Overview of User Interface

The MindMate mobile application is designed with a user-friendly and intuitive interface to ensure ease of use for all university students, including those with minimal technical experience. The app's UI focuses on clarity, accessibility, and emotional comfort, aligning with its purpose to support mental well-being.

MindMate provides a structured navigation flow that begins with login and registration. Upon successful authentication, users are directed to the main dashboard, where they can access core features such as emotion tracking, chatbot interaction, personal insights and gamified rewards. Each screen is clearly labeled, uses meaningful icons, and provides responsive feedback.

For example, users receive confirmation messages when emotions are logged, pop-ups when badges are unlocked, and error messages if inputs are incomplete or invalid. Graphs are used to present emotion trends, and simple sliders and toggles allow customization of personal settings. The application also supports dark mode and language preferences.

All interfaces are developed using Flutter and tested on Android Studio Emulator to ensure responsive design, proper touch interactions, and support for a wide range of device screen sizes.

6.2 Screen Images

Subsystem: Authentication

Screen Name	View Layer Functionality	User Experience and Feedback
Login/Signup Screen	Users enter credentials or register.	Displays success message and redirects to dashboard. Invalid input prompts error notification.
Forgot Password Screen	Users can reset their password.	Displays password reset link upon successful entry, with error handling for invalid email.

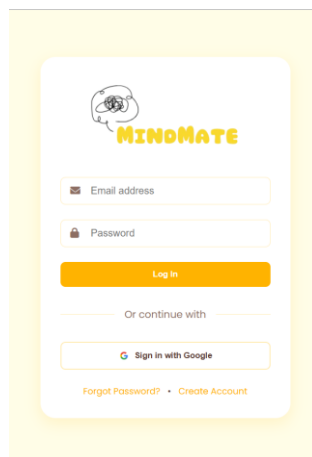


Figure 6.2.1: Login Screen

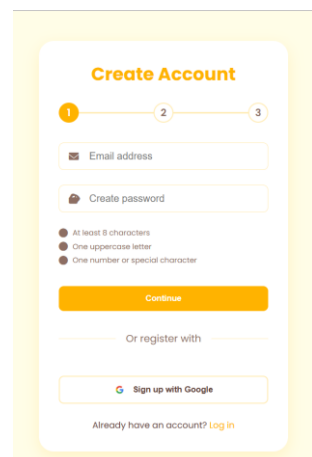


Figure 6.2.2: Sign Up Screen

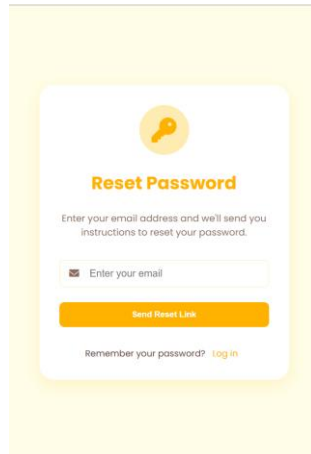


Figure 6.2.3: Forgot Password Screen

Subsystem: Profile

Screen Name	View Layer Functionality	User Experience and Feedback
View Profile Screen	Users view their account details.	Shows current information. Confirmation when information is up to date.
Edit Profile Screen	Users update personal or organizational information.	Confirms updates or shows field-specific error messages.
Settings Screen	Users manage notification preferences.	Toggles and checkboxes for preferences; confirmation shown on changes.

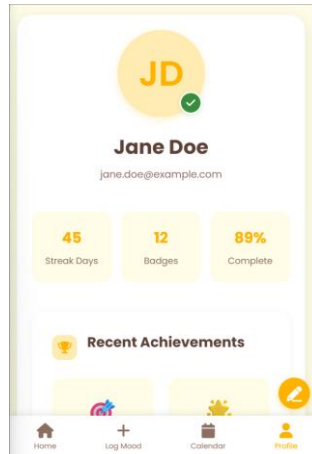


Figure 6.2.4: View Profile Screen

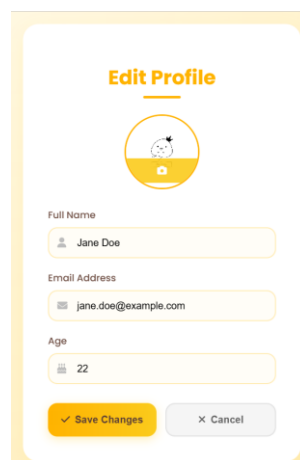


Figure 6.2.5: Edit Profile Screen

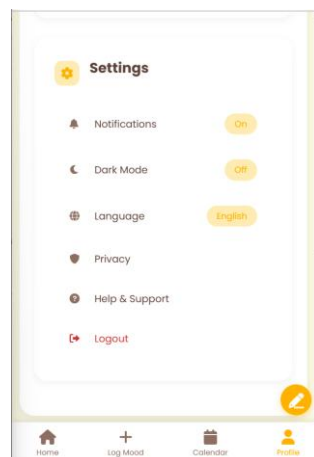


Figure 6.2.6: Settings Screen

Subsystem: Mental Health Support

Screen Name	View Layer Functionality	User Experience and Feedback
Emotion Log Screen	Users select and log their emotional state.	Displays mood icons, a slider for intensity, and a notes field. Success message when saved.
Insights Screens	Users view their emotional trends and analysis.	Graphs and charts showing mood trends; success message when data is loaded.
Chatbot Screen	Users interact with the AI chatbot.	Message input box and chat history. Response time visual indicators, fallback messages if AI cannot respond.

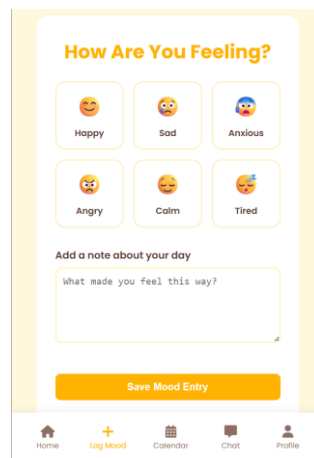


Figure 6.2.7: Emotion Log Screen

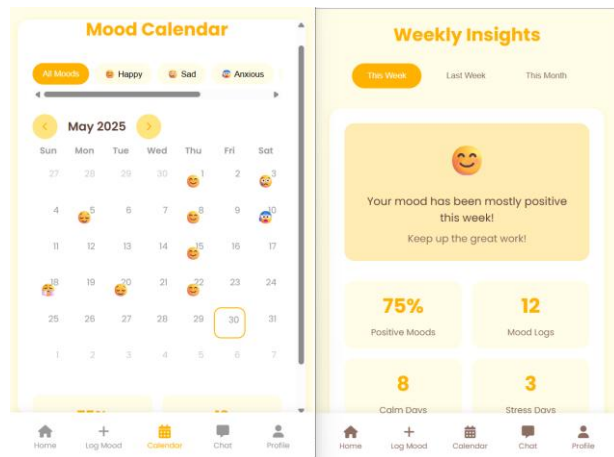


Figure 6.2.8: Insights Screens

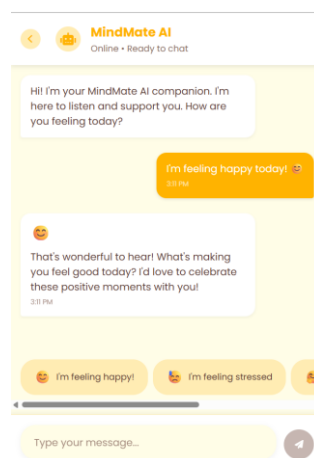


Figure 6.2.9: Chatbot Screen

Subsystem: Engagement

Screen Name	View Layer Functionality	User Experience and Feedback
Gamification Screen	Users view badges, points, and achievements.	Displays points and badges earned. Animations and sounds for unlocking achievements.
Activity Progress Screen	Users track daily activities and self-care goals.	Progress bars and goal completion status;

		motivational feedback for task completion.
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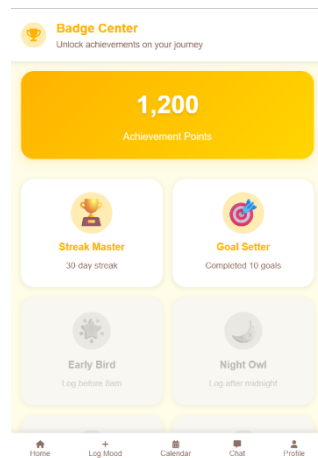


Figure 6.2.10: Gamification Screen

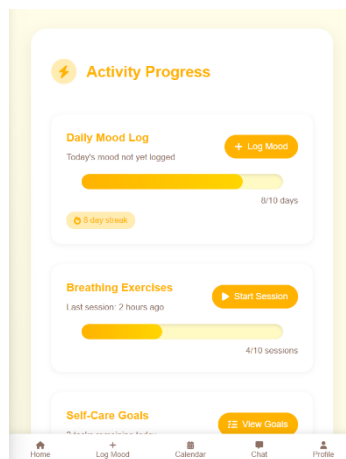


Figure 6.2.11: Activity Progress Screens

Subsystem: Admin

Screen Name	View Layer Functionality	User Experience and Feedback
Admin Dashboard Screen	Admins manage user accounts and view system reports.	Table view of user data, buttons for update and delete actions.

		Success/error messages for actions.
User Management Screen	Admins add, modify, or remove users.	Form-driven input with success confirmation and error prompts for invalid input.
Usage Reports Screen	Admins generate and view system usage reports.	Graphs and tables summarizing user interactions, points, and system engagement metrics.

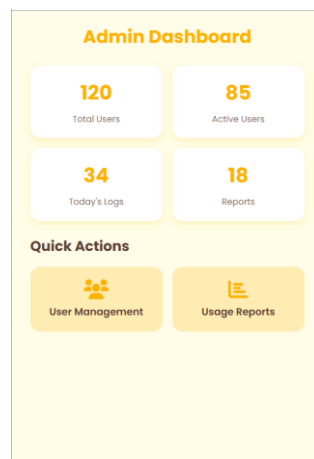


Figure 6.2.12: Admin Dashboard Screen

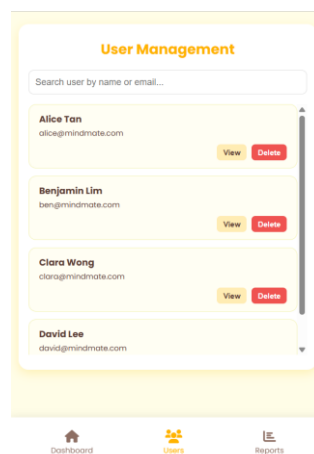


Figure 6.2.13: User Management Screens

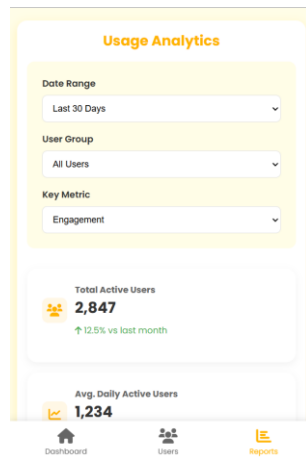


Figure 6.2.14: Usage Report Screen

Appendix F Software Testing Documentation (STD)



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 3032: Software Engineering FYP1
Semester 02, 2024/2025

Software Test Documentation

**MindMate: A Flutter-Based Mental Health App for
University Students with Emotion Tracking and AI
Powered Virtual Assistant**

Version 1.0

Prepared by: Lau Yun Xi

Revision Page

a. Overview

This document presents the Software Test Documentation (STD) for the MindMate App. It provides a list of test cases that are mapped to each system use case, describing the input data, expected output, and testing steps for functional verification.

b. Target Audience

The target audience includes:

- Software testers involved in verifying the implementation of system features.
- Developers using the STD to understand expected behaviors.
- Project supervisors and evaluators from Universiti Teknologi Malaysia (UTM) reviewing the software's quality assurance process.

c. Project Team Members

Name	Role	Assigned tasks	Status
Lau Yun Xi	Sole Developer	Documentation of STD	Completed

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Lau Yun Xi	Initial draft of test cases and STD documentation	16 June 2025

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1. Introduction

1.1 Purpose

This STD explains the testing for the MindMate App. It shows all the test cases used to check if the app works correctly based on the Software Requirements Specification (SRS). This document is meant for people involved in the project such as the developer, testers, supervisor, and evaluator. It helps them understand the parts of the system that have been tested.

1.2 Scope

The software product is MindMate, a Flutter-based mobile application designed for university students to monitor their mental well-being. It allows users to log their moods, view mood trends, and receive personalized motivational content. The app integrates with Google Dialogflow via API to provide chatbot functionality for real-time virtual assistance, and Firebase is used to manage user authentication and store emotional data securely in the cloud. MindMate aims to help university students become more aware of their mental health through daily check-ins, simple reflections, and mood-related suggestions. It also includes gamification features such as streak counters and badges to promote consistent engagement. The app is developed as a prototype project and will focus on emotion tracking and mental health support rather than clinical diagnosis or therapy.

There are two types of users in the system:

- End users (students): Can interact with the app features such as logging moods, chatting with the virtual assistant, and reviewing mood history.
- Admin (developer): Has backend access to monitor data structure, manage application updates, and perform system testing.

The application will focus on supporting six core emotion states: happy, sad, stressed, anxious, frustrated, and focused. The system does not include professional counseling modules, emergency services, real-time human therapist chat, or any payment integration. Additionally, the chatbot content is powered by external APIs and will not be developed manually from scratch.

1.3 Definitions, Acronyms and Abbreviation

Terms / Acronyms / Abbreviation	Definitions
AI	Artificial Intelligence. In MindMate, used to support the chatbot virtual assistant for mental health conversations.
API	Application Programming Interface. Allows communication between software modules or with backend services like Firebase.
Dialogflow	A Google tool that helps the app understand and reply to user messages in the chatbot.
SDS / SDD	Software Design Specification / Software Design Description. A document that outlines how software will be designed and structured.
SRS	Software Requirements Specification.
STD	Software Test Documentation. A document that describes how the app is tested to make sure it works correctly.
UI	User Interface. The screen and buttons users see and use in the app.

1.4 References

IEEE Computer Society. (1998). *IEEE recommended practice for software design descriptions (IEEE Std 1016-1998)*. Institute of Electrical and Electronics Engineers.

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Sommerville, I. (2016). Software engineering (10th ed.). Pearson Australia Pty Limited.

UTM Faculty of Computing. (2024). *FYP1 software test document template* (Version 2.0). Universiti Teknologi Malaysia.

1.5 System Overview

This STD contains all the test cases written to check each feature in the MindMate app. The tests are grouped by module and use case. For each test case, it includes the test ID, what is tested, input data, and the expected result.

The rest of the document includes:

- A list of all test cases
- A traceability matrix that links test cases to the system's diagrams

2. Test Cases, Data and Expected Results

2.1 Test TC001 for Module Authentication: UC001 – Sign In

This test contains the following test cases:

2.1.1 TC001_01: Sign in with valid email and password

Test Case ID:	TC001_01		Test Case Description:	Ensure the user can successfully log in using correct email and password.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is already registered with valid credentials.		1	Email: abc@gmail.com	
			2	Password: abc123	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to login screen		Login screen is displayed		
2	Enter registered email and password		Input is accepted		
3	Click Login		User redirected to dashboard		

Figure 2.1.1.1: Test Case TC001_01

2.1.2 TC001_02: Sign in with invalid password

Test Case ID:	TC001_02		Test Case Description:	Ensure the system prevents login when an incorrect password is entered.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User account already exists.		1	Email: abc@gmail.com	
			2	Password: wrong password	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to login screen		Login screen is displayed		
2	Enter registered email and wrong password		Input is accepted		
3	Click Login		Error message "Invalid credentials" is displayed		

Figure 2.1.2.1: Test Case TC001_02

2.1.3 TC001_03: Sign in when network is down

Test Case ID:	TC001_03		Test Case Description:	Check how the system responds if the network is unavailable during login.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User account already exists.		1	Email: abc@gmail.com	
2	Device is disconnected from the internet.		2	Password: abc123	
Test Conditions:		Internet connection is available.			
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to login screen		Login screen is displayed		
2	Enter registered email and wrong password		Input is accepted		
3	Click Login		Error message "No internet connection" is displayed		

Figure 2.1.3.1: Test Case TC001_03

2.2 Test TC002 for Module Authentication: UC002 - Register Account

This test contains the following test cases:

2.2.1 TC002_01: Register with valid details via email

Test Case ID:	TC002_01		Test Case Description:	Register a new user with valid details via email.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	The user must not have an existing account in the system		1	Email: johndoe@gmail.com	
			2	Password: johndoe123	
			3	Name: John Doe	
			4	Age: 22	
Test Conditions:	All fields are filled correctly, internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to registration screen		Registration screen is displayed		
2	Fill in the registration form with valid data		Form fields are populated correctly		
3	Click Register		System processes registration		
4	System validates data and creates a new account		User is successfully registered and redirected to the dashboard		

Figure 2.2.1.1: Test Case TC002_01

2.2.2 TC002_02: Register with already used email

Test Case ID:	TC002_02		Test Case Descripton:	Ensure the system prevents registration with an already used email.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	Email johndoe@gmail.com is already registered		1	Email: johndoe@gmail.com
			2	Password: johndoe123
			3	Name: John Doe
			4	Age: 22
Test Conditions:		All fields are filled correctly, internet connection is available.		
Step#	Step Details		Expected Result	Pass/Fail
1	Navigate to registration screen		Registration screen is displayed	
2	Fill in the registration form with an already registered email		Form fields are populated correctly	
3	Click Register		System shows an error message: "Email already in use. Please log in or use a different email."	

Figure 2.2.2.1: Test Case TC002_02

2.2.3 TC002_03: Google sign-up fails due to network

Test Case ID:	TC002_03		Test Case Description:	Ensure the system handles network issues during Google sign-up.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is attempting Google Sign-up and has network issues		1	Email: alice@gmail.com	
Test Conditions:	Internet connection is unstable.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to registration screen		Registration screen is displayed		
2	Click "Sign up with Google"		Google sign-up option is selected		
3	System attempts Google authentication		System waits for Google authentication		
4	Network error occurs		System shows an error message: "Network issue. Please try again."		

Figure 2.2.3.1: Test Case TC002_03

2.3 Test TC003 for Module Authentication: UC003 - Reset Password

This test contains the following test cases:

2.3.1 TC003_01: Reset password with registered email

Test Case ID:	TC003_01		Test Case Description:	Ensure the user can successfully reset their password with a registered email.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User has a registered email in the system		1	Email: johndoe@gmail.com
Test Conditions: Internet connection is available.				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to the "Forgot Password" screen	Forgot password screen is displayed		
2	Enter registered email address	Input is accepted		
3	Click "Send Reset Link"	System sends a password reset email		
4	Check the inbox for the password reset email	User receives the password reset email		
5	Click the reset link and set a new password	User is able to set a new password		

Figure 2.3.1.1: Test Case TC003_01

2.3.2 TC003_02: Reset with unregistered email

Test Case ID:	TC003_02		Test Case Description:	Ensure the system shows an error when attempting to reset password with an unregistered email.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	Email nonexistent@gmail.com is not registered		1	Email: nonexistent@gmail.com
Test Conditions: Internet connection is available.				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to the "Forgot Password" screen	Forgot password screen is displayed		
2	Enter unregistered email address	Input is accepted		
3	Click "Send Reset Link"	System shows an error message: "Email not found. Please check your email address."		

Figure 2.3.2.1: Test Case TC003_02

2.3.3 TC003_03: Sign in when network is down

Test Case ID:	TC003_03		Test Case Description:	Ensure the system handles network issues during the password reset process.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User has a registered email in the system		1	Email: johndoe@gmail.com
Test Conditions: Internet connection is unstable				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to the "Forgot Password" screen	Forgot password screen is displayed		
2	Enter registered email address	Input is accepted		
3	Click "Send Reset Link"	System shows an error message: "Network error. Please try again later."		

Figure 2.3.3.1: Test Case TC003_03

2.4 Test TC004 for Module Profile: UC004 - Manage Profile

This test contains the following test cases:

2.4.1 TC004_01: Update profile info with valid input

Test Case ID:	TC004_01		Test Case Description:	Ensure the user can successfully update their profile with valid input.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has existing profile data		1	Name: John Doe
			2	Age: 22
			3	Email: johndoe@gmail.com
Test Conditions:	Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail	
1	Navigate to profile screen	Profile screen is displayed		
2	Edit fields with valid data	Fields accept the new input		
		Profile is saved		
3	Click "Save"	System shows confirmation message: "Profile updated successfully"		

Figure 2.4.1.1: Test Case TC004_01

2.4.2 TC004_02: Submit empty required fields

Test Case ID:	TC004_02		Test Case Description:	Ensure the system prevents submission of profile with empty required fields.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has existing profile data		1	Name: John Doe
			2	Age: 22
			3	Email: johndoe@gmail.com
Test Conditions:	Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail	
1	Navigate to profile screen	Profile screen is displayed		
2	Leave required fields empty (e.g., Name or Email)	Fields are empty and form is not submitted		
3	Click "Save"	System shows an error message: "This field is required"		

Figure 2.4.2.1: Test Case TC004_02

2.4.3 TC004_03: Cancel profile editing

Test Case ID:	TC004_03		Test Case Description:	Ensure the user can cancel profile editing without saving changes.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has existing profile data		1	Name: John Doe
			2	Age: 22
			3	Email: johndoe@gmail.com
Test Conditions:	Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail	
1	Navigate to profile screen	Profile screen is displayed		
2	Edit profile fields	Fields are populated with new data		
3	Click "Cancel"	Profile editing is canceled, no changes are saved User returns to previous screen		

Figure 2.4.3.1: Test Case TC004_03

2.4.4 TC004_04: Update fails due to database error

Test Case ID:	TC004_04		Test Case Description:	Ensure the system handles errors during profile update due to database failure.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has existing profile data		1	Name: John Doe
			2	Age: 22
			3	Email: johndoe@gmail.com
Test Conditions:	Internet connection is available, but there is a database issue.			
Step#	Step Details	Expected Result	Pass/Fail	
1	Navigate to profile screen	Profile screen is displayed		
2	Edit profile fields	Fields accept the new input		
3	Click "Save"	System shows an error message: "Unable to save changes. Please try again later."		

Figure 2.4.4.1: Test Case TC004_04

2.5 Test TC005 for Module Profile: UC005 - Manage Notifications

This test contains the following test cases:

2.5.1 TC005_01: Enable reminders and save

Test Case ID:	TC005_01		Test Case Description:	Ensure the user can enable reminders and save the settings.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has access to notification settings		1	Enable reminders: Yes
Test Conditions: Internet connection is available				
Step#	Step Details		Expected Result	Pass/Fail
1	Navigate to notification settings screen		Notification settings screen is displayed	
2	Toggle reminders setting to "On"		Reminder setting is enabled Settings are saved.	
3	Click "Save"		System shows confirmation message: "Settings updated successfully"	

Figure 2.5.1.1: Test Case TC005_01

2.5.2 TC005_02: Change setting but exit without saving

Test Case ID:	TC005_02		Test Case Description:	Ensure the user can change settings but exit without saving.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has access to notification settings		1	Enable reminders: No
Test Conditions: Internet connection is available				
Step#	Step Details		Expected Result	Pass/Fail
1	Navigate to notification settings screen		Notification settings screen is displayed	
2	Toggle reminders setting to "On"		Reminder setting is toggled to "On" Changes are not saved.	
3	Click "Back" or exit without clicking "Save"		User is returned to the previous screen without any changes being saved	

Figure 2.5.2.1: Test Case TC005_02

2.5.3 TC005_03: Fail to save due to connectivity issue

Test Case ID:	TC005_03		Test Case Description:	Ensure the system handles failure to save due to connectivity issues	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is logged in and has access to notification settings		1	Enable reminders: Yes	
Test Conditions:		Internet connection is available			
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to notification settings screen		Notification settings screen is displayed		
2	Toggle reminders setting to "On"		Reminder setting is enabled		
3	Click "Save"		System shows an error message: "Failed to save. Please check your connection."		

Figure 2.5.3.1: Test Case TC005_03

2.6 Test TC006 for Module Mental Health Support: UC006 - Track Emotions

This test contains the following test cases:

2.6.1 TC006_01: Log emotion with intensity and note

Test Case ID:	TC006_01		Test Case Description:	Ensure the user can successfully log an emotion with intensity and optional notes.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is logged in and has access to emotion tracking		1	Emotion: Sad	
			2	Intensity: 3	
			3	Note: "Feeling low today"	
Test Conditions:		Internet connection is available			
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to emotion tracking screen		Emotion tracking screen is displayed		
2	Select emotion (e.g., Sad) and set intensity		Emotion and intensity are selected correctly		
3	Enter a note (optional)		Note is entered into the system		
4	Click "Submit"		Emotion log is saved successfully.		
			System shows confirmation: "Emotion log saved successfully"		

Figure 2.6.1.1: Test Case TC006_01

2.6.2 TC006_02: Submit without selecting emotion

Test Case ID:	TC006_02		Test Case Description:	Ensure the system shows an error when the user submits without selecting an emotion.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is logged in and has access to emotion tracking		1	No emotion selected	
Test Conditions:	Internet connection is available				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to emotion tracking screen		Emotion tracking screen is displayed		
2	Do not select any emotion		No emotion selected		
3	Click "Submit"		System shows an error message: "Please select an emotion"		

Figure 2.6.2.1: Test Case TC006_02

2.6.3 TC006_03: Skip emotion tracking popup

Test Case ID:	TC006_03		Test Case Description:	Ensure the user can skip the emotion tracking popup.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has access to emotion tracking		1	No emotion logged yet
Test Conditions: Internet connection is available				
Step#	Step Details	Expected Result	Pass/Fail	
1	Launch the app	Emotion tracking popup is displayed		
2	Click "Skip"	Emotion tracking prompt is dismissed User is returned to the app home screen		

Figure 2.6.3.1: Test Case TC006_03

2.6.4 TC006_04: Fail to save log due to connectivity issue

Test Case ID:	TC006_04		Test Case Description:	Ensure the system handles failure to save the emotion log due to a connectivity issue.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has access to emotion tracking		1	Emotion: Happy
			2	Intensity: 4
			3	Note: "Feeling great today"
Test Conditions: Internet connection is available				
Step#	Step Details	Expected Result	Pass/Fail	
1	Navigate to emotion tracking screen	Emotion tracking screen is displayed		
2	Select emotion (e.g., Happy) and set intensity	Emotion and intensity are selected correctly		
3	Enter a note (optional)	Note is entered into the system		
4	Click "Submit"	System shows an error message: "Failed to save. Please check your connection."		

Figure 2.6.4.1: Test Case TC006_04

2.7 Test TC007 for Module Mental Health Support: UC007 - View Insights

This test contains the following test cases:

2.7.1 TC007_01: View weekly emotion trends

Test Case ID:	TC007_01	Test Case Description:	Ensure the user can view the weekly emotion trends.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	User is logged in and has existing emotion data	1	Emotion logs for the past 7 days
Test Conditions: Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to Insights screen	Insights screen is displayed	
2	Select "Weekly" option to view emotion trends	Weekly trends for the past 7 days are displayed	

Figure 2.7.1.1: Test Case TC007_01

2.7.2 TC007_02: Filter insights by custom date range

Test Case ID:	TC007_02	Test Case Description:	Ensure the user can filter emotion insights by a custom date range.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	User is logged in and has existing emotion data	1	Emotion logs for the past 30 days
Test Conditions: Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to Insights screen	Insights screen is displayed	
2	Select the "Custom" option for date range	Custom date range option is selected	
3	Input a custom date range (e.g., last 10 days)	Emotion data for the selected date range is displayed	

Figure 2.7.2.1: Test Case TC007_02

2.7.3 TC007_03: View insights with no data

Test Case ID:	TC007_03		Test Case Description:	Ensure the system shows an appropriate message when there are no emotion logs to view.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in, but has not logged any emotions		1	No emotion data available
Test Conditions: Internet connection is available				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to Insights screen	Insights screen is displayed		
2	Select "Weekly" or "Custom" date range	System shows message: "No data available. Start tracking your emotions to view insights."		

Figure 2.7.3.1: Test Case TC007_03

2.8 Test TC008 for Module Mental Health Support: UC008 - Access AI Chatbot

This test contains the following test cases:

2.8.1 TC008_01: Chat with AI assistant using a mood prompt

Test Case ID:	TC008_01	Test Case Description:	Ensure the user can chat with the AI assistant using a mood prompt.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	User is logged in and has access to the AI chatbot	1	Mood prompt: "I feel stressed"
Test Conditions: Internet connection is available			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to chatbot screen	Chatbot screen is displayed	
2	Type a mood prompt (e.g., "I feel stressed")	System processes the input	
3	Click "Send"	AI assistant responds with a relevant reply System displays the assistant's reply in the chat window	

Figure 2.8.1.1: Test Case TC008_01

2.8.2 TC008_02: Chatbot response fails due to API issue

Test Case ID:	TC008_02	Test Case Description:	Ensure the system handles chatbot response failure due to an API issue.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	User is logged in and has access to the AI chatbot	1	Mood prompt: "I feel anxious"
Test Conditions: Internet connection is available, but the AI API is down.			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to chatbot screen	Chatbot screen is displayed	
2	Type a mood prompt (e.g., "I feel anxious")	System attempts to send the input to the AI API	
3	Click "Send"	System fails to get a response from the AI System shows an error message: "Chatbot service unavailable, please try again later."	

Figure 2.8.2.1: Test Case TC008_02

2.9 Test TC009 for Module Engagement: UC009 - Gamification

This test contains the following test cases:

2.9.1 TC009_01: Earn badge after 3-day check-in streak

Test Case ID:	TC009_01		Test Case Description:	Ensure the user earns a badge after a 3-day check-in streak.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User has logged in for 3 consecutive days		1	3 consecutive emotion logs
Test Conditions: Internet connection is available.				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to Gamification screen	Gamification screen is displayed Streak counter shows 3 days		
2	Check if the 3-day streak is active	User earns a badge for the 3-day streak Badge is displayed and notification shown		

Figure 2.9.1.1: Test Case TC009_01

2.9.2 TC009_02: View badge progress

Test Case ID:	TC009_02		Test Case Description:	Ensure the user can view their badge progress.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User has logged in and earned at least one badge		1	Badge: 3-day streak badge
Test Conditions: Internet connection is available.				
Step#	Step Details	Expected Result		Pass/Fail
1	Navigate to Gamification screen	Gamification screen is displayed Badge progress screen is displayed		
2	Click on "View Progress"	User can see earned badges and progress toward other badges		

Figure 2.9.2.1: Test Case TC009_02

2.9.3 TC009_03: Fail to unlock badge due to save error

Test Case ID:	TC009_03		Test Case Description:	Ensure the system handles failure to unlock a badge due to save error.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User has completed the criteria to unlock a badge (e.g., 3-day streak)		1	Badge: 3-day streak badge	
			2	Streak count: 3	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result	Pass/Fail	
1	Navigate to Gamification screen		Gamification screen is displayed		
2	Click on "View Progress"		Badge progress screen is displayed		
			Badge progress shows 3-day streak		
3	Attempt to unlock badge		System attempts to unlock badge		
			System shows an error message: "Failed to unlock badge. Please try again later."		

Figure 2.9.3.1: Test Case TC009_03

2.10 Test TC010 for Module Engagement: UC010 - View Progress History

This test contains the following test cases:

2.10.1 TC010_01: View emotion and activity logs

Test Case ID:	TC010_01		Test Case Description:	Ensure the user can view their emotion and activity logs from the progress history screen.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has historical data available (e.g., emotion logs, chatbot sessions)		1	Emotion log: "Happy"
			2	Activity: "Breathing exercise"
Test Conditions: Internet connection is available.				
Step#	Step Details		Expected Result	Pass/Fail
1	Navigate to Progress History screen		Progress history screen is displayed	
2	System fetches emotion logs, chatbot sessions, and achievements		System retrieves historical data from the database	
3	System displays a categorized list of entries		Entries are categorized (e.g., Emotion logs, Chat sessions, Achievements)	
4	The user browses through the timeline or filters by date/type		Logs are displayed by date or category as selected	
5	The user selects an entry (e.g., chat session or achievement)		Entry details are displayed	

Figure 2.10.1.1: Test Case TC010_01

2.10.2 TC010_02: Filter history by date

Test Case ID:	TC010_02		Test Case Description:	Ensure the user can filter emotion and activity logs by date.
Created by:	Lau Yun Xi		Version:	1.0
No.	Prerequisites		No.	Test Data
1	User is logged in and has historical data available (e.g., emotion logs, chatbot sessions)		1	Emotion log: "Happy"
Test Conditions: Internet connection is available.				
Step#	Step Details		Expected Result	Pass/Fail
1	Navigate to Progress History screen		Progress history screen is displayed	
2	Select the date filter option (e.g., "Last 7 days", "Custom range")		Available filters are shown and selectable	
3	Choose a custom date range or predefined filter (e.g., "Last 30 days")		Emotion and activity logs are filtered by the selected date range	
4	System displays logs based on the selected date range		Logs displayed match the selected date range	

Figure 2.10.2.1: Test Case TC010_02

2.10.3 TC010_03: Open history with no data

Test Case ID:	TC010_03		Test Case Description:	Ensure the system shows a proper message when there is no historical data available.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	User is logged in but has not logged any emotions or performed activities		1	No emotion or activity logs available	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to Progress History screen		Progress history screen is displayed		
			System checks for available logs		
			System shows a message: "No data available. Start tracking your emotions and activities."		

Figure 2.10.3.1: Test Case TC010_03

2.11 Test TC011 for Module Admin: UC011 - Manage User Accounts

This test contains the following test cases:

2.11.1 TC011_01: Reset user password from admin panel

Test Case ID:	TC011_01	Test Case Description:	Ensure the admin can reset a user's password from the admin panel.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	Admin is logged in and has access to the user management panel	1	Email: admin@gmail.com
		2	Password: admin123
Test Conditions: Internet connection is available.			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to the Admin Panel	Admin Panel is displayed	
2	Search for the user by email	System shows the user's details	
3	Click "Reset Password"	System initiates password reset for the selected user Admin receives a success message: "Password reset successfully"	

Figure 2.11.1.1: Test Case TC011_01

2.11.2 TC011_02: Search user by email

Test Case ID:	TC011_02	Test Case Description:	Ensure the admin can search for a user by their email.
Created by:	Lau Yun Xi	Version:	1.0
No.	Prerequisites	No.	Test Data
1	Admin is logged in and has access to the user management panel	1	Email: admin@gmail.com
		2	Password: admin123
Test Conditions: Internet connection is available.			
Step#	Step Details	Expected Result	Pass/Fail
1	Navigate to the Admin Panel	Admin Panel is displayed	
2	Enter the user email in the search bar	Input is accepted	
3	Click "Search"	System displays the user details The user's information is shown correctly	

Figure 2.11.2.1: Test Case TC011_02

2.11.3 TC011_03: Fail to load user list due to network issue

Test Case ID:	TC011_03		Test Case Description:	Ensure the system handles failure to load the user list due to network issues.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	Admin is logged in and has access to the user management panel		1	Email: admin@gmail.com	
			2	Password: admin123	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to the Admin Panel		Admin Panel is displayed		
2	Click on "View Users"		System attempts to fetch user list		
			System shows an error message: "Unable to load user list. Please check your connection."		
			System shows an error message: "Unable to load user list. Please check your connection."		

Figure 2.11.3.1: Test Case TC011_03

2.11.4 TC011_04: Delete user account from admin panel

Test Case ID:	TC011_04		Test Case Description:	Ensure the admin can delete a user account from the admin panel.	
Created by:	Lau Yun Xi		Version:	1.0	
No.	Prerequisites		No.	Test Data	
1	Admin is logged in and has access to the user management panel		1	Email: admin@gmail.com	
			2	Password: admin123	
Test Conditions:	Internet connection is available.				
Step#	Step Details		Expected Result		Pass/Fail
1	Navigate to the Admin Panel		Admin Panel is displayed		
2	Click on "View Users"		System attempts to fetch user list		
3	Click "Delete"		System asks for confirmation before deleting the account		
4	Admin confirms the deletion		User account is successfully deleted from the system		
			System shows a confirmation message: "User account deleted successfully"		

Figure 2.11.4.1: Test Case TC011_04

Appendix A: Traceability Matrix

Test Case ID	Use Case ID/ Sequence Diagram ID	Package ID
TC001 for Authentication Subsystem • TC001_01 • TC001_02 • TC001_03 • TC001_04	UC001 • SD001	P001
TC002 for Authentication Subsystem • TC002_01 • TC002_02 • TC002_03	UC002 • SD002	P001
TC003 for Authentication Subsystem • TC003_01 • TC003_02 • TC003_03	UC003 • SD003	P001
TC004 for Profile Subsystem • TC004_01 • TC004_02 • TC004_03 • TC004_04	UC004 • SD004	P002
TC005 for Profile Subsystem • TC005_01 • TC005_02 • TC005_03	UC005 • SD005	P002
TC006 for Mental Health Support Subsystem • TC006_01 • TC006_02 • TC006_03 • TC006_04	UC006 • SD006	P003
TC007 for Mental Health Support Subsystem	UC007	P003

• TC007_01 • TC007_02 • TC007_03	• SD007	
TC008 for Mental Health Support Subsystem • TC008_01 • TC008_02 • TC008_03	UC008 • SD008	P003
TC009 for Gamification Subsystem • TC009_01 • TC009_02 • TC009_03	UC009 • SD009	P004
TC010 for Gamification Subsystem • TC010_01 • TC010_02 • TC010_03	UC010 • SD010	P004
TC011 for Admin Subsystem • TC011_01 • TC011_02 • TC011_03	UC011 • SD011	P005